Class 12th

PHYSICS

Short Notes

JEE MAIN | JEE ADVANCED | NEET

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PURPOSE OF PHYSICS SHORT NOTES

These hand written short notes are prepared by Mohit Goenka, Founder of Eduniti and Alumnus of IIT Kharagpur. It is prepared with an aim to help JEE and NEET aspirants to revise physics in best possible manner.

It has been prepared considering all concepts and referring to past year's questions. You can also map your notes with this one.

To make best use of this file you must complete '**Revision Series Playlist from our YouTube channel Eduniti or our Android App EDUNITI on Play store**'. This series contains 60+ videos completing in 27hrs.

****Link of Revision Series Playlist:* https://bit.ly/3eBbib9

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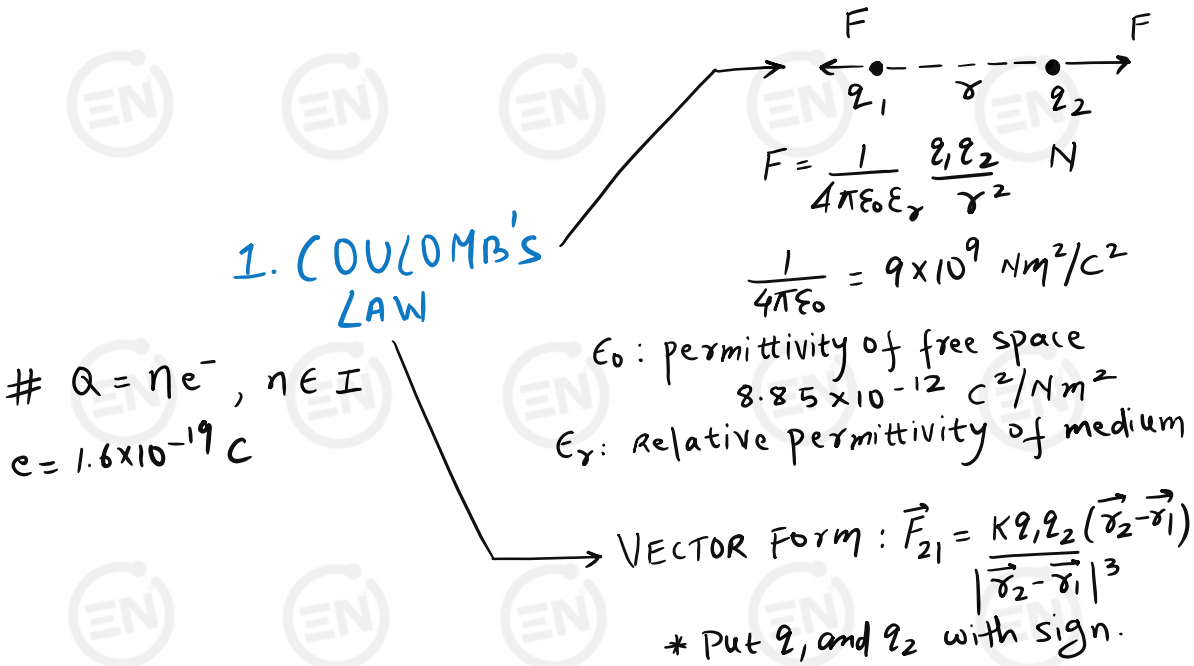
Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

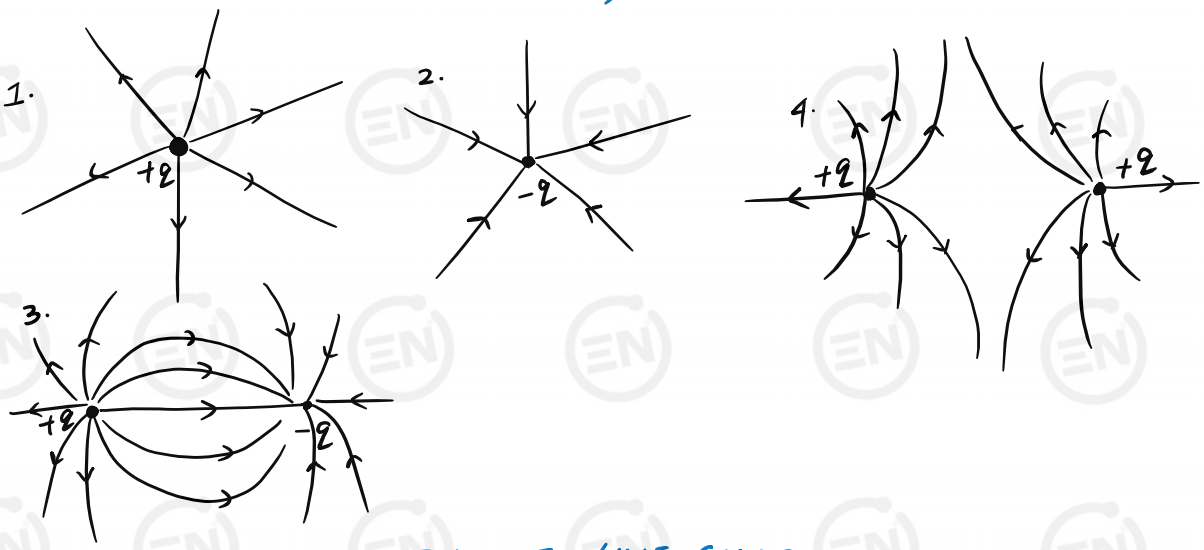
Topics to cover in ELECTROSTATICS – PART 1 (ELECTRODYNAMICS)

1. Coulomb's Law
2. Electric Field & Standard Line diagram
3. Electric Field due to line charge
4. Electric Field due to a charged ring
5. Electric Field due to a charged disc
6. Electric Field due to a charged sphere
7. Electric Field due to Non-Uniform Charge Distribution
8. Electrostatic Potential (ring & sphere)
9. Electrostatic Potential Energy (self energy)
10. Relation between E & V
11. Electric Dipole
12. Electric Flux
13. Conductors

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel



2. ELECTRIC FIELD ($E = KQ/r^2$), N/C

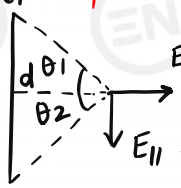


3. ELECTRIC FIELD DUE TO LINE CHARGE

(CHARGE IS UNIFORMLY DISTRIBUTED)

FINITE LENGTH

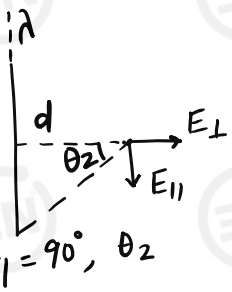
λ C/m



$$E_\perp = \frac{K\lambda}{d}(\sin\theta_1 + \sin\theta_2)$$

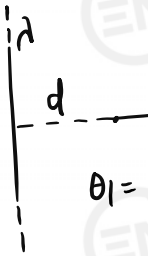
$$E_{||} = \frac{K\lambda}{d}(\cos\theta_2 - \cos\theta_1)$$

SEMI-INFINITE



$\theta_1 = 90°$, θ_2

IN FINITE



$\theta_1 = \theta_2 = 90°$

$$E_\perp = \frac{2K\lambda}{d} = \frac{\lambda}{2\pi\varepsilon_0 d}$$

$E_{||} = 0$

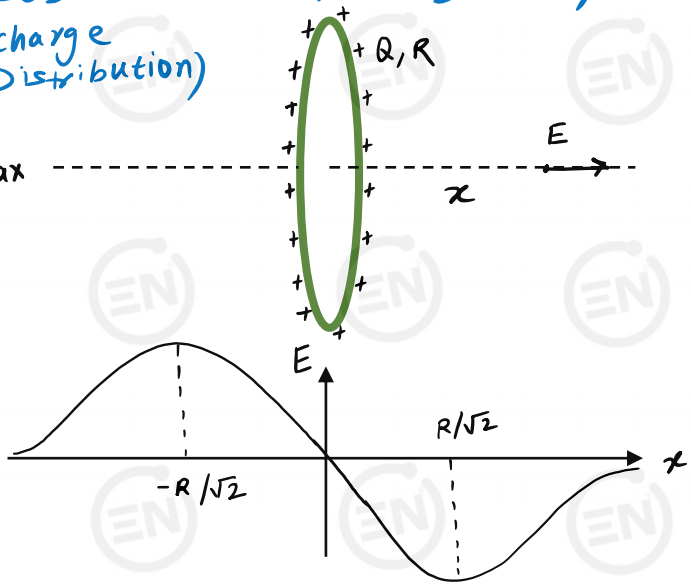
4. ELECTRIC FIELD DUE TO CHARGED RING

(uniform charge Distribution)

$$E = \frac{KQx}{(x^2+R^2)^{3/2}}$$

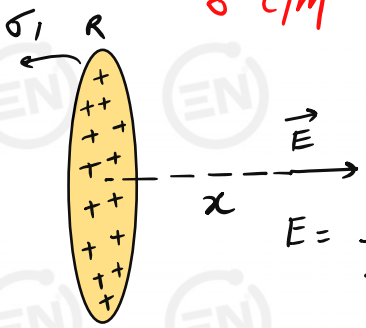
↳ at $x = \pm R/\sqrt{2}$, E is Max

↳ at $x = 0$, $E = 0$ (center)



5. ELECTRIC FIELD DUE TO CHARGED DISC

σ C/m^2 (UNIFORM CHARGE Distribution)



$$E = \frac{\sigma}{2\epsilon_0}\left(1 - \frac{x}{\sqrt{x^2+R^2}}\right)$$

Case If Disc is very large ($x \ll R$)

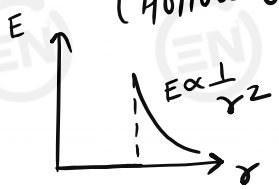
$$\boxed{E = \frac{\sigma}{2\epsilon_0}}$$

↳ for infinite sheet

6. ELECTRIC FIELD DUE TO CHARGED SPHERE

CONDUCTOR

(Hollow or solid, Q is on surface)

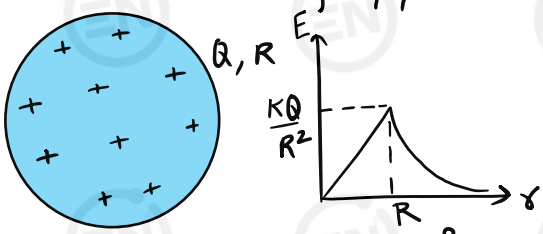


(1) For $r < R$, $E = 0$

(2) For $r \geq R$, $E = \frac{KQ}{r^2}$

NON-CONDUCTOR

(uniformly in solid) ρ C/m^3

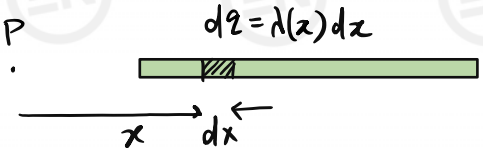


(1) $r < R$, $E = \frac{KQr}{R^3}$ or $\frac{\rho r}{3\epsilon_0}$

(2) $r \geq R$, $E = \frac{KQ}{r^2}$

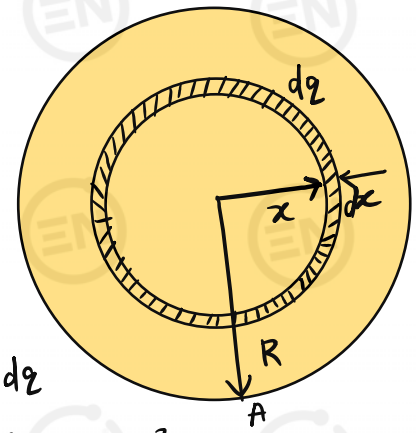
7. ELECTRIC FIELD (NON-UNIFORM CHARGE Distribution)

1.



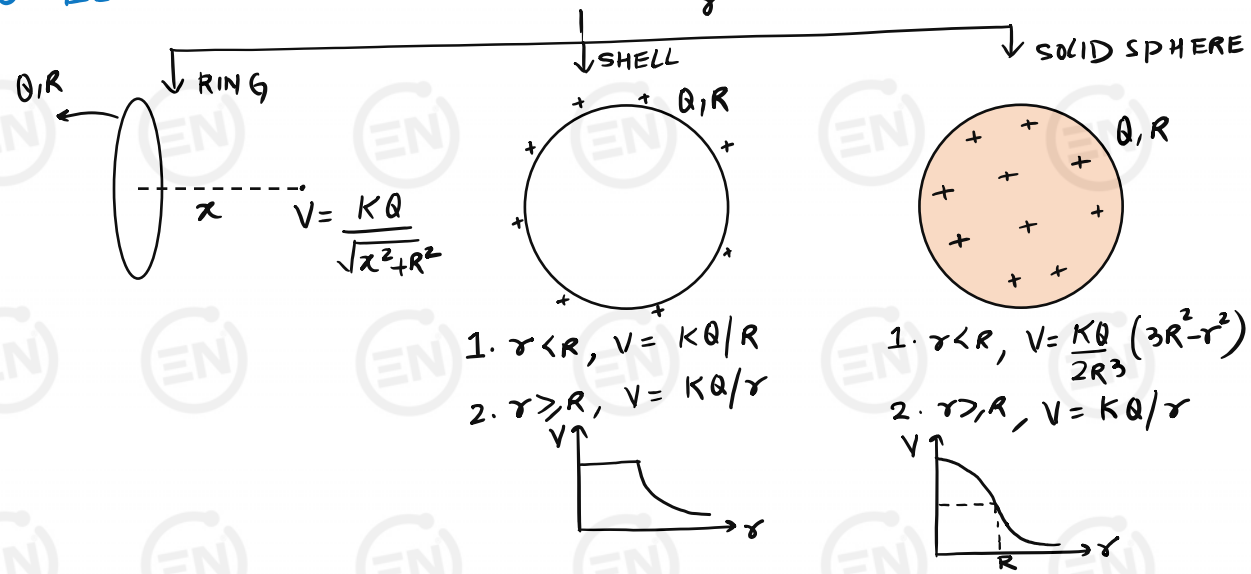
$$E = \int_a^b \frac{K\lambda(x)\,dx}{x^2}$$

2.



$$E_A = \frac{KQ_{in}}{R^2} \qquad Q_{in} = \int dq = \int_0^R \rho(x) \times 4\pi x^2\,dx$$

8. ELECTROSTATIC POTENTIAL $V = \frac{Kq}{r}$, Put q with sign.

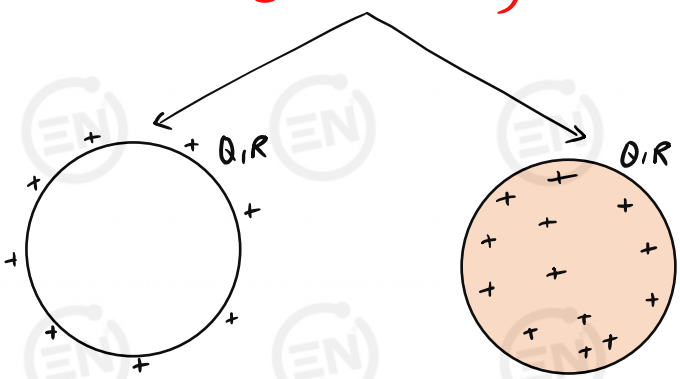


9. ELECTROSTATIC POTENTIAL ENERGY

$U = Kq_1q_2/r$

↳ put q_1 and q_2 with sign

SELF ENERGY



10. RELATION BETWEEN E and V

(1.) $\vec{E} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k}$

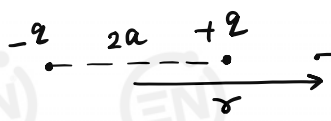
(2.) $\Delta V = -\int \vec{E} \cdot d\vec{r}$

Here $\frac{\partial V}{\partial x}$ means differentiate V w.r.t x keeping y and z constant.

11. ELECTRIC DIPOLE ($-q$ ---- $+q$, $P = qd$, direction from -ve to +ve)

(a) ELECTRIC FIELD

AXIAL POINT



$$E = \frac{2PKr}{(r^2-a^2)^2}$$

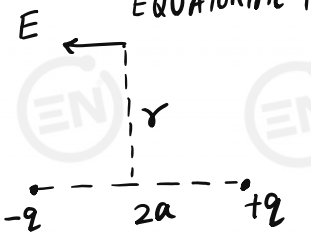
* SHORT DIPOLE ($r >> a$)

$$E_{axial} = 2PK/r^3$$

Direction : Along $\vec{P}$

$P = 2aq$

EQUATORIAL POINT



$$E = \frac{PK}{(r^2+a^2)^{3/2}}$$

* SHORT DIPOLE ($r >> a$)

$$E_{eq} = PK/r^3$$

Dirn : opp $\vec{P}$

(b) POTENTIAL

AXIAL POINT

$$V = \frac{PK}{r^2-a^2}$$

SHORT DIPOLE ($r >> a$)

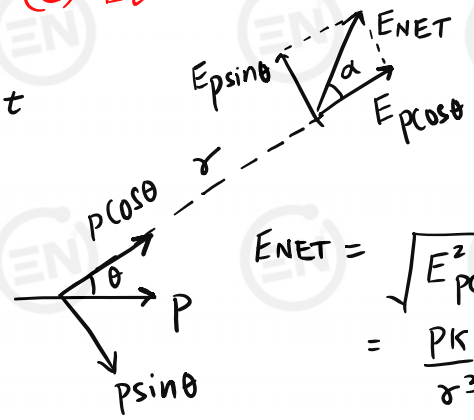
$$V = \frac{PK}{r^2}$$

$P = 2aq$

Equatorial Point

$V = 0$

(c) ELECTRIC FIELD AT GENERAL POINT



$$E_{p\cos\theta} = \frac{2p\cos\theta K}{r^3}$$

$$E_{p\sin\theta} = \frac{p\sin\theta K}{r^3}$$

$$E_{NET} = \sqrt{E^2_{p\cos\theta} + E^2_{p\sin\theta}}$$

$$= \frac{PK}{r^3}\sqrt{1+3\cos^2\theta}$$

$$\boxed{\tan\alpha = \frac{\tan\theta}{2}}$$

(d) DIPOLE IN E (uniform)

TORQUE

$\vec{\tau} = \vec{P} \times \vec{E}$

↳ SHM based Question

Potential Energy

$U = -\vec{P}.\vec{E}$

STABLE ↳ $\theta = 0°$, $U_{min} = -PE$

UNSTABLE ↳ $\theta = 180°$, $U_{max} = PE$

↳ $\theta = 90°$, $U = 0$

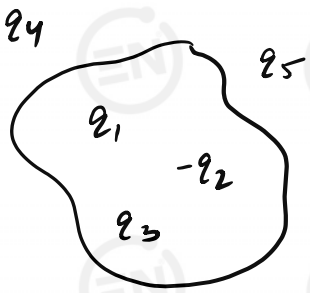
12. ELECTRIC FLUX ($\phi = \vec{E} \cdot \vec{A}$)

GAUSS's LAW

↳ $\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\varepsilon_0}$

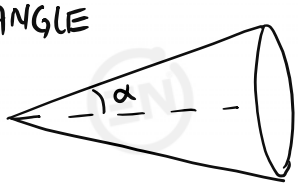
① q_{in} : charge enclosed

② E : Electric field is due to all the charges.



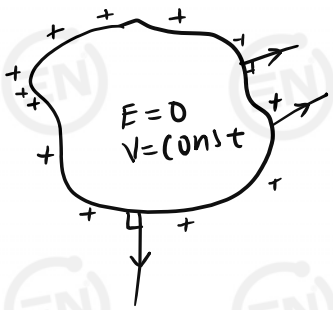
$\phi = \frac{q_1 - q_2 + q_3}{\varepsilon_0}$

Solid ANGLE



$\Omega = 2\pi(1 - \cos\alpha)$

13. CONDUCTOR



(1.) charge remains on surface

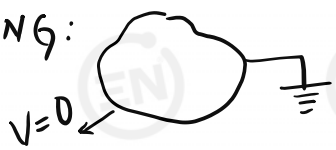
(2.) Electric field inside is zero

(3.) V is constant

(4.) Field Lines are ⊥ to surface

(5.) CONNECTING TWO CONDUCTORS
↳ they share charge untill V of both bodies are same.

(6.) EARTHING:



V of body will always be zero

Space to add concepts learnt from PYQs if any

Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

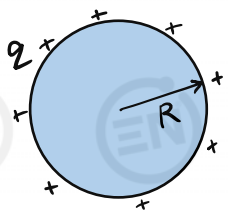
Topics to cover in CAPACITORS – PART 2 (ELECTRODYNAMICS)

1. Spherical and Cylindrical Capacitors
2. Parallel Plate capacitors
3. Charge, Energy Stored, Work done by Battery, Heat
4. Force Between Plates
5. Series / Parallel
6. Alternative figure for plate arrangement
7. Wheatstone bridge (Balanced & Unbalanced)
8. Charge Sharing & Heat generated
9. Dielectric in Electric Field (Induced Charge)
10. Dielectric Slab in Capacitor
11. Effect of slab insertion in a capacitor (at Const. V and Q)
12. Capacitance for Multiple Dielectric Medium
13. Capacitance for Variable K
14. Capacitance for variable dimensions
15. RC – Charging and Discharging

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. CAPACITANCE (unit: Farad)

ISOLATED SPHERE

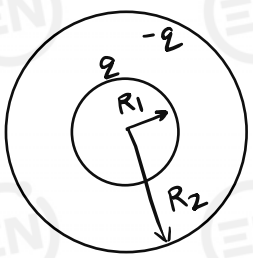


$$C = \frac{q}{V} = \frac{q}{Kq/R}$$

$$C = 4\pi\varepsilon_0 R$$

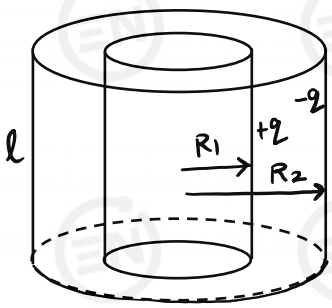
TWO CONDUCTORS

Spherical



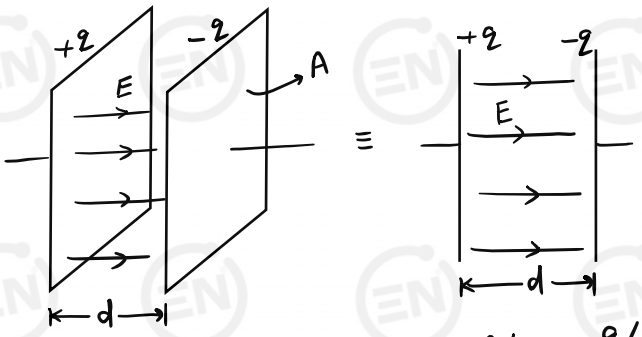
$$C = \frac{4\pi\varepsilon_0 R_1 R_2}{R_2 - R_1}$$

Cylindrical



$$C = \frac{2\pi\varepsilon_0 l}{\ln(R_2/R_1)}$$

2. PARALLEL PLATE CAPACITOR



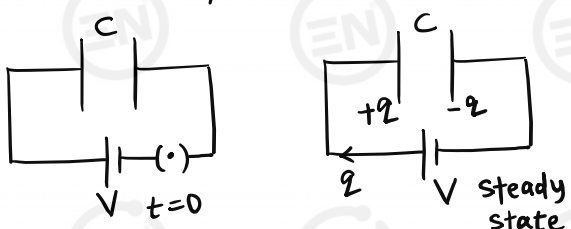
$C = q/V = q/Ed$

$(E = \frac{\sigma}{\varepsilon_0} = \frac{q}{A\varepsilon_0})$

$$\Rightarrow \boxed{C = \frac{A\varepsilon_0}{d}}$$

3. CHARGE / ENERGY STORED

$W_{battery}$ / HEAT DISSIPATION



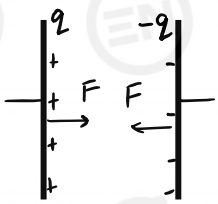
(i) $q = CV$

(ii) $W_b = q_{flown} \times V = CV^2$

(iii) $U = \frac{1}{2}CV^2$ $\{q^2/2C$

(iv) Heat Dissipated
$= W_b - \Delta U$
$= CV^2 - \frac{1}{2}CV^2 = \frac{1}{2}CV^2$

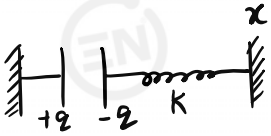
4. FORCE BETWEEN PLATES



$F = q \times E_{-q} = q \times \frac{q}{2A\varepsilon_0}$

$\Rightarrow F = q^2/2A\varepsilon_0$

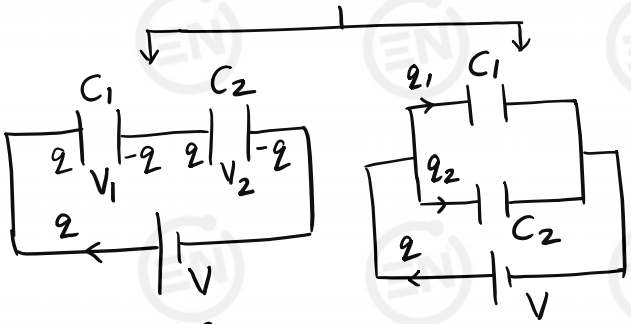
Ex:



x = spring elongation

$\therefore Kx = \frac{q^2}{2A\varepsilon_0}$

5. COMBINATION OF CAPACITOR



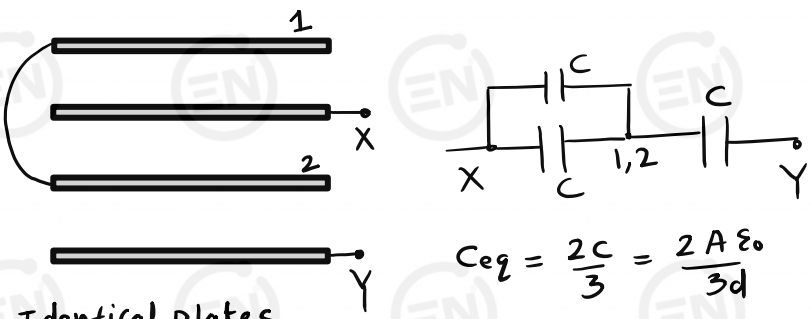
(i) $C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$ $\quad C_{eq} = C_1 + C_2$

(ii) $q = C_{eq} V$ $\quad q_1 = C_1 V,\ q_2 = C_2 V$

NOTE: (a) n identical capacitor in series, $C_{eq} = C/n$

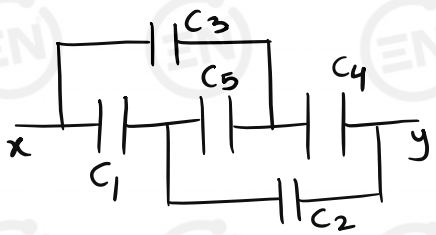
(b) If in parallel, $C_{eq} = nC$

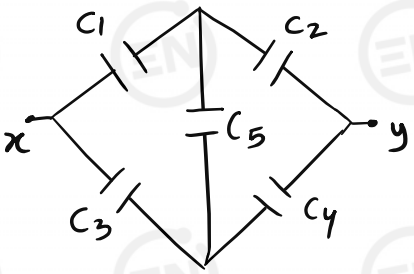
Alternatively:

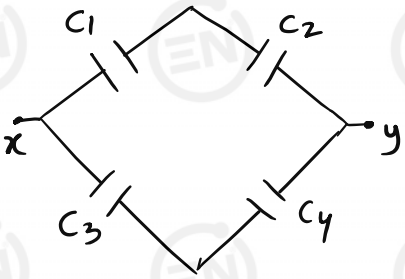


$C_{eq} = \frac{2C}{3} = \frac{2A\varepsilon_0}{3d}$

6. WHEATSTONE BRIDGE (BALANCED)







If $C_1C_4 = C_2C_3$

7. TECHNIQUE FOR UNBALANCED WHEATSTONE BRIDGE ($C_1C_4 \neq C_2C_3$)

(Point Potential + Junction Rule)

At x : $C_1(x-V) + C_5(x-y) + C_2(x-0) = 0$ — (1)

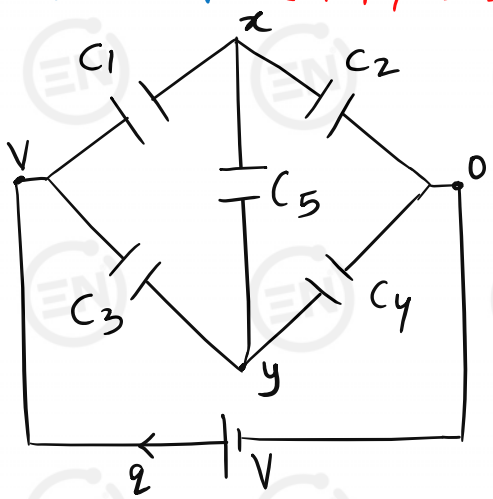
At y : $C_3(y-V) + C_5(y-x) + C_4(y-0) = 0$ — (2)

(a) solve (1) and (2) to find x and y.

(b) Then we can find q_1, q_2, q_3, q_4 and q_5

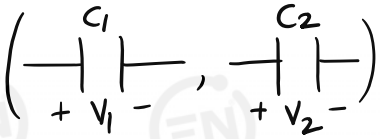
(c) $q = q_1 + q_3$

(d) $C_{eq} = \frac{q}{V}$

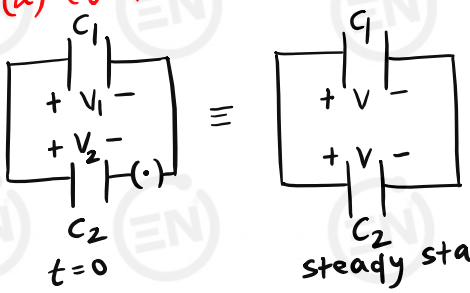


#NOTE : This method can be used to solve any kind of circuit.

8. CHARGE SHARING AND HEAT GENERATED

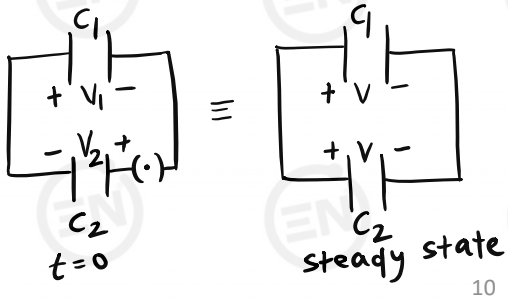


(a) CONNECTED SAME POLARITY



(i) $V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$ (ii) $H = \frac{C_1C_2}{2(C_1+C_2)}(V_1 - V_2)^2$

(b) CONNECTED OPPOSITE POLARITY

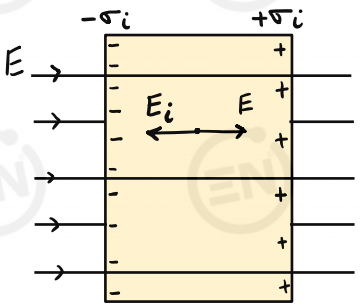


(i) $V = \frac{C_1V_1 - C_2V_2}{C_1 + C_2}$

(ii) $H = \frac{C_1C_2}{2(C_1+C_2)}(V_1 + V_2)^2$

9. DIELECTRIC IN EXTERNAL ELECTRIC FIELD

- Insulators (gets polarized in E)
- Dielectric Constant (K or ε_r)
  - for air/vacuum $K = 1$
  - for metal $K \to \infty$



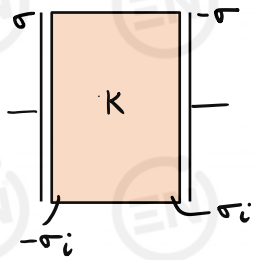
$E_{net} = E - E_i$

$\Rightarrow \frac{E}{K} = E - \frac{\sigma_i}{\varepsilon_0}$

$\Rightarrow \sigma_i = \varepsilon_0 E\left(1 - \frac{1}{K}\right)$

Induced charge density, $\sigma_i = \frac{Q_i}{A}$

10. SLAB IN CAPACITOR



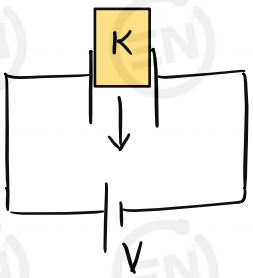
(i) $C = KA\varepsilon_0/d$

(ii) $\sigma_i = \sigma\left(1 - \frac{1}{K}\right)$

$Q_i = Q\left(1 - \frac{1}{K}\right)$

11. EFFECT OF INSERTING DIELECTRIC IN CAPACITOR

(a) At Constant V (Battery connected)



(i) $C \to KC$ (C↑)

(ii) $Q \to KQ$ (Q↑)

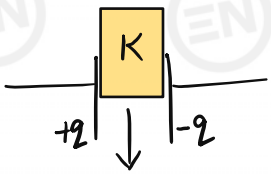
(iii) V is const.

(iv) E is const. $\left(E = \frac{V}{d}\right)$

(v) $U \to KU$ (U↑)

$\left(U = \frac{1}{2}CV^2\right)$

(b) At Constant charge (Battery removed)



(i) Q is const.

(ii) $C \to KC$ (C↑)

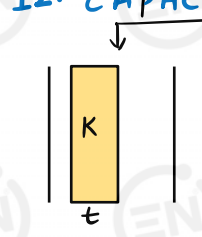
(iii) $V \to \frac{V}{K}$ (V↓)

$(V = Q/C)$

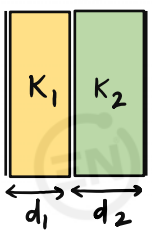
(iv) $E \to E/K$ (E↓)

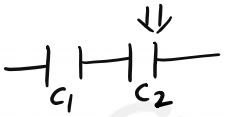
(v) $U \to U/K$ (U↓) $U = \frac{Q^2}{2C}$

12. CAPACITANCE FOR MULTIPLE DIELECTRIC MEDIUM

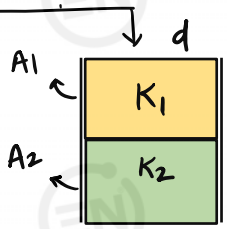


$C = \frac{A\varepsilon_0}{d - t + t/K}$

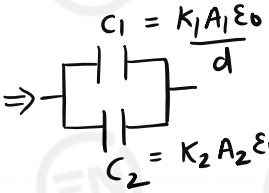


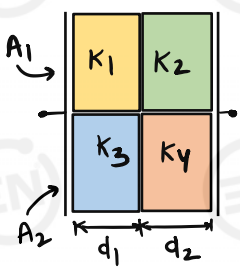


$C_1 = \frac{K_1 A\varepsilon_0}{d_1}$, $C_2 = \frac{K_2 A\varepsilon_0}{d_2}$

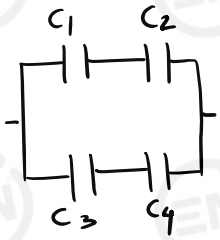


$\Rightarrow$



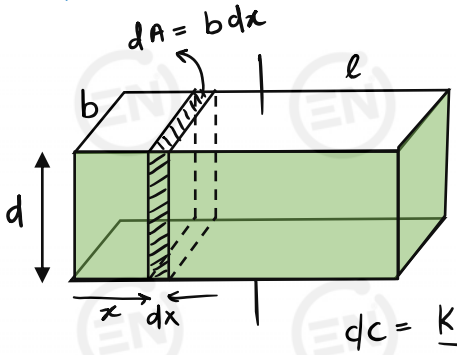


$\Rightarrow$



13. CAPACITANCE WITH VARIABLE K

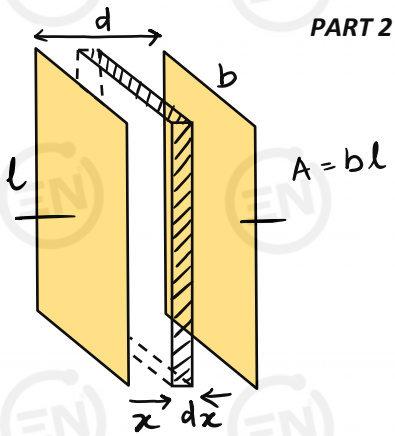
(a)



$$dC = \frac{K(x)\, b\,dx\, \varepsilon_0}{d}$$

$$\Rightarrow C = \frac{b\varepsilon_0}{d}\int_0^{\ell} K(x)\,dx$$

(b)

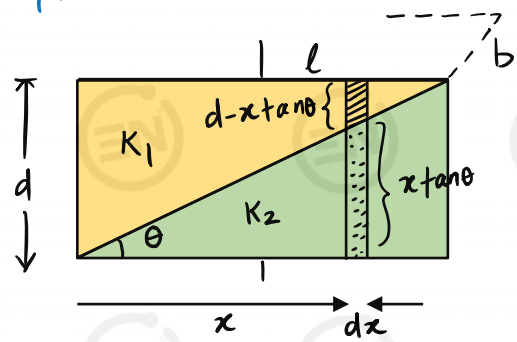


$$dC = \frac{K(x)\, A\, \varepsilon_0}{dx}$$

$$\Rightarrow \frac{1}{C} = \int \frac{1}{dC} = \frac{1}{A\varepsilon_0}\int_0^{d} \frac{dx}{K(x)}$$

14. CAPACITANCE WITH VARIABLE DIMENSIONS

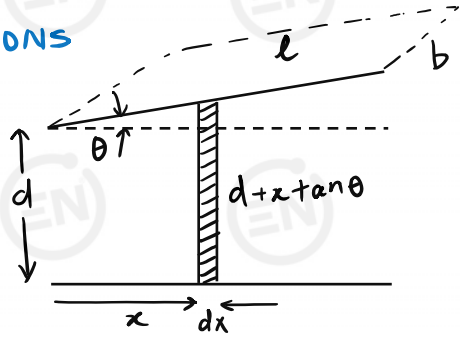
(a)



$$dC_1 = \frac{K_1\, b\,dx\, \varepsilon_0}{d - x\tan\theta}, \quad dC_2 = \frac{K_2\, b\,dx}{x\tan\theta}$$

$$dC_{eq} = \frac{dC_1 \times dC_2}{dC_1 + dC_2} \qquad C_{eq} = \int_0^{\ell} dC_{eq}$$

(b)

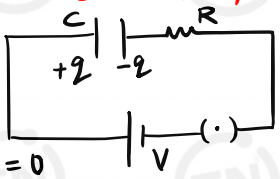


$$dC = \frac{b\,dx\,\varepsilon_0}{d + x\tan\theta}$$

$$\therefore C = b\varepsilon_0 \int_0^{\ell} \frac{dx}{d + x\tan\theta}$$

15. RC, CHARGING AND DISCHARGING

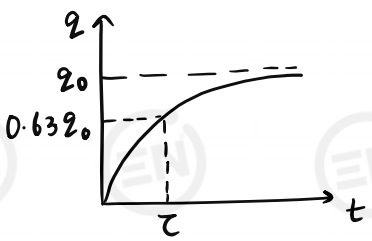
CHARGING



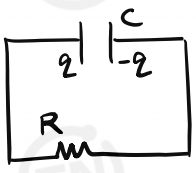
At $t=0$, $q=0$

At $t=t$, $q = q_0(1 - e^{-t/RC})$ $\quad q_0 = CV$

$RC = \tau$, time const.

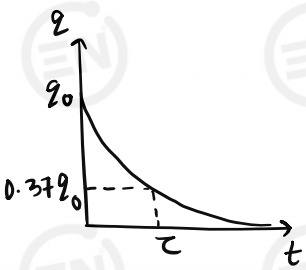


DISCHARGING



at $t=0$, $q = q_0$

at $t=t$, $q = q_0 e^{-t/RC}$



NOTE:

(1.) At $t=0$, capacitor behaves as conducting wire

(2) At steady state it acts as open circuit.

Space to add concepts learnt from PYQs if any

Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in CURRENT ELECTRICITY – PART 3 (ELECTRODYNAMICS)

1. Charge Flow
2. Important current parameters
3. Resistance (dependency and variation with T)
4. Color Code
5. Cell (emf & internal resistance
6. Combination of Cell
7. Kirchhoff's Law (KVL & KCL)
8. Circuit analysis techniques
9. Combination of resistors
10. Wheatstone Bridge
11. Cube resistors
12. Infinite ladder
13. Thermal effect of current
14. Maximum power transfer theorem
15. Concept of power rating
16. Galvanometer to Ammeter & Voltmeter
17. Meter bridge
18. Potentiometer

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. CHARGE FLOW

$$q_{flown} = \int_{t_1}^{t_2} i(t)\,dt$$

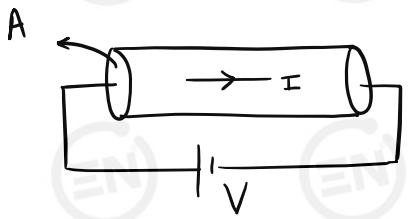
Ex: $i(t) = 2\sin 50\pi t$

$i(t) = 3t^2$

2. IMPORTANT CURRENT PARAMETERS

→ DRIFT VELOCITY, $V_d = \frac{eE\tau}{m}$ "τ" Relaxation time (avg time elapsed between two collisions)

↳ $I = neAV_d$



n: no of free e^- per unit volume

→ MOBILITY, $\mu = \frac{V_d}{E} = \frac{eE\tau/m}{E} = \frac{e\tau}{m}$ (m^2/Vs)

→ CURRENT DENSITY, $J = I/A \Rightarrow I = \vec{J}\cdot\vec{A}$ (A/m^2)

$\boxed{E = \rho J}$ ρ: Resistivity

3. RESISTANCE $\left(R = \underbrace{\frac{m}{ne^2\tau}}_{\rho}\frac{l}{A}\right)$, ohm (Ω)

ρ : resistivity, $\sigma = \frac{1}{\rho}$ → conductivity

R DEPENDS ON :

- $R \propto l$
- $R \propto 1/A$
- If temperature increases, Resistance also increases. $\left\{\rho = \frac{m}{ne^2\tau}\right.$, If $T\uparrow \Rightarrow \tau\downarrow \Rightarrow \rho\uparrow \Rightarrow R\uparrow$

For small variation in temp°

$R_{T_2} = R_{T_1}(1+\alpha\Delta T)$,

$\rho_{T_2} = \rho_{T_1}(1+\alpha\Delta T)$

↳ FOR SEMICONDUCTORS

If $T\uparrow \Rightarrow \rho\downarrow \Rightarrow R\downarrow$

(on $\uparrow T$, $n\uparrow$ dominating $\downarrow$ in τ)

SOURCE : NCERT

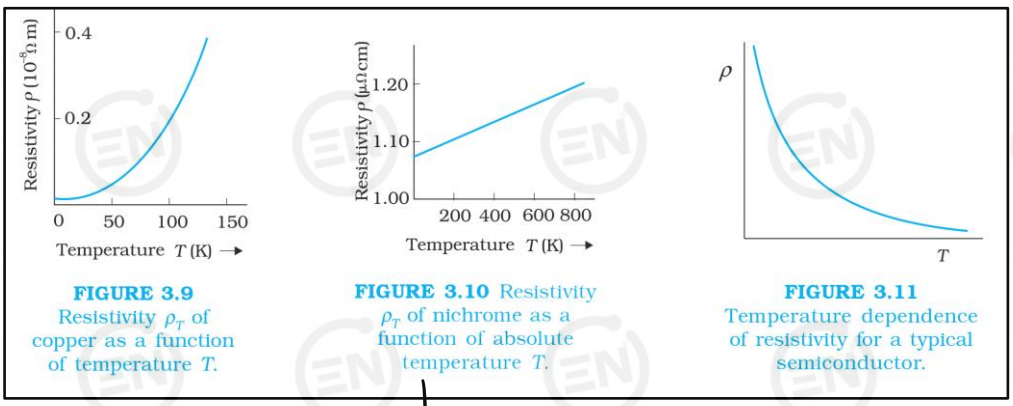


FIGURE 3.9 Resistivity ρ_T of copper as a function of temperature T.

FIGURE 3.10 Resistivity ρ_T of nichrome as a function of absolute temperature T.

FIGURE 3.11 Temperature dependence of resistivity for a typical semiconductor.

↳ Manganin and Constantan

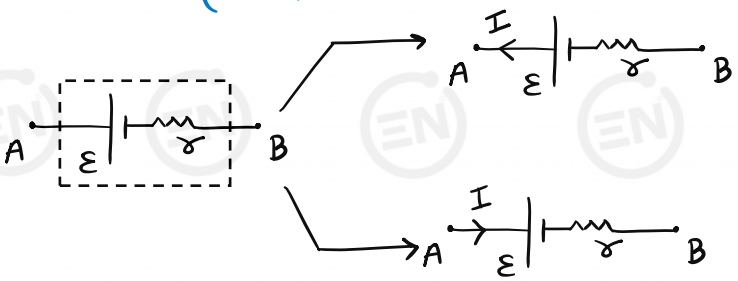
4. COLOUR CODE

BROWN 1 | BLACK 0 | RED 2 | GOLDEN ±5%

$R = 10 \times 10^2 \pm 5\%$

Resistor colour codes			
Colour	Number	Multiplier	Tolerance (%)
Black	0	10^0	
Brown	1	10^1	
Red	2	10^2	
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	
Blue	6	10^6	
Violet	7	10^7	
Gray	8	10^8	
White	9	10^9	
Gold		10^{-1}	5
Silver		10^{-2}	10
No colour			20

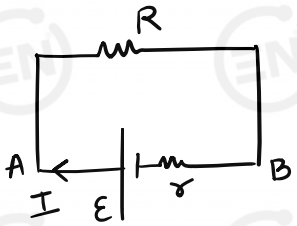
5. CELL (Emf, Internal Resistance)



$V_{AB} = \varepsilon - Ir$, Work done by battery

$V_{AB} = \varepsilon + Ir$, Work done on the battery (charging)

Alternative:

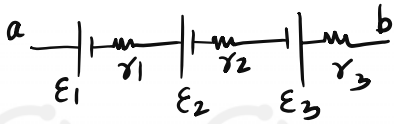


$V_{AB} = IR$

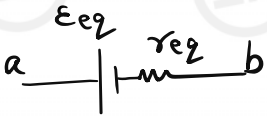
$$V_{AB} = \frac{\varepsilon R}{r+R}$$

6. COMBINATION OF CELL

SERIES



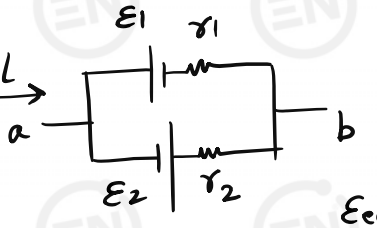
$\Rightarrow$



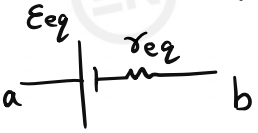
$\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2 - \varepsilon_3$

$r_{eq} = r_1 + r_2 + r_3$

PARALLEL



$\Rightarrow$



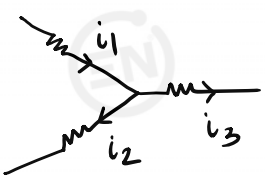
$$\varepsilon_{eq} = \frac{\frac{\varepsilon_1}{r_1} - \frac{\varepsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}}, \quad \frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2}$$

Here we took ⊣ as $+\varepsilon$ (ε)

7. KIRCHHOFF'S LAW (KVL and KCL)

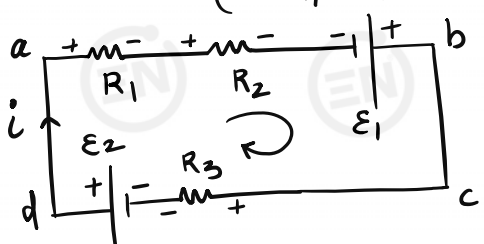
LOOP RULE → KCL ($\Sigma i_n = 0$), KVL ($\Sigma V_n = 0$) In a Loop

JUNCTION RULE



$i_1 - i_2 - i_3 = 0$

(i towards junction is taken +VE)

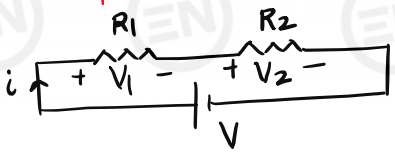


In Loop abcda

$iR_1 + iR_2 - \varepsilon_1 + iR_3 - \varepsilon_2 = 0$

8. CIRCUIT ANALYSIS MORE TECHNIQUES

p.d. distribution



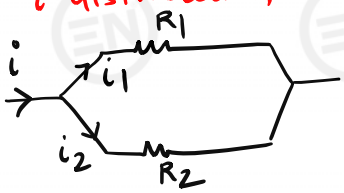
$\because i$ is same

$\Rightarrow V \propto R$

$V_1 = \frac{VR_1}{R_1+R_2}$

$V_2 = \frac{VR_2}{R_1+R_2}$

i distribution



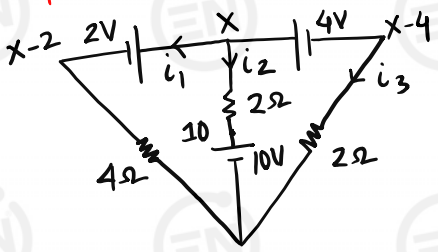
$\because V$ is same

$\Rightarrow i \propto 1/R$

$i_1 = \frac{iR_2}{R_1+R_2}$

$i_2 = \frac{iR_1}{R_1+R_2}$

Point Potential Method



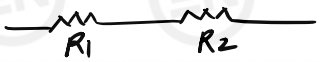
$i_1 + i_2 + i_3 = 0$

$\Rightarrow \frac{x-2}{4} + \frac{x-10}{2} + \frac{x-4}{2} = 0$

$x = 6V$

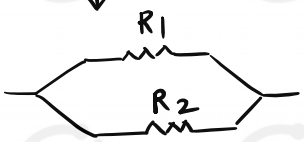
9. COMBINATION OF RESISTORS

SERIES



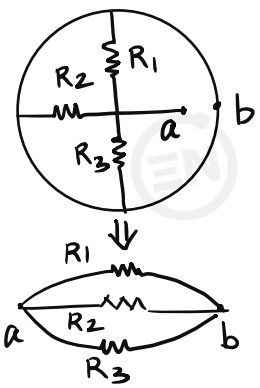
$R_{eq} = R_1 + R_2$

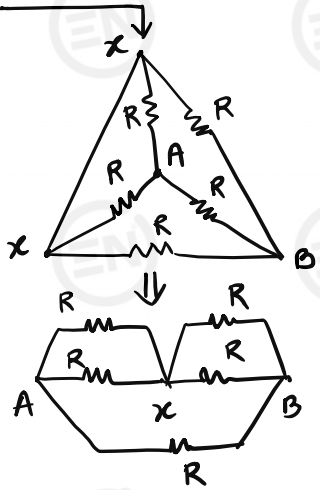
PARALLEL



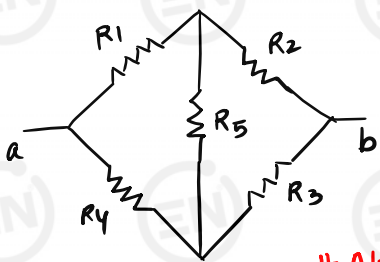
$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

$\Rightarrow R_{eq} = \frac{R_1R_2}{R_1+R_2}$

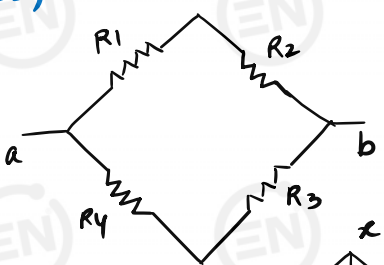




10. WHEATSTONE BRIDGE (BALANCED)

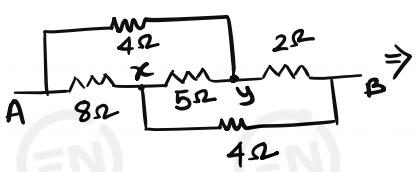


If $R_1R_3 = R_2R_4$ ⟶

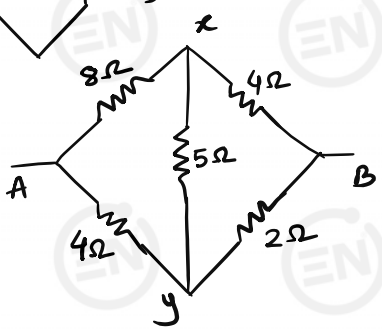


$\left\{ \frac{R_1}{R_2} = \frac{R_4}{R_3} \right\}$

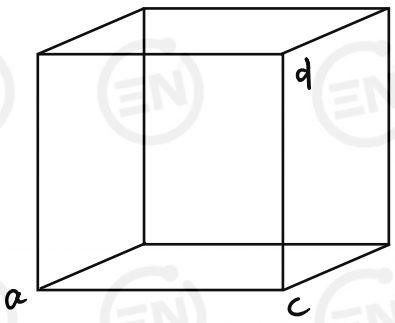
#Alternative fig.



⇒



11. CUBE RESISTORS

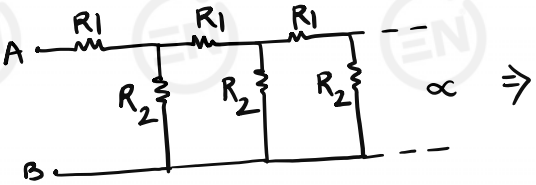


$R_{eq,ab} = 5R/6$ (body diagonal)

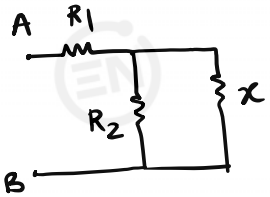
$R_{eq,ac} = 7R/12$ (edge)

$R_{eq,ad} = 3R/4$ (face diagonal)

12. INFINITE LADDER



$\propto \Rightarrow$



$R_{AB} = R_1 + \frac{xR_2}{x+R_2}$

x

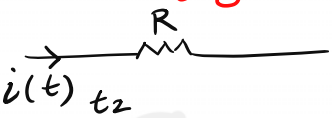
13. THERMAL EFFECT OF CURRENT (JOULES HEATING EFFECT)

CONSTANT CURRENT

$P = i^2R = \frac{V^2}{R} = Vi$ (Watt)

$H = i^2Rt = \frac{V^2}{R}t = Vit$ (Joules)

Time Varying Current

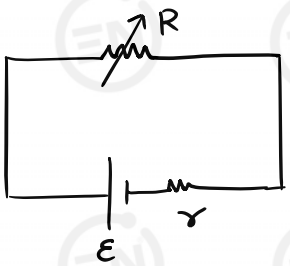


$$H = \int_{t_1}^{t_2} i^2 R\,dt$$

$$P_{av} = \frac{\int i^2 R\,dt}{\int dt}$$

Ex: $i(t) = i_0 \sin\omega t$

14. MAX POWER TRANSFER THEOREM



CONDITION: $\boxed{R = r}$

for maximum power transfer, External resistance must be equall to internal resistance.

15. CONCEPT OF POWER RATING

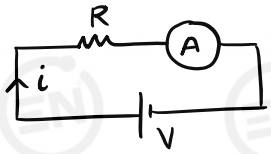
BULB specifications 220V, 50W
220V → Rated Voltage
50W → Rated Power

NOTE:
(i) Means bulb will consume 50 W if 220V is across it.
(ii) $R_{bulb} = \frac{V^2}{P} = \frac{220^2}{50} = 968\,\Omega$
(iii) If $V > 220V$ is across bulb, it will fuse
(iv) More power ⇒ More Bright

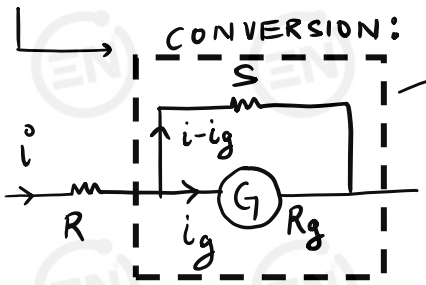
16. GALVANOMETER TO AMMETER AND VOLTMETER

(a) AMMETER

→ connected in series
→ IDEAL AMMETER has zero resistance
(Practically it has very Low resistance)



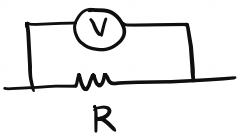
→ CONVERSION:

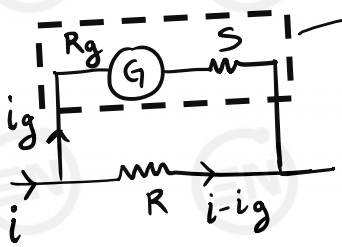


(i) i_g is max current that can pass through G for full deflection
(ii) $S << R_g$ (S: shunt)
(iii) $(i-i_g)S = i_g R_g \Rightarrow i = i_g\left(1+\frac{R_g}{S}\right)$

(b) Voltmeter

→ Connected in parallel
→ IDEAL VOLTMETER has infinite Resistance
(Practically it has Very resistance)





So max p.d. measured by voltmeter is,

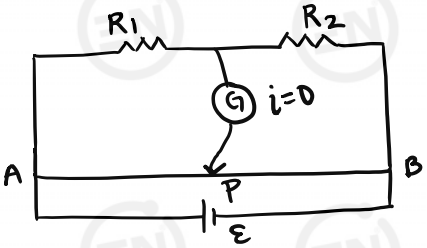
$$V = i_g(R_g + S)$$

↳ i_g : galvanometer current for full deflection

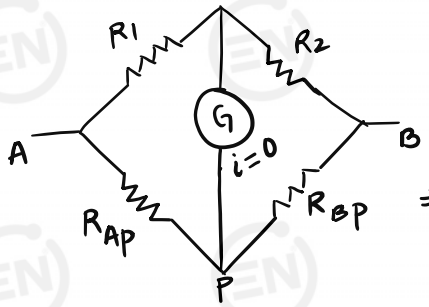
17. METER BRIDGE

→ AIM: to find resistance of R_2

→ Based on Balanced WHEATSTONE BRIDGE



$\Rightarrow$



$\frac{R_1}{R_2} = \frac{R_{AP}}{R_{BP}}$

$\Rightarrow \boxed{\frac{R_1}{R_2} = \frac{l_1}{l_2}}$

OR

$R_2 = R_1 \times \frac{l_2}{l_1}$

$AP = l_1$

$BP = l_2$

P is "Null-Deflection" point.

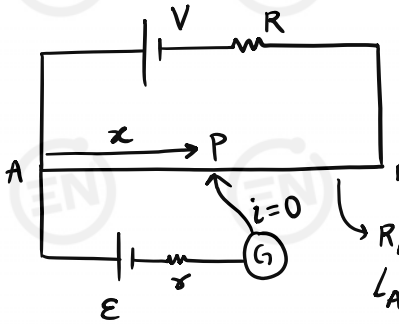
Generally $l_1 + l_2 = 100\,cm$

18. POTENTIOMETER

→ AIM: To find emf of a cell and its internal resistance

→ Potential gradient (K): $K = \frac{P.d.}{l}$ V/m (Potential difference per unit Length)

(a) FINDING EMF OF A CELL



$R_{AB} = R_0$

$l_{AB} = l_0$

(i) P is null-deflection pt. or balance pt.

(ii) $k = V_{AB}/l_0$, $V_{AB} = \frac{VR_0}{R_0+R}$

(iii) $\varepsilon = V_{AP}$

$\Rightarrow \boxed{\varepsilon = kx}$

NOTE: MAX value of ε that can be measured is "$\varepsilon_{max} = V_{AB}$"

(b) FINDING Internal resistance (r)

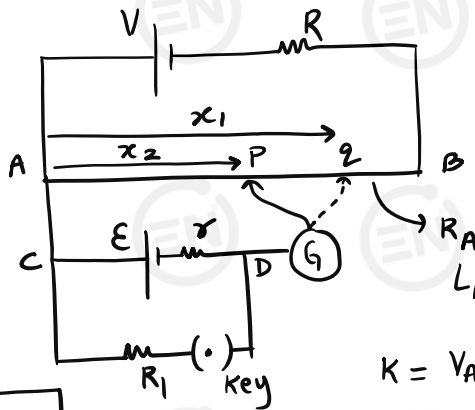
(i) When key is open: (Null deflection at Q)

$\varepsilon = kx_1$ --- (1)

(ii) When key is closed (Null deflection at P)

$V_{CD} = kx_2 \Rightarrow \frac{\varepsilon R_1}{R_1+r} = kx_2$ --- (2)

(1)/(2): $\frac{R_1+r}{R_1} = \frac{x_1}{x_2} \Rightarrow \boxed{r = R\left(\frac{x_1-x_2}{x_2}\right)}$



$R_{AB} = R_0$

$l_{AB} = l_0$

$K = V_{AB}/l_0$

$= \left(\frac{VR_0}{R+R_0}\right)\frac{1}{l_0}$

Space to add concepts learnt from PYQs if any

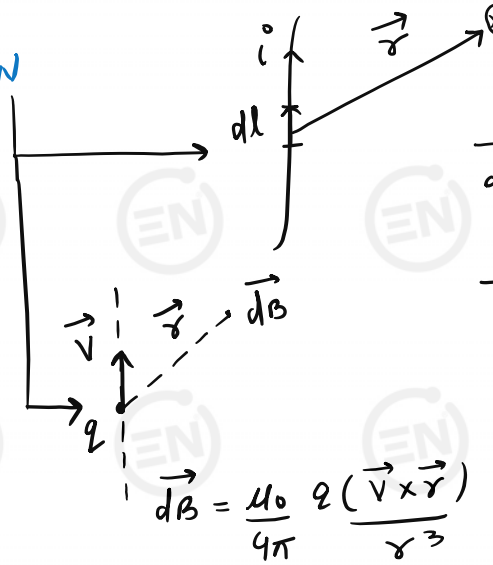
Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in MOVING CHARGES & MAGNETIC EFFECT OF CURRENT – PART 4 (ELECTRODYNAMICS)

1.Biot Savart's Law
2.B due to current carrying straight wire
3.B due to current carrying circular wire
4.B due to Solenoid & Toroid
5.Ampere circuital Law
FORCE ON CHARGE MOVNIG IN B
6.If velocity is Perpendicular to B
7.If not complete circle
8.Angle of deviation
9.Velocity at angle with B (helical path)
10.Special case (V, E & B are mutually perpendicular)
11.Force on current carrying conductor in B
12.Magnetic Moment
13. Torque on Loop in Magnetic Field
14.Potential of Loop in Magnetic Field

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. BIOT SAVART'S LAW



$\otimes\ d\vec{B} = \frac{\mu_0}{4\pi} \frac{i(d\vec{l} \times \vec{r})}{r^3}$

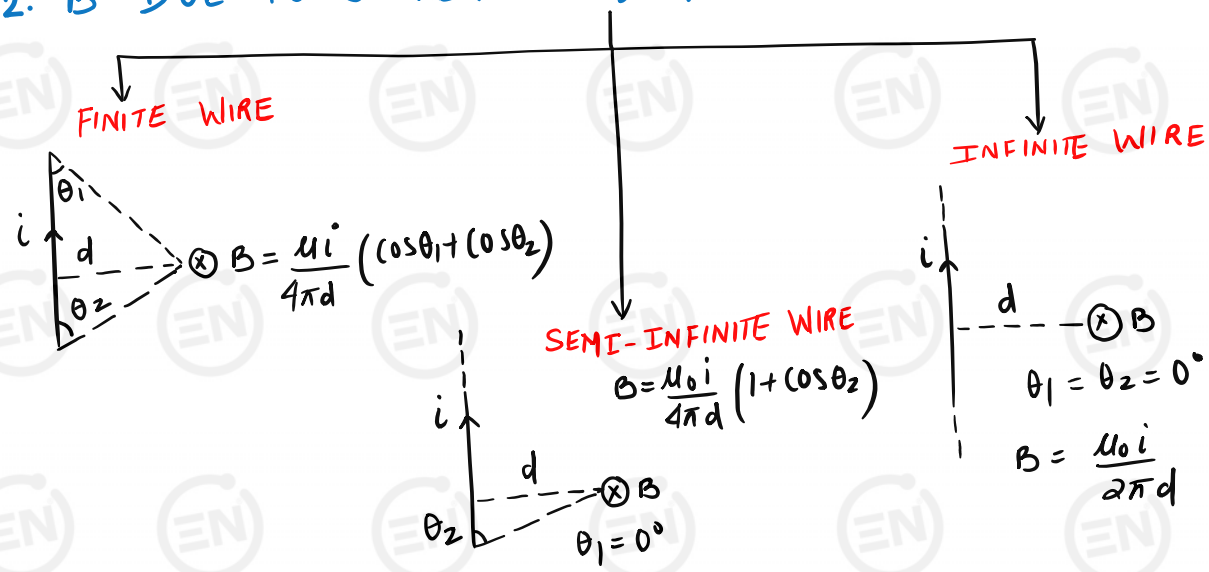
→ direction of $d\vec{B}$ is decided by dirn of $d\vec{l} \times \vec{r}$.

→ μ_0 : Permeability of free space $4\pi \times 10^{-7}$ Tm/A

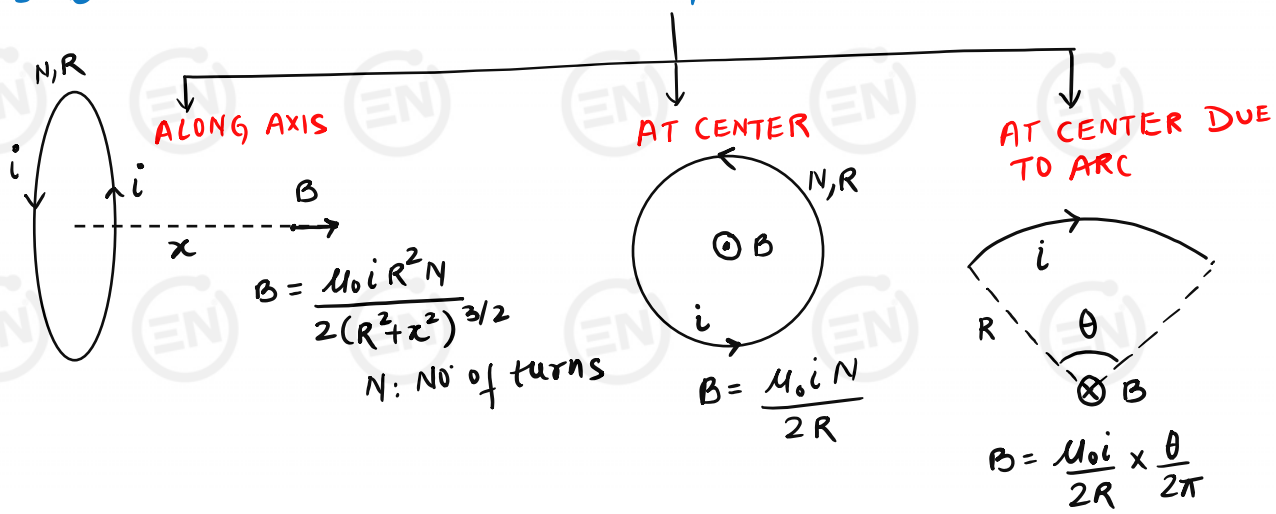
→ $d\vec{l}$ is along current

$d\vec{B} = \frac{\mu_0}{4\pi} \frac{q(\vec{V} \times \vec{r})}{r^3}$

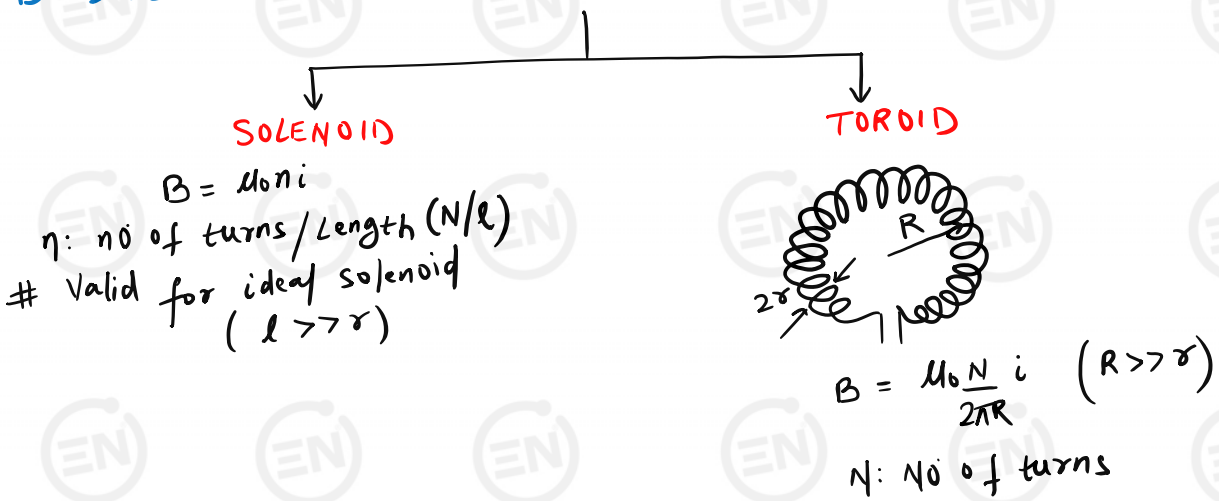
2. B DUE TO CURRENT CARYING STRAIGHT WIRE



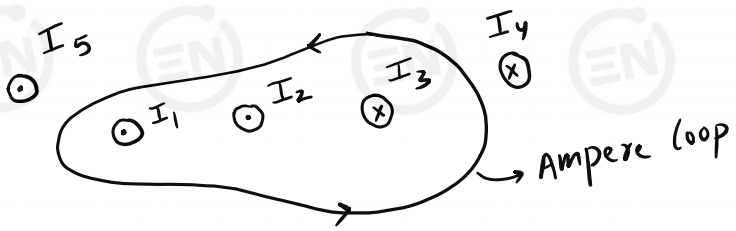
3. B DUE TO CURRENT CARYING CIRCULAR WIRE



4. B DUE TO SOLENOID AND TOROID



5. AMPERE CIRCUITAL LAW



$$\oint \vec{B}\cdot d\vec{l} = \mu_0 (I_1 + I_2 - I_3)$$

NOTE: (i) only I enclosed by Ampere Loop considered

(ii) B is due to all current

NOTE: Outward I is +VE.

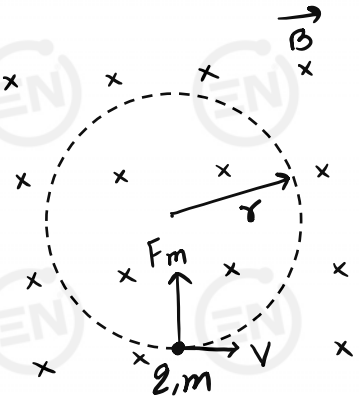
FORCE ON q MOVING IN B, $\vec{F}_m = q(\vec{v}\times\vec{B})$

$\vec{F}_m$ is always perpendicular to $\vec{v}$

⇒ Thus no Work done.

⇒ no change in speed or K.E.

6. If V is ⊥ to B



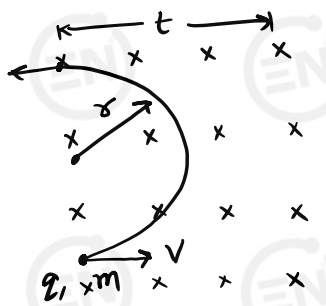
(i) $r = \frac{mv}{qB}$ or $\sqrt{\frac{2mK}{qB}}$ NOTE $K = q\Delta V$

(ii) $T = \frac{2\pi m}{qB}$, time period

(iii) $\omega = \frac{qB}{m}$

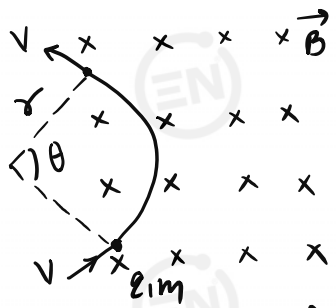
7. IF NOT COMPLETE CIRCLE

(i)



$r < t$, $T = \frac{\pi m}{qB}$

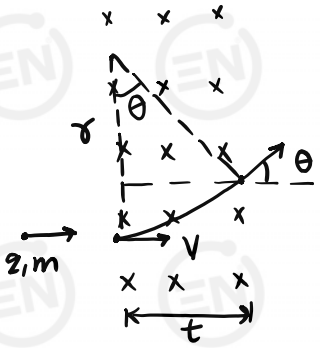
(ii)



$T = \frac{2\pi m}{qB} \times \frac{\theta}{2\pi} = \frac{\theta m}{qB}$

↳ Duration of time inside B

8. ANGLE OF DEVIATION



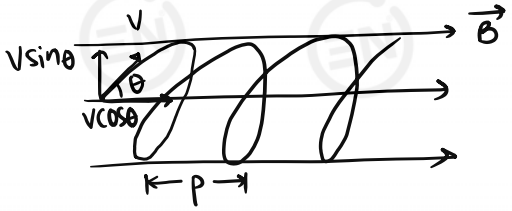
$r > t$

$\sin\theta = t/r$

$= \frac{tqB}{mv}$

$\therefore \boxed{\theta = \sin^{-1}\left(\frac{qBt}{mv}\right)}$

9. V AT ANGLE θ to B (Helical path)



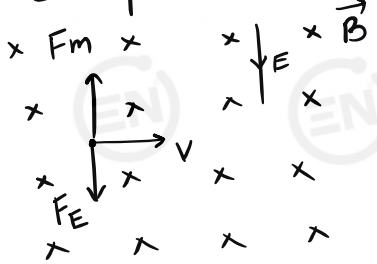
(i) $r = \frac{mv\sin\theta}{qB}$

(ii) $T = 2\pi m/qB$

(iii) Pitch, $P = V\cos\theta \times T$

10. SPECIAL CASE $E \perp B \perp V$

IN this situation possibility of q moving undeviated.



COND^n:

$F_E = F_M$

$\Rightarrow qE = qVB$

$\Rightarrow \boxed{V = \frac{E}{B}}$

11. FORCE ON I CARYING CONDUCTOR IN B (UNIFORM FIELD)

$\vec{F} = I(\vec{l} \times \vec{B})$

STRAIGTH WIRE

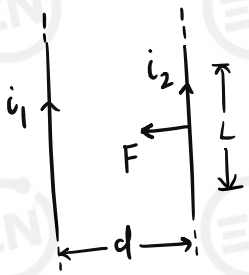
$F_m = IlB$

C/closed LOOP

$F_{Net} = 0$

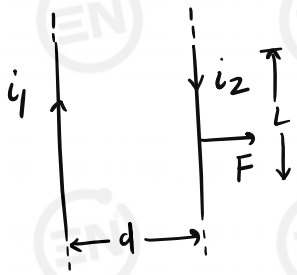
Parallel wire

Parallel i



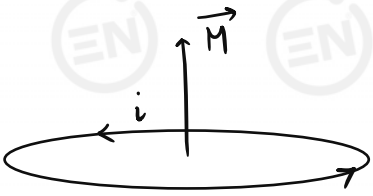
$F = \frac{\mu_0 i_1 i_2 L}{2\pi d}$

ANTI-PARALLEL i



$F = \frac{\mu_0 i_1 i_2 L}{2\pi d}$

12. MAGNETIC MOMENT



$\vec{M} = N i \vec{A}$

NOTE:

(i) N : No of turns

(ii) A = Loop area

(iii) Dirn of M using right hand thumb rule.

13. TORQUE ON LOOP in B

$$\vec{\tau} = \vec{M} \times \vec{B}$$

14. POTENTIAL ENERGY OF LOOP in B

$$U = -\vec{M} \cdot \vec{B}$$

STABLE → $\theta = 0^\circ \Rightarrow U_{min} = -MB$

UNSTABLE → $\theta = 180^\circ \Rightarrow U_{max} = MB$

15. FORCE ON LOOP IN B (NON-UNIFORM B)

$$F = M \frac{dB}{dx}$$ [USE IF VARIATION OF B IS SMALL]

M : Magnetic moment of Loop

Space to add concepts learnt from PYQs if any

Space to add concepts learnt from PYQs if any

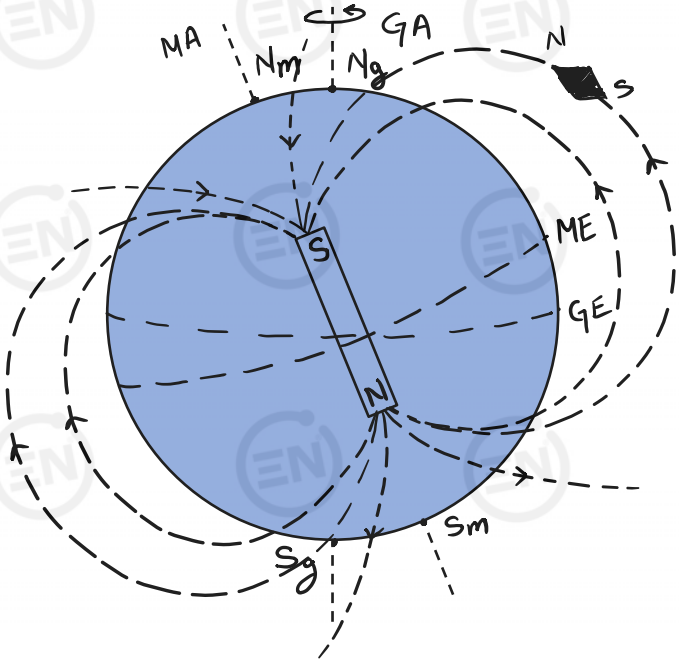
Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in EARTH'S MAGNETISM – PART 5 (ELECTRODYNAMICS)

1-5. Geographical axis, poles & Geomagnetic Poles
6.Geographical and Magnetic Meridian
7.Elements of Magnetic Field
8.True and Apparent angle of dip
9. Oscillation of compass needle

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

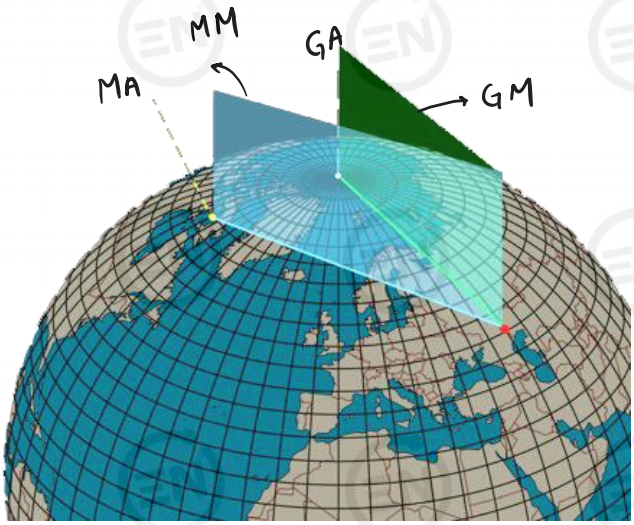
(1) GA : Geographical axis
(2) MA: Magnetic axis
(3) Ng, Sg: Geographical North, south
(4.) Nm, Sm: Geomagnetic North, soutch
(5.) GE, ME: Geographic, Magnetic Equator



6. Geographical and Magnetic Meridian

(a) **Geographical Meridian**:
A Vertical plane at any place on earth which passes through GA

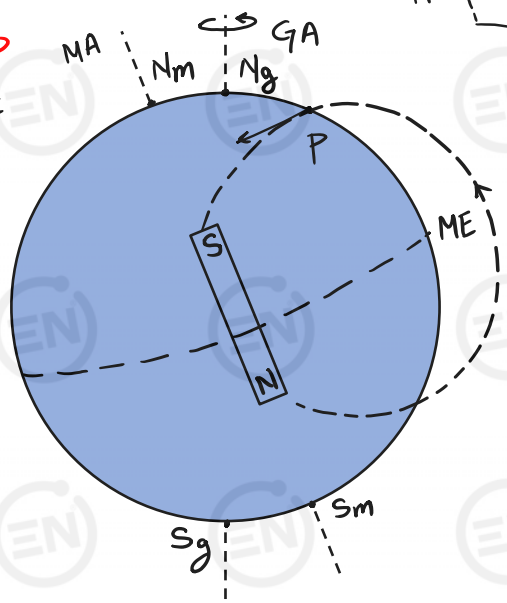
(b) **Magnetic Meridian**
A Vertical plane on earth that Passes through magnetic axis.
or
Vertical plane that Contains magnetic field lines.

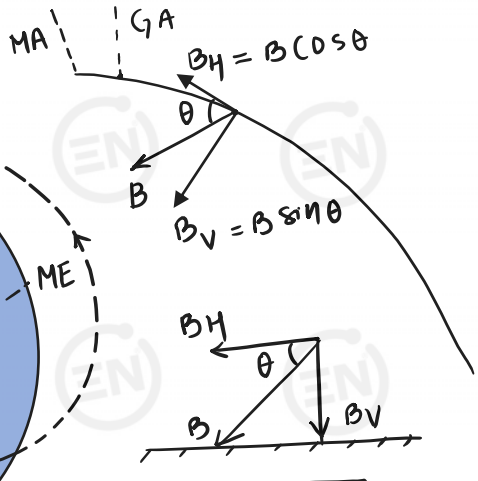


7. ELEMENTS OF MAGNETIC FIELD

(a) ANGLE OF DIP

↳ Angle which magnetic field makes with horizontal in Magnetic Meridian.

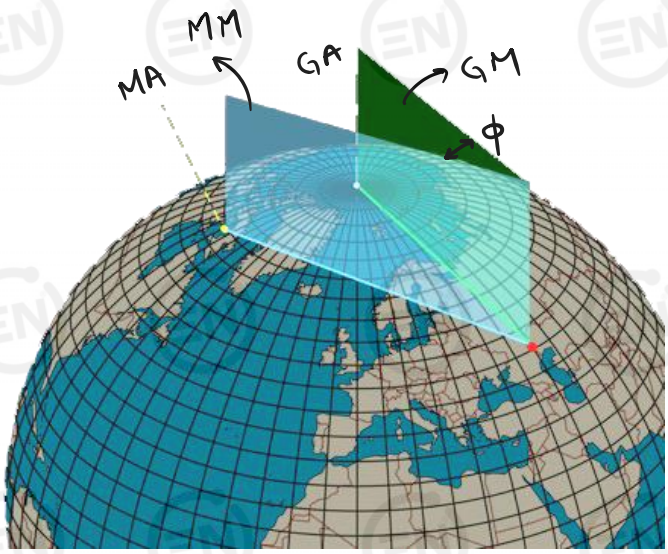


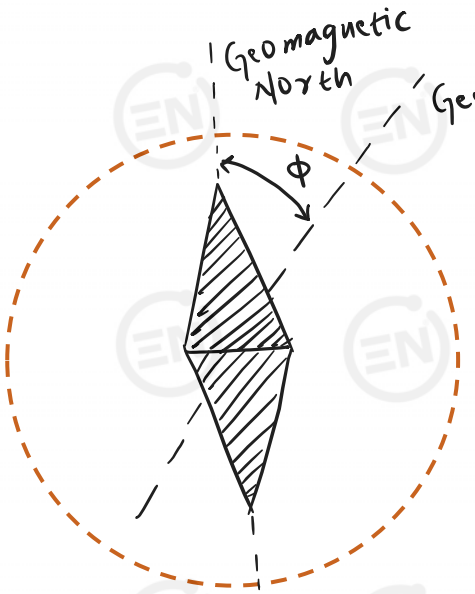


$$\tan\theta = \frac{B_V}{B_H}$$

(b) ANGLE OF DECLINATION (ϕ)

ϕ is angle between Geographical meridian and magnetic meridian at a point





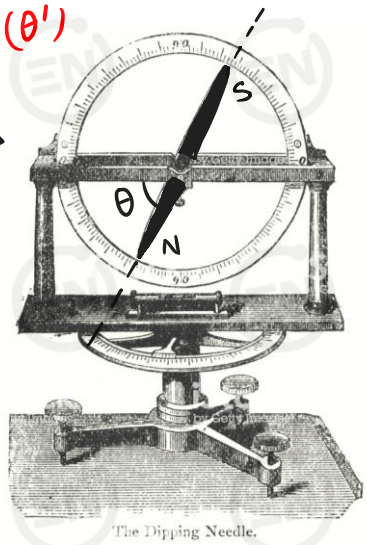
↳ Horizontal plane

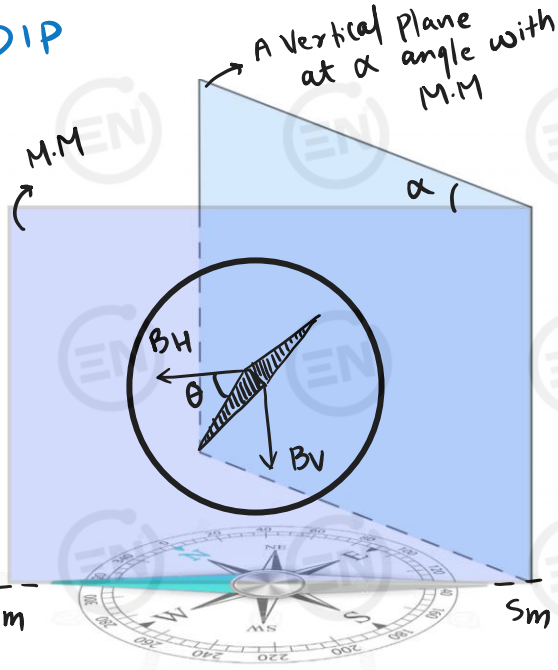
NOTE: Compass in Horizontal Plane always gets aligned along $B_H = B\cos\theta$ and is in Magnetic Meridian.

8. TRUE AND APPARENT ANGLE OF DIP

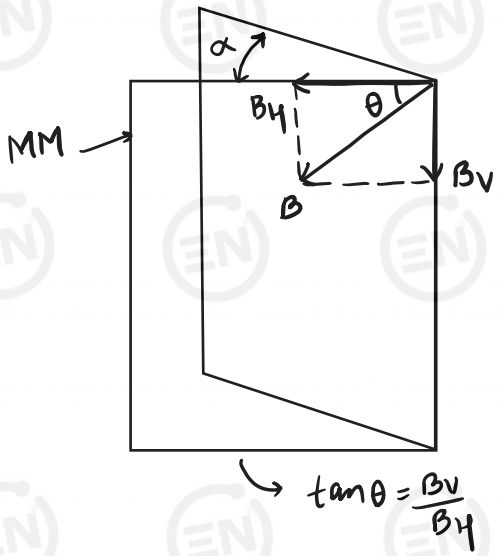
(θ) (θ')

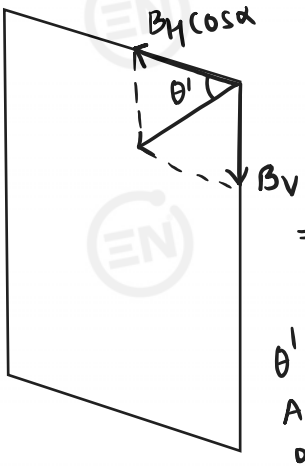
DIP CIRCLE →





IF DIP CIRCLE IS NOT MAGNETIC MERIDIAN ?





$\tan\theta' = \frac{B_V}{B_H\cos\alpha}$

$\Rightarrow \tan\theta' = \frac{\tan\theta}{\cos\alpha}$

θ' is called Apparent angle of DIP.

9. OSCILLATION OF COMPASS NEEDLE

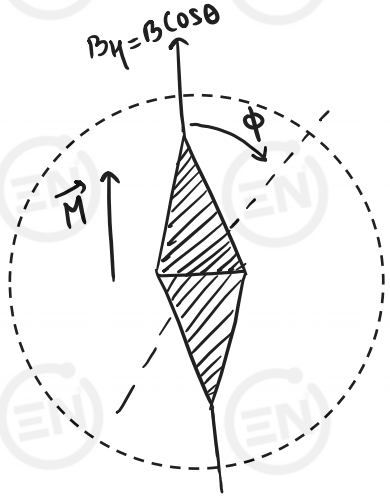
$\vec{M}$: Magnetic moment (md)

$\tau = MB_H \sin\phi$

$\Rightarrow I\alpha = MB_H \sin\phi$

$\Rightarrow \alpha = \frac{MB_H}{I}\phi$ { ϕ is small

$T = 2\pi\sqrt{\frac{I}{MB_H}}$



Space to add concepts learnt from PYQs if any

Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in MAGNETIC PROPERTIES OF MATTER – PART 6 (ELECTRODYNAMICS)

1. Magnetizing Field, Intensity of Magnetization, Susceptibility
2. Diamagnetic and Paramagnetic Material
3. Ferromagnetic Material
4. Curie's Law
5. Magnetic Hysteresis
6. Hysteresis Curve of Soft Iron and Steel

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. $\vec{H}$, $\vec{I}$, $\vec{B}_{net}$, χ

(a) $\vec{H}$, Magnetizing Field

For external magnetic field $\vec{B}_{ext}$, $\vec{H}$ is defined.

$\vec{B}_{ext} = \mu_0 \vec{H}$, unit of $\vec{H}$ is A/m.

(b) $\vec{I}$, (A/m) Intensity of Magnetization

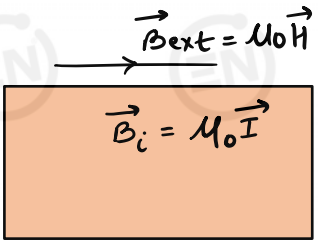
Total magnetic moment of material per unit volume.

$\vec{I} = \frac{\vec{M}}{V}$ (It tells how much a material is magnetized)

(c) χ, Magnetic Susceptibility

Tells about the ease with which a material can be magnetized.

$\chi = I/H$



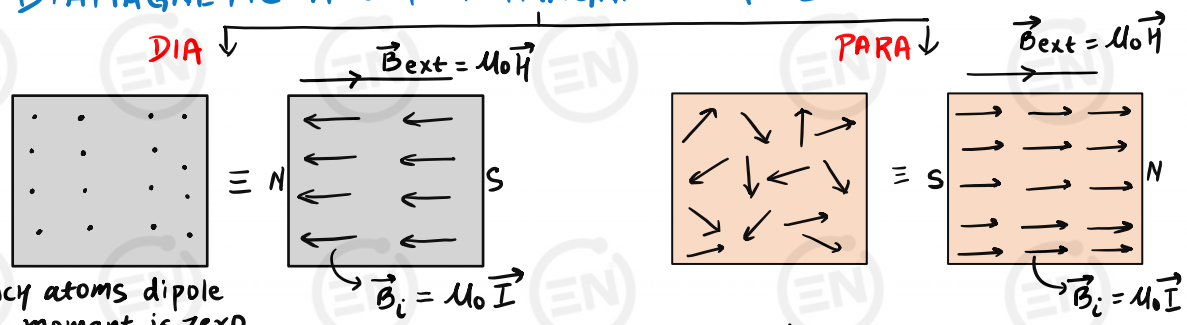
$\vec{B}_{net} = \vec{B}_{ext} + \vec{B}_i$

$\Rightarrow \mu_0 \mu_r \vec{H} = \mu_0 \vec{H} + \mu_0 \vec{I}$

$\Rightarrow \mu_r \vec{H} = \vec{H} + \chi \vec{H}$

$\boxed{\mu_r = 1 + \chi}$

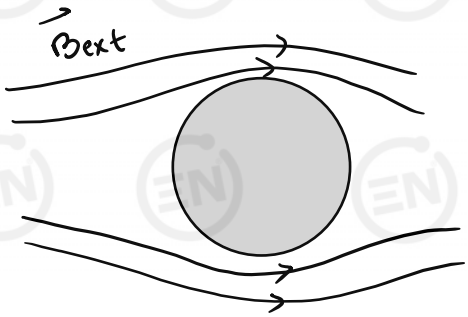
2. DIAMAGNETIC AND PARAMAGNETIC MATERIAL



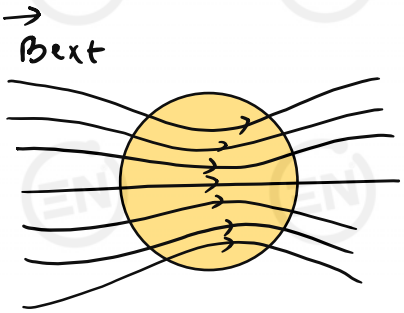
Each atoms dipole moment is zero

(i) Weakly magnetized
(ii) Material repels $\vec{B}_{ext}$
(iii) $\mu_r = 1 + \chi$, $\mu_r < 1$ and χ is -ve
(iv) Graphite, Bismuth

(i) Weakly magnetized
(ii) Material gets weakly attracted to $\vec{B}_{ext}$
(iii) $\mu_r = 1 + \chi$, $\mu_r > 1$ and χ is +ve
(iv) Al, Li, Mg



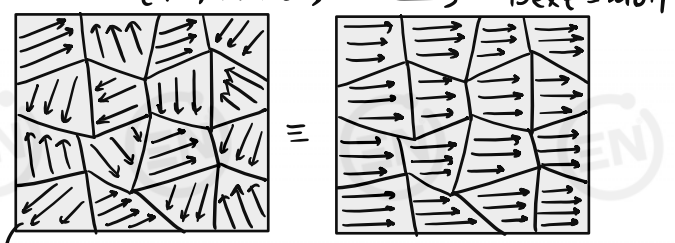
DIA



PARA

3. FERROMAGNETIC MATERIAL

(Fe, Ni, Co)



$\vec{B}_{net} = \vec{B}_{ext} + \vec{B}_i$

Here, $B_i >> B_{ext}$

$\Rightarrow I >> H$

$\therefore \chi >> 1$

4. CURIE'S LAW

If $T\uparrow$, due to thermal agitation, the alignment of dipoles gets disturbed and overall $I\downarrow$ for a given H.

$\because I\downarrow \Rightarrow \chi\downarrow$

(a) FOR PARAMAGNETIC MATERIAL

$\chi = \frac{C}{T}$ C: Curie const. T: Abs Temp°

(b) For Ferromagnetic material on heating it to T_c, it changes to Paramagnetic. on further $\uparrow T$

$\chi = \frac{C'}{T - T_c}$ T_c: Curie Temp°

5. MAGNETIC HYSTERESIS

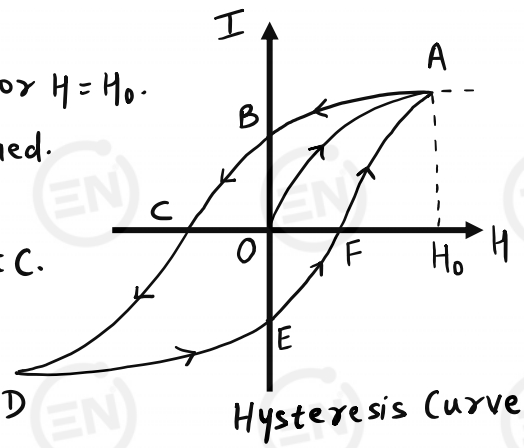
(i) $O \to A$: on $\uparrow H$, $I\uparrow$ and gets saturated for $H = H_0$.

(ii) $A \to B$: on $\downarrow H$ to zero, still I is retained. OB is called "RETENTIVITY".

(iii) $B \to C$: $\uparrow H$ in reverse direction, $I = 0$ at C. OC is called "COERCIVITY".

(iv) $C \to D$: Further $\uparrow H$ and I again gets saturated at D.

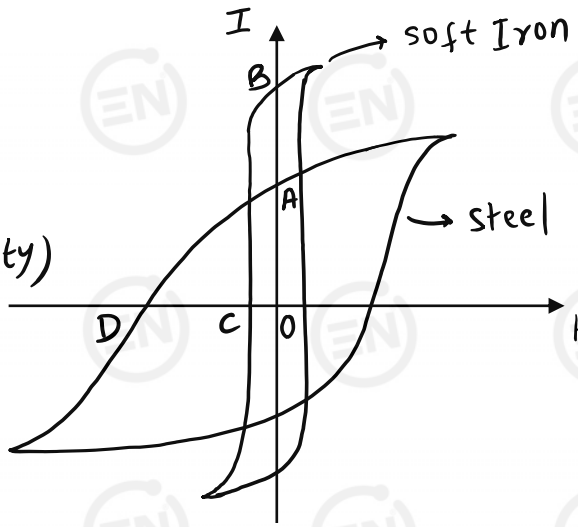
and so on...



Hysteresis Curve

NOTE: (a) Area of LOOP ABCDEFA shows thermal energy or heat produced / Volume in 1 cyc.

6. HYSTERESIS CURVE: SOFT IRON VS STEEL



(i) ∵ Soft Iron gets easily magnetized and looses almost all magnetism easily (Low coercivity) It is used for "Electromagnets".

(ii) ∵ Steel is difficult to demagnetize (OD = coercivity) it is used to make "Permanent Magnets".

Space to add concepts learnt from PYQs if any

Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in ELECTROMAGNETIC INDUCTION – PART 7 (ELECTRODYNAMICS)

1. Faraday's Law
2. Direction of Induced Current : Lenz's law
3. Charge Flown (due to induced emf)
4. Coil rotation in uniform magnetic field
5. Motional emf
6. Motional emf in random shaped wire
7. Motional emf of rotating conductors
8. Parallel rail track problems
9. Induced Electric Field in Time Varying Magnetic Field (TVMF)
10. Eddy Currents
11. Self Induction
12. How to find L ?
13. Battery Polarity induced in an Inductor
14. Growth and Decay of Current
15. Magnetic energy Stored
16. Behavior of L at t = 0 and at steady state
17. Mutual Inductance
18. How to find M ?
19. LC Oscillations

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. FARADAY'S LAW, $\varepsilon = -\frac{d\phi}{dt}$

(a) Flux, $\phi = \vec{B}\cdot\vec{A} = BA\cos\theta$

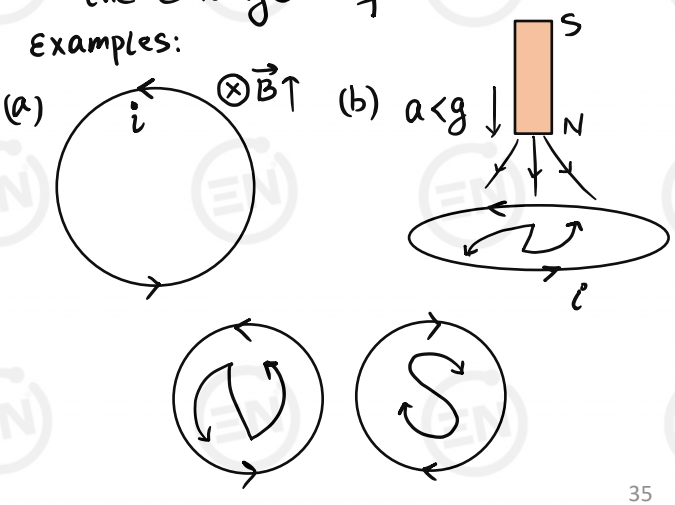
(b) Magnitude is given by $\left|\frac{d\phi}{dt}\right|$

(c) Direction of induced emf (ε) or induced current ($i = \varepsilon/R$) is given by LENZ'S LAW

2. DIRECTION OF INDUCED CURRENT

LENZ'S LAW : Direction of i will such that its effect will oppose the change in flux.

Examples:

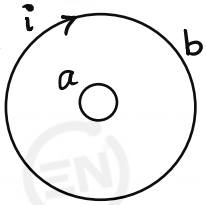


3. CHARGE FLOWN, $q_{flown} = \frac{\Delta\phi}{R}$

(a) $\Delta\phi$ is change in flux, $|\phi_f - \phi_i|$

(b) R is resistance of coil

Example:



$a << b$

If direction of i is reversed,

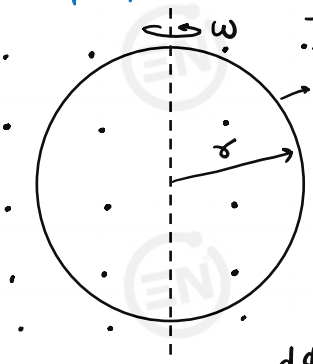
$\Delta\phi = 2 \times BA$

$= 2 \times \frac{\mu_0 i}{2b} \times \pi a^2$

q_{flown} in small coil $= \frac{\Delta\phi}{R}$

R is resistance of small coil

4. COIL ROTATION IN UNIFORM MAGNETIC FIELD



At $t=0$, $\vec{A} \parallel \vec{B}$

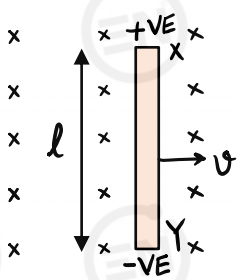
At $t=t$, $\theta = \omega t$

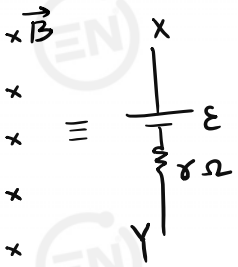
$\phi = NBA\cos\omega t$

$\mathcal{E} = -\frac{d\phi}{dt} = \underbrace{NBA\omega}_{\mathcal{E}_0} \sin\omega t$

$i = \underbrace{\frac{\mathcal{E}_0}{R}}_{i_0} \sin\omega t$

5. MOTIONAL EMF



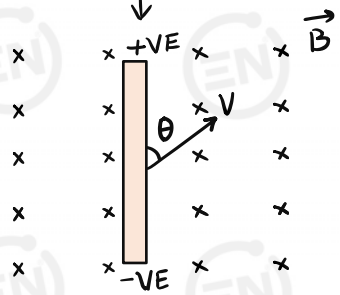


(a) $E = VB$

(b) $\mathcal{E} = Blv$

(c) applicable only if V, B and l are mutually perpendicular

NOTE: If any two are parallel $\mathcal{E} = 0$

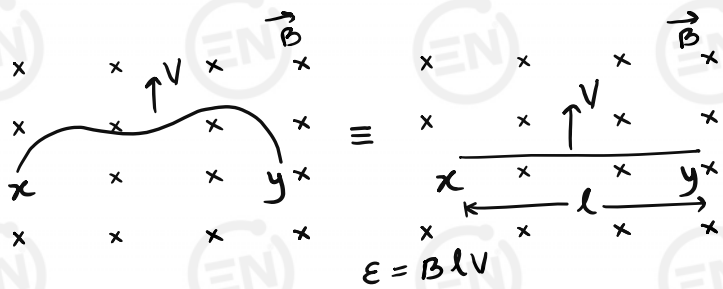


$\mathcal{E} = Bl$ (component of V ⊥ to l)

$= Blv\sin\theta$

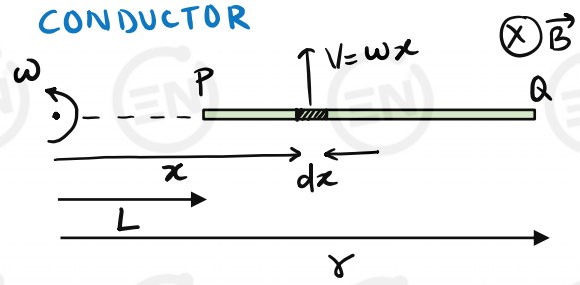
+VE polarity is towards direction of $\vec{V} \times \vec{B}$.

6. MOTIONAL EMF IN RANDOM SHAPED WIRE



$\mathcal{E} = Blv$

7. MOTIONAL EMF OF ROTATING CONDUCTOR



$d\epsilon = B\,dx \times \omega x$

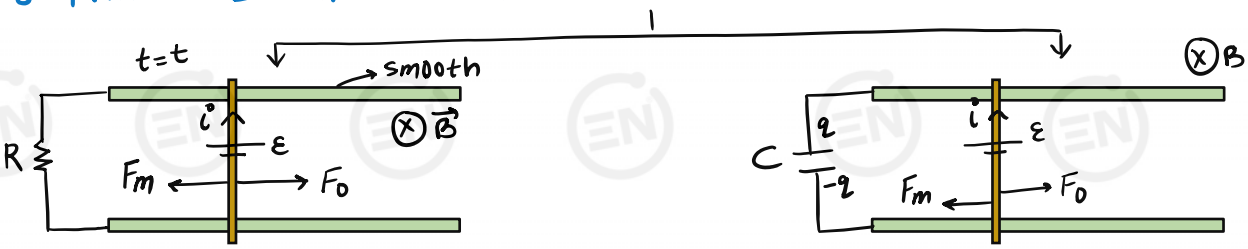
$\Rightarrow \epsilon = B\omega \int_L^r x\,dx$

$\Rightarrow \epsilon = \frac{1}{2}B\omega(r^2 - L^2)$

NOTE: (a) If $L=0$, $\epsilon = \frac{1}{2}B\omega r^2$

(b) For above fig. P at higher Potential.

8. PARALLEL RAIL TRACK PROBLEMS



$F_m = ilB = \frac{BlV}{R} \times lB$

(i) $\frac{dV}{dt} = \frac{F_0 - F_m}{m}$

$V = \frac{F_0 R}{B^2 l^2}\left(1 - e^{-\frac{B^2 l^2 t}{mR}}\right)$

At $t \to \infty$ (steady state)

Terminal Vel, $V_t = \frac{F_0 R}{B^2 l^2}$

$P = F_0 V_t$ = Power across R

$q = C\epsilon \Rightarrow q = CBlV$

$\Rightarrow i = \frac{dq}{dt} = CBla$

$F_0 - F_m = ma$

$\Rightarrow F_0 - (CBla)lB = ma$

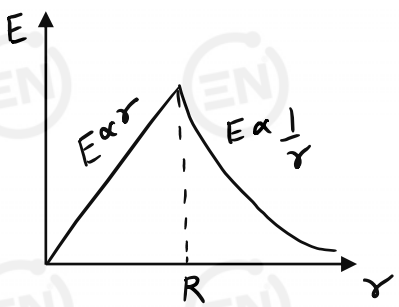
$\therefore a = \frac{F_0}{m + B^2 l^2 C}$

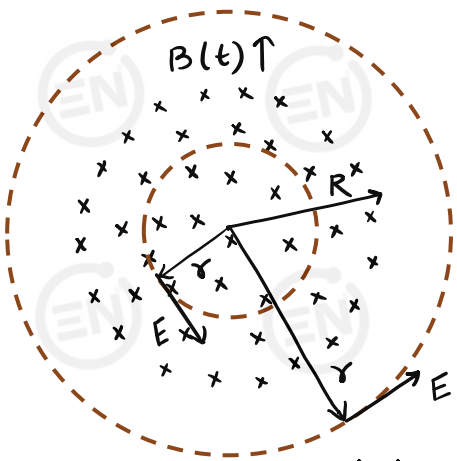
↳ a is const.

9. INDUCED ELECTRIC FIELD IN TIME VARYING MAGNETIC FIELD (Cylindrical region)

For $r < R$, $E = \frac{r}{2}\frac{dB}{dt}$

For $r \geq R$, $E = \frac{R^2}{2r}\frac{dB}{dt}$

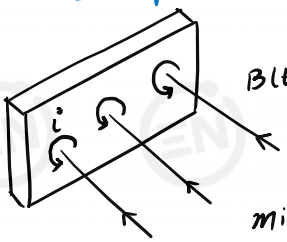




(i) E is in closed Loop

(ii) Here it is Non-conservative in nature.

10. EDDY CURRENTS



When a metal body is placed in TVMF, miniature currents are induced.

Due to this current heat is dissipated.

11. SELF INDUCTION

↳ A property of coil by which it opposes the change in itself.

(a) For any coil carrying current I,

$\phi \propto I$ (self flux linkage due to own I)

$\Rightarrow \phi = LI$

↳ Inductance

UNIT: Henry, H

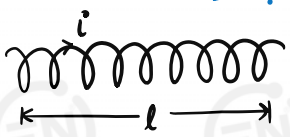
(b) If I varies, emf induced

$\varepsilon = -L\frac{dI}{dt}$ or $\varepsilon = \left|L\frac{dI}{dt}\right|$

↳ Polarity of ε can be found by LENZ'S LAW

12. HOW TO FIND L ?

Example: INDUCTOR or SOLENOID



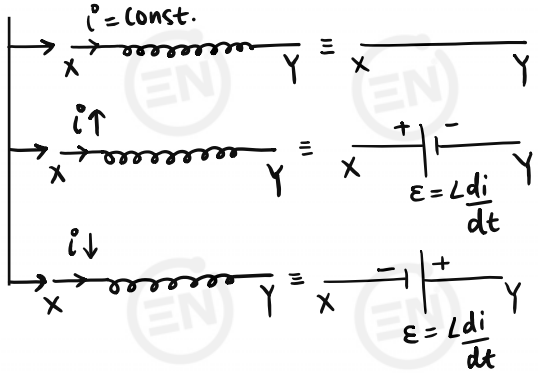
$\phi = NBA \Rightarrow \phi = N \times \frac{\mu_0 N}{\ell} i A$

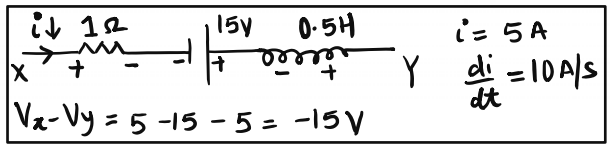
$\Rightarrow \phi = \left(\frac{\mu_0 N^2 A}{\ell}\right) i \quad \therefore L = \frac{\mu_0 N^2 A}{\ell}$

NOTE: If any medium inside

$L = \frac{\mu_0 \mu_r N^2 A}{\ell}$

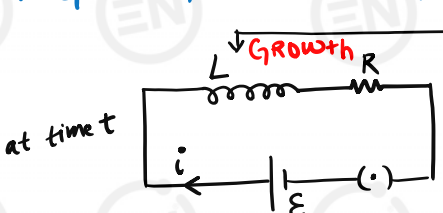
13. BATTERY POLARITY INDUCED IN INDUCTOR





14. GROWTH AND DECAY OF CURRENT (LR)

GROWTH

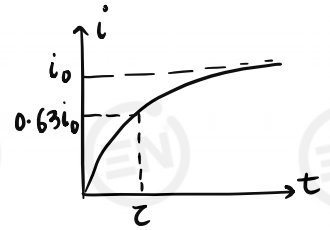


at $t = 0$, $i = 0$

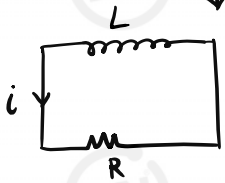
at $t = t$, $i = i_0(1 - e^{-Rt/L})$

$i_0 = \varepsilon/R$

$\tau = L/R$ → Time const.



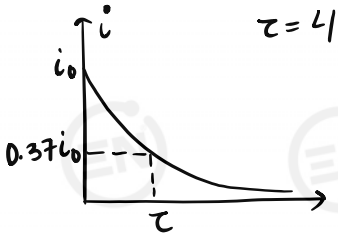
DECAY



at $t = 0$, $i = i_0$

at $t = t$, $i = i_0 e^{-Rt/L}$

$\tau = L/R$

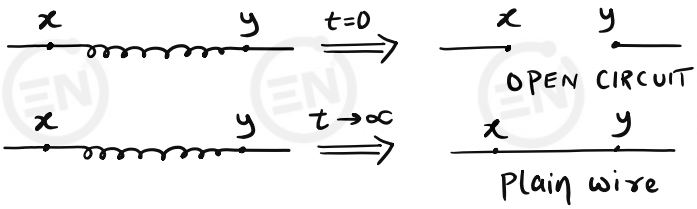


15. ENERGY STORED

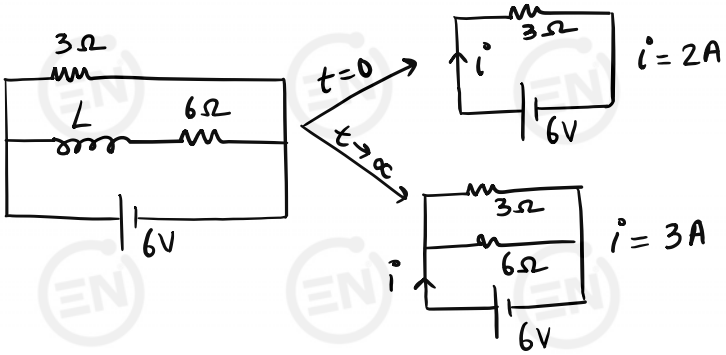
Magnetic energy stored is,

$$U = \frac{1}{2}Li^2$$

16. BEHAVIOR OF L AT $t=0$ AND $t\to\infty$



 # Example:



17. MUTUAL INDUCTION

Property of pair of coils due to which a change in current in 1 coil is opposed by emf induced in other coil because of ϕ LINKAGE.

$\phi_2 = Mi_1$

↳ M is Mutual Inductance

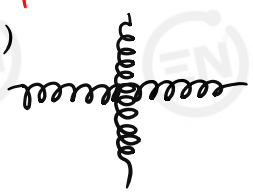
↳ $\mathcal{E}_2 = -M\frac{di_1}{dt}$

 # NOTE $M = K\sqrt{L_1L_2}$ K: coupling factor $0 \le K \le 1$

L_1 and L_2 are self inductance of coil 1 and coil 2

→ K gives idea of % flux linkage.

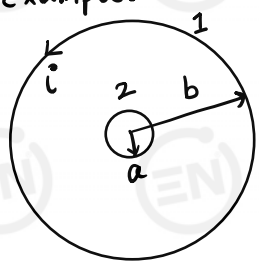
Example:

(i)  K = 0

(ii) K = 1

18. HOW TO FIND M ?

Example:



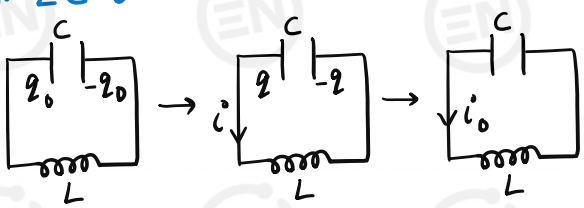
$a \ll b$

$\phi_2 = \frac{\mu_0 i}{2b} \times \pi a^2$

$\Rightarrow \phi_2 = \left(\frac{\mu_0 \pi a^2}{2b}\right) i$

$\therefore M = \frac{\mu_0 \pi a^2}{2b}$

19. LC OSCILLATION



(a) Total energy is const.

$$\frac{q_0^2}{2C} = \frac{q^2}{2C} + \frac{1}{2}Li^2 = \frac{1}{2}Li_0^2$$

(b) $\omega = \frac{1}{\sqrt{LC}}$, $T = 2\pi\sqrt{LC}$

(c) General equation

$q = q_0 \sin(\omega t + \phi)$, $i = i_0 \cos(\omega t + \phi)$

Space to add concepts learnt from PYQs if any

Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in ALTERNATING CURRENT – PART 8 (ELECTRODYNAMICS)

1. Average and RMS current
2. R, L, C taken individually in circuit (Phase relation)
3. LR Circuit across AC source
4. RC Circuit across AC source
5. LC Circuit across AC Source
6. RLC Series
7. Power in AC Circuit and Power factor
8. Resonant Frequency in RLC Series
9. Current variation with frequency, Bandwidth, Quality factor
10. RLC in Parallel
11. Transformer

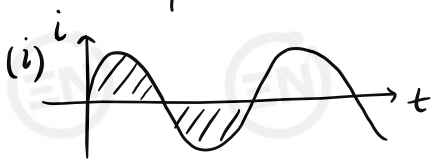
Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. I_{av} AND I_{rms}

$$I_{av} = \frac{\int_{t_1}^{t_2} i(t)\,dt}{t_2 - t_1}$$

$$I_{rms} = \sqrt{\frac{\int_{t_1}^{t_2} i^2(t)\,dt}{t_2 - t_1}}$$

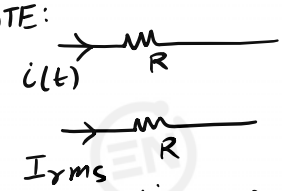
NOTE: $\int i\,dt$ is area under i v/s t Curve



for $i = i_0 \sin \omega t$

$i_{av} = 0$

NOTE:



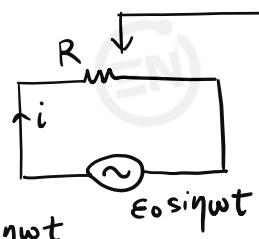
I_{rms}

In same time, same heat is dissipated

Ex: If $i = i_0 \sin \omega t$

$i_{rms} = i_0/\sqrt{2}$

2. PHASE RELATION BETWEEN $\mathcal{E}$ AND i (R, L, C TAKEN INDIVIDUALLY)

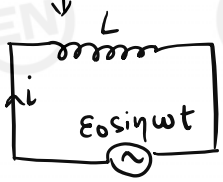


$i = i_0 \sin\omega t$

(a) $i_0 = \mathcal{E}_0 / R$

(b) i and $\mathcal{E}$ in phase

(c) R doesn't depend on ω



$i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right)$

(a) $i_0 = \frac{\mathcal{E}_0}{X_L}$, $X_L = \omega L$ → Inductive Reactance

(b) i lags by $\pi/2$

(c) At $\omega \to 0$, $X_L \to 0$
$\omega \to \infty$, $X_L \to \infty$

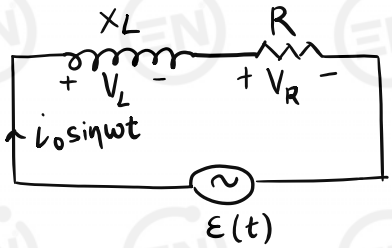
$i = i_0 \sin\left(\omega t + \frac{\pi}{2}\right)$

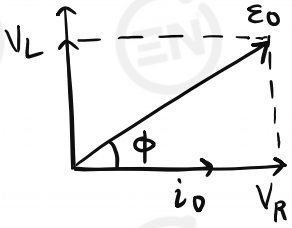
(a) $i_0 = \mathcal{E}_0 / X_C$
$X_C = 1/\omega C$ → CAPACITIVE REACTANCE

(b) i Leads by $\pi/2$

(c) At $\omega \to 0$, $X_C \to \infty$
$\omega \to \infty$, $X_C \to 0$

3. LR CIRCUIT (PHASOR DIG.)





(a) $Z = \sqrt{X_L^2 + R^2}$

(b) $\tan\phi = X_L / R$

(c) $\mathcal{E}$ Leads i by ϕ
$\therefore \mathcal{E} = \mathcal{E}_0 \sin(\omega t + \phi)$
↳ $\mathcal{E}_0 = i_0 Z$

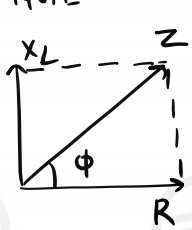
(d) $v_L = V_L \sin(\omega t + \pi/2)$
↳ $V_L = i_0 X_L$

$\mathcal{E}_0 = \sqrt{V_L^2 + V_R^2} \Rightarrow i_0 Z = \sqrt{(i_0 X_L)^2 + (i_0 R)^2}$

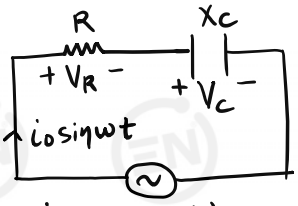
$Z = \sqrt{X_L^2 + R^2}$ → Impedance

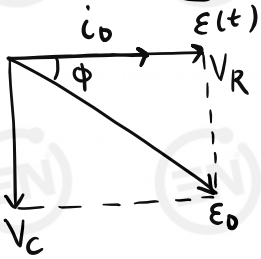
$\tan\phi = \frac{V_L}{V_R} = \frac{X_L}{R}$

NOTE:



4. RC CIRCUIT





$\mathcal{E}_0 = \sqrt{V_R^2 + V_C^2}$

$i_0 Z = \sqrt{(i_0 R)^2 + (i_0 X_C)^2}$

$\Rightarrow Z = \sqrt{R^2 + X_C^2}$
↳ Impedance

$\tan\phi = \frac{V_C}{V_R} = \frac{X_C}{R}$

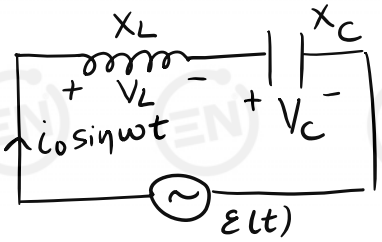
(a) $Z = \sqrt{X_C^2 + R^2}$, $X_C = \frac{1}{\omega C}$

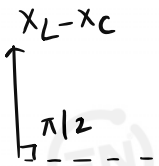
(b) $\tan\phi = X_C / R$

(c) $\mathcal{E}$ lags i by ϕ
$\therefore \mathcal{E} = \mathcal{E}_0 \sin(\omega t - \phi)$
↳ $\mathcal{E}_0 = i_0 Z$

(d) $v_C = V_C \sin\left(\omega t - \frac{\pi}{2}\right)$

5. LC CIRCUIT



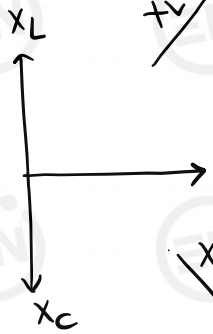


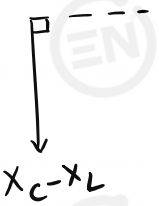
(a) $Z = X_L - X_C$

(b) $\mathcal{E}$ Leads i by $\pi/2$

(c) $\mathcal{E} = \mathcal{E}_0 \sin\left(\omega t + \frac{\pi}{2}\right)$

↳ $i_0 Z$



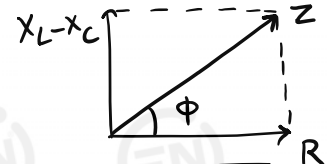


(a) $Z = X_C - X_L$

(b) $\mathcal{E}$ Lags i by $\pi/2$

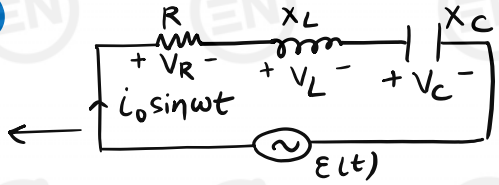
(c) $\mathcal{E} = \mathcal{E}_0 \sin\left(\omega t - \frac{\pi}{2}\right)$

6. RLC (SERIES)



$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$\tan\phi = \frac{X_L - X_C}{R}$



(a) If $X_L > X_C$ (Inductive) ⇒ $\mathcal{E}$ Leads by ϕ

If $X_C > X_L$ (capacitive) ⇒ $\mathcal{E}$ Lags by ϕ

$\mathcal{E} = \mathcal{E}_0 \sin(\omega t + \phi)$
↳ $i_0 Z$

↳ $\mathcal{E} = \mathcal{E}_0 \sin(\omega t - \phi)$

(b) $v_L = V_L \sin(\omega t + \pi/2)$
↳ $V_L = i_0 X_L$

$v_C = V_C \sin(\omega t - \pi/2)$
↳ $V_C = i_0 X_C$

7. POWER IN AC CIRCUIT

$P_{av} = \mathcal{E}_{rms}\, i_{rms} \cos\phi$

NOTE: In questions If nothing mentioned, take given supply Voltage as $\mathcal{E}_{rms}$.

Power factor, $\cos\phi = R/Z$

- only C, $\cos\phi = 0 \Rightarrow P_{av} = 0$
- only L, $\cos\phi = 0, \Rightarrow P_{av} = 0$
- only R, $\cos\phi = 1, \Rightarrow P_{av}$ is Max
- RL, $\cos\phi = \frac{R}{\sqrt{X_L^2 + R^2}}$
- RC, $\cos\phi = \frac{R}{\sqrt{X_C^2 + R^2}}$
- RLC, $\cos\phi = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$

8. RESONANT FREQUENCY (SERIES RLC)

↳ That value of ω for which impedance is purely resistive. ($Z = R$)

(a) $Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$

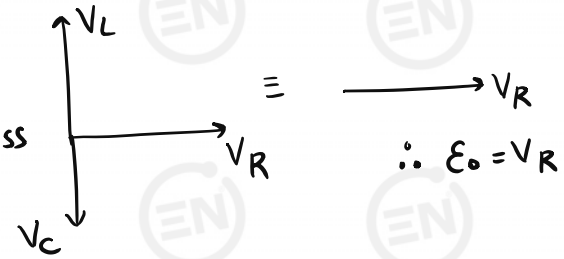
$\therefore$ for $\omega L = \frac{1}{\omega C}$, $Z = R \Rightarrow \omega_R = \frac{1}{\sqrt{LC}}$

(b) At resonance, $P_{av} = i^2_{rms} R = \frac{\varepsilon^2_{rms}}{R}$ $(\cos\phi = 1)$

(c) $\because X_L = X_C$

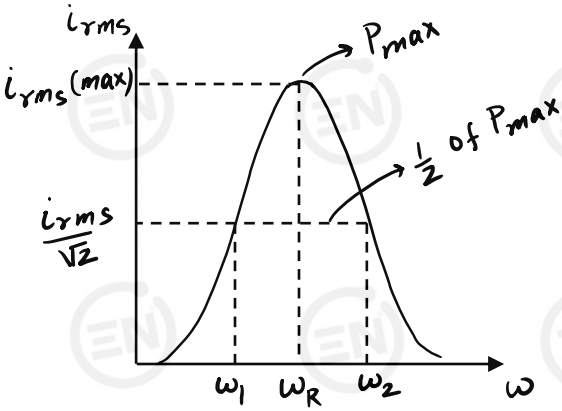
$\Rightarrow V_L = V_C$

$\therefore$ All supply is across Resistor.



9. CURRENT VARIATION WITH ω (SERIES RLC)

$$i_{rms} = \frac{\varepsilon_{rms}}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$



(a) For ω_1 and ω_2 P_{av} is $\frac{1}{2}$ of P_{max}.

(b) BANDWIDTH $\Delta\omega = \omega_2 - \omega_1 = R/L$

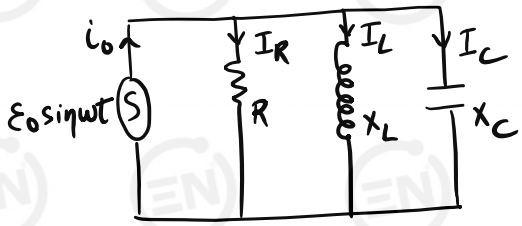
(c) QUALITY FACTOR (Q factor)

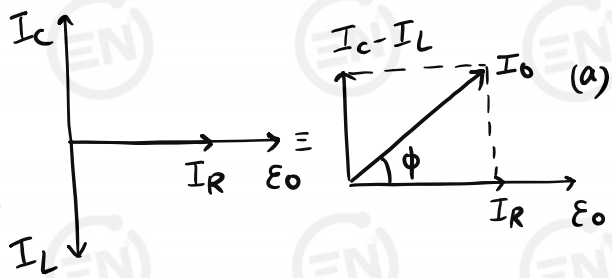
$$Q = \frac{\omega_R}{\Delta\omega} = \frac{1}{\sqrt{LC}} \times \frac{L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

(d) If $\omega < \omega_R \Rightarrow X_C > X_L$ (i leads)

(e) $\omega > \omega_R \Rightarrow X_L > X_C$ (i lags)

10. RLC (PARALLEL)





(a) $\tan\phi = \frac{I_C - I_L}{I_R} = \frac{\frac{\varepsilon_0}{X_C} - \frac{\varepsilon_0}{X_L}}{\frac{\varepsilon_0}{R}}$

(b) $I_0 = \sqrt{I_R^2 + (I_C - I_L)^2}$

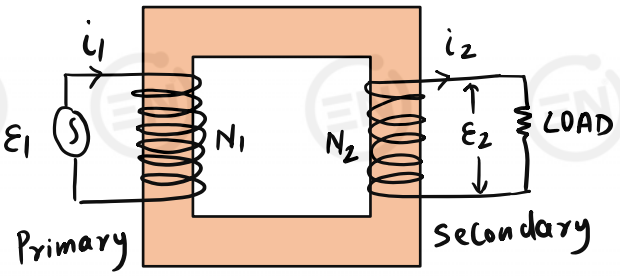
$\Rightarrow \frac{\varepsilon_0}{Z} = \sqrt{\left(\frac{\varepsilon_0}{R}\right)^2 + \left(\frac{\varepsilon_0}{X_C} - \frac{\varepsilon_0}{X_L}\right)^2}$

$\Rightarrow \frac{1}{Z} = \sqrt{\frac{1}{R^2} + \left(\frac{1}{X_C} - \frac{1}{X_L}\right)^2}$

(c) $\omega_R = \frac{1}{\sqrt{LC}}$

(d) At ω_R, Z is max $\Rightarrow i$ is min.

11. TRANSFORMER



N_1 and N_2 are number of turns.

(a) $\frac{\mathcal{E}_1}{\mathcal{E}_2} = \frac{N_1}{N_2}$

- $N_1 > N_2 \Rightarrow \mathcal{E}_1 > \mathcal{E}_2$ (step down transformer)
- $N_2 > N_1 \Rightarrow \mathcal{E}_2 > \mathcal{E}_1$ (step up transformer)

(b) For ideal transformer (No Losses)

$\mathcal{E}_1 i_1 = \mathcal{E}_2 i_2$

(c) LOSSES

- Cu loss (Joules heating)
- Eddy current (Heat due to it)

Space to add concepts learnt from PYQs if any

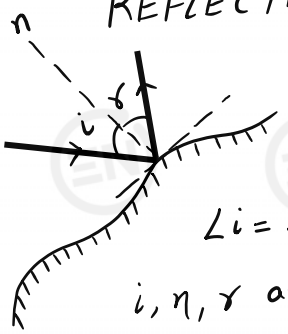
Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel

Topics to cover in RAY OPTICS – PART 1

1.Law of reflection
2.Vector form of reflected ray
3.Minimum mirror length to see ones image
4.Angle of deviation
5.Mirror rotation
6.Image velocity
7.Spherical Mirror – Important Terms
8.Mirror Formulae
9.Magnification (Transverse & Longitudinal)
10. Image formation in Concave Mirror (2 imp case of magnification)
11.Diameter of image of Sun (distant object)
12.Image velocity in spherical mirrors
13.Refraction – Snell's Law
14.Apparent Depth (image formation due to Plane surface)
15.Shifting due to Slab
16.Lateral displacement
17.Spherical Refraction Image Formulae
18.Thin Lenses: Lens Formulae & Lens Makers Formulae
19.Magnification
20.Optical Power
21.Combination of thin lenses
22.Combination of Lenses and Mirrors (equivalent focal length)
23.Displacement method to measure focal length of convex lens
24.Critical angle and TIR
25.Prism
26.Dispersive Power
27.Condition of deviation without dispersion
28.Condition of dispersion without deviation

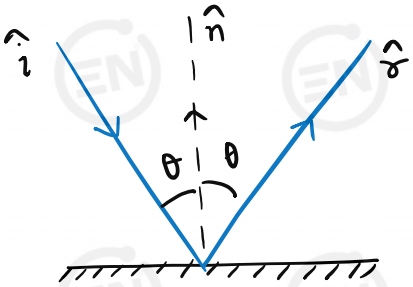
Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. REFLECTION



i, n, r are in same plane

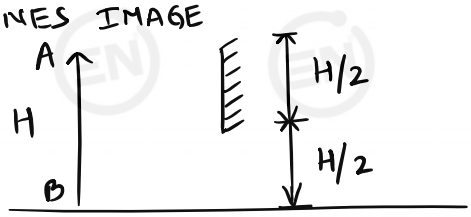
2. VECTOR FORM OF REFLECTED RAY



$$\hat{r} = \hat{i} - 2(\hat{i}\cdot\hat{n})\cdot\hat{n}$$

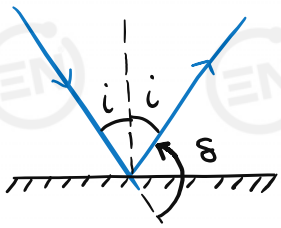
$\hat{i}$, $\hat{r}$ and $\hat{n}$ are unit vectors along incident ray, reflected ray and normal.

3. MINIMUM MIRROR LENGTH TO SEE ONES IMAGE



* Assuming eye level at A

4. ANGLE OF DEVIATION (δ)

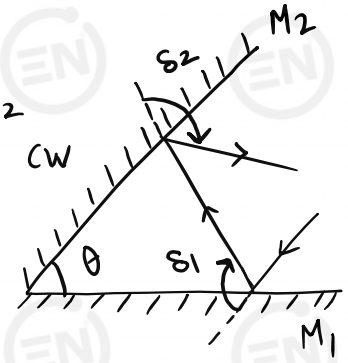


$\delta = \pi - 2i$

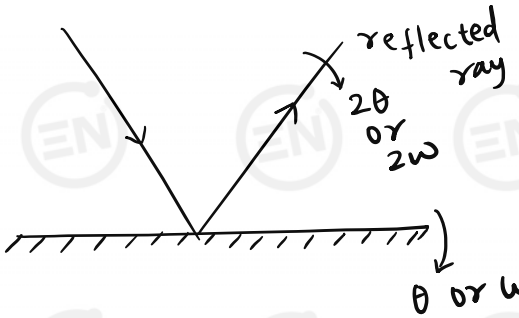
Here it is ACW rotation

$\delta_{NET} = \delta_1 + \delta_2$

$= 2\pi - 2\theta$ CW

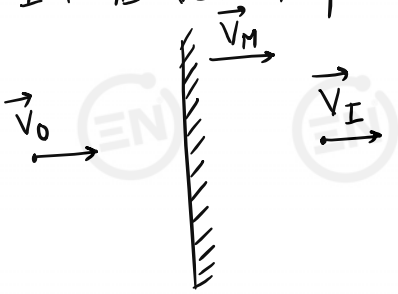


5. MIRROR ROTATION



↳ If mirror rotated by angle θ, reflected ray rotates by 2θ. (ω is angular vel)

6. IMAGE VELOCITY



$$\vec{V}_{O/M} = -\vec{V}_{I/M}$$

$$\Rightarrow \vec{V}_O - \vec{V}_M = -(\vec{V}_I - \vec{V}_M)$$

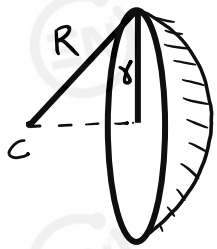
$$\Rightarrow \boxed{\vec{V}_I = 2\vec{V}_M - \vec{V}_O}$$

* these are all Velocities Normal to mirror surface.

SPHERICAL MIRROR

7. MIRROR IMPORTANT TERMS

(a)



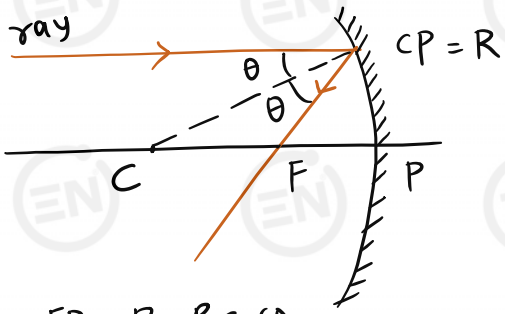
r: radius of aperture

2r: Aperture size

C: center of Curvature

R: Radius of Curvature

(b)



$$FP = R - \frac{R}{2}\sec\theta$$

NOTE:

If rays were PARAXIAL (close to axis) then θ is very small $\Rightarrow \sec\theta \simeq 1$

$\Rightarrow$ F is FOCUS

and $\boxed{FP = \frac{R}{2}}$

8. MIRROR FORMULAE

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \quad \text{or} \quad \boxed{v = \frac{uf}{u-f}}$$

– Put u, f with sign

– f is -VE for CONCAVE and +VE for CONVEX Mirror

9. MAGNIFICATION

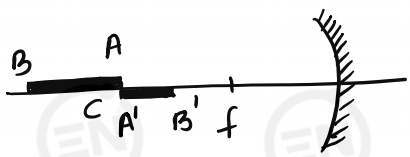
TRANSVERSE, $m = \frac{h_i}{h_o} = \frac{-v}{u} = -\frac{f}{u-f}$

* Put terms with sign
* Erect image $\Rightarrow$ m is +VE
* INVERTED $\Rightarrow$ m is -VE

LONGITUDNAL

$$m = \frac{\text{Image Length along P.A.}}{\text{Object Length along PA}}$$

EX:



$$m = \frac{A'B'}{AB}$$

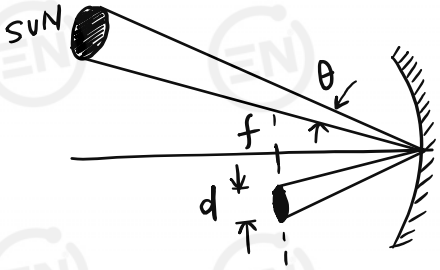
10. Six standard cases of image formation in CONCAVE MIRROR.

→ Must remember them all

→ Magnified image

- Object between f and C (Real image) m is -VE
- Object between pole and focus (Virtual image) m is +VE.

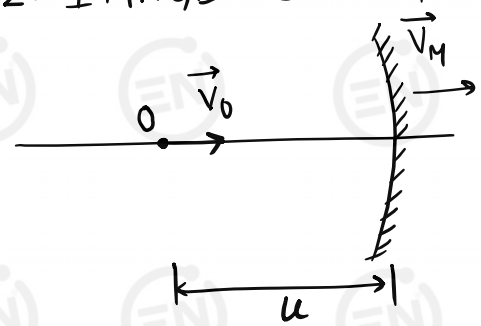
11. Diameter of image of SUN (Distant object)



$d = \theta f$

↳ θ in radians

12. IMAGE VELOCITY



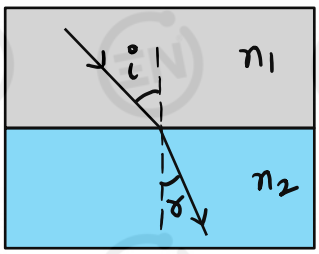
$$V_{I/M} = -\left(\frac{v}{u}\right)^2 V_{O/M}$$

If $V_M = 0$ then, $V_I = -\left(\frac{f}{u-f}\right)^2 V_0$

↳ put u, f, V_0 with sign

REFRACTION

13. SNELL'S LAW



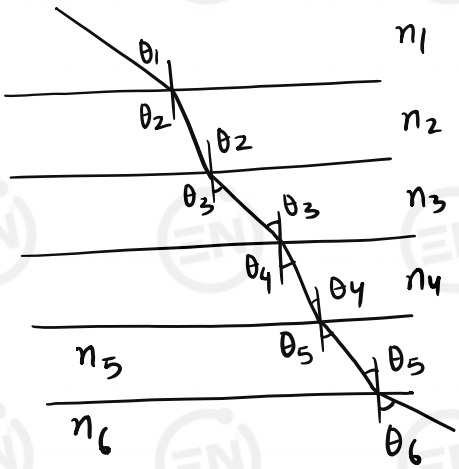
(a.) $n_1 \sin i = n_2 \sin r$

(b) As per this fig. $n_2 > n_1$

(c) $\frac{n_2}{n_1} = \frac{V_1}{V_2} = \frac{\lambda_1}{\lambda_2}$

#V_1 and V_2 are speed of light in medium 1 and 2.

#

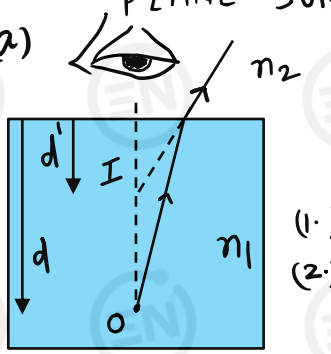


$n_1 \sin\theta_1 = n_2 \sin\theta_2 = n_3 \sin\theta_3 = \ldots = n_6 \sin\theta_6$

$\therefore$ $n \sin\theta = \text{constant}$

14. IMAGE FORMATION DUE TO PLANE SURFACE

(a)



$$d' = d\frac{n_2}{n_1}$$

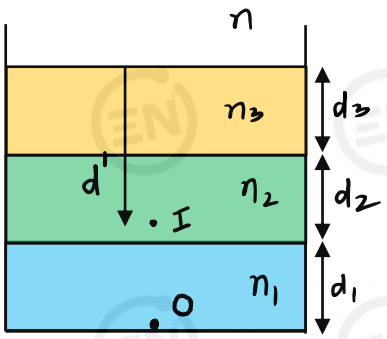
(1.) d' is Apparent depth

(2.) $d' < d$ If $n_1 > n_2$

$d' > d$ If $n_1 < n_2$

(3.) Paraxial rays or observer is looking from above.

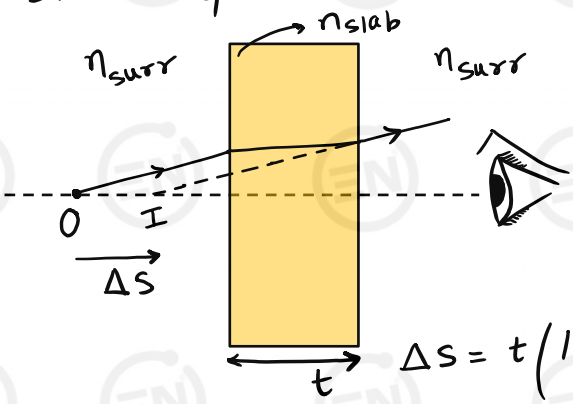
(b)



$$d' = n\left(\frac{d_1}{n_1} + \frac{d_2}{n_2} + \frac{d_3}{n_3}\right)$$

15. SHIFTING DUE TO SLAB

(a)

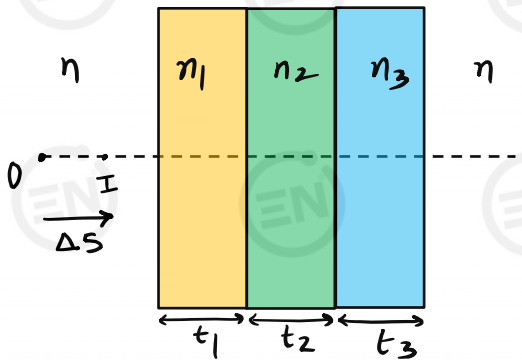


$$\Delta S = t\left(1 - \frac{n_{surr}}{n_{slab}}\right)$$

(1.) valid for paraxial

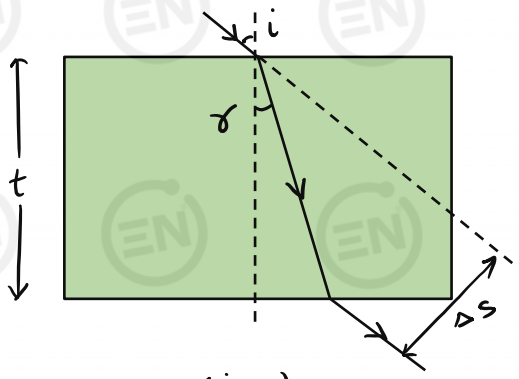
(2.) valid only if medium around slab is same

(b)



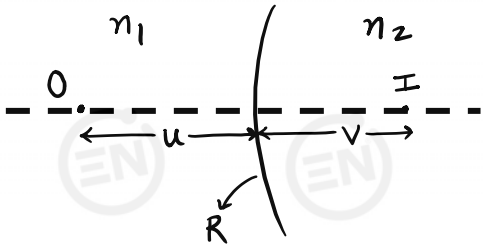
$$\Delta S = t_1\left(1 - \frac{n}{n_1}\right) + t_2\left(1 - \frac{n}{n_2}\right) + t_3\left(1 - \frac{n}{n_3}\right)$$

16. LATERAL DISPLACEMENT



$$\Delta S = \frac{t\sin(i-r)}{\cos r}$$

17. SPHERICAL REFRACTION IMAGE FORMULAE



$$\frac{n_2}{V} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

(1.) Put u, v, R with sign

(2.) n_2 is medium where rays are going

18. THIN LENSES: LENS FORMULAE

$$\frac{1}{V} - \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad V = \frac{uf}{u+f}$$

(1.) f is +VE for converging lens and -VE for diverging lens

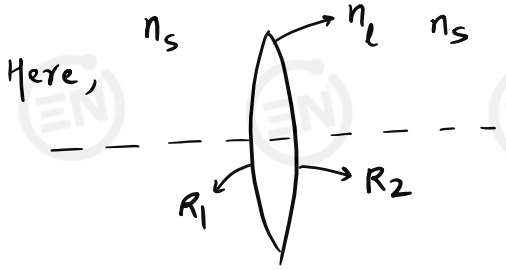
(2.) Valid for paraxial rays

19. MAGNIFICATION

$$m = \frac{h_i}{h_o} = \frac{v}{u} = \frac{f}{u+f}$$

↳ Transverse mag.

LENS MAKER'S FORMULAE



Here,

$$\frac{1}{f} = \left(\frac{n_\ell}{n_s} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

(1.) Valid only if Lens is surrounded by only one medium.

(2.) Put R_1, R_2 with sign

20. OPTICAL POWER

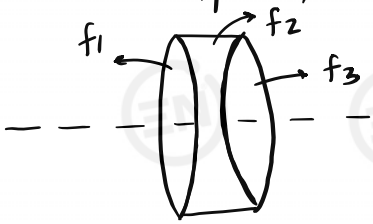
$$P = \frac{1}{f_\ell}$$

↳ UNIT: DIOPTER

↳ Put f_ℓ with sign

↳ Put f_ℓ in meters

21. Combination of thin Lenses, f_{eq}



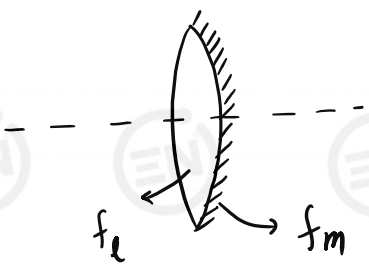
$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$

(1.) Put f_1, f_2, f_3 with sign

(2.) Lenses must be in contact

(3.) f_1, f_2, f_3 are individual focal lengths w.r.t surrounding medium.

22. COMBINATION OF LENS and MIRRORS (f_{eq})



$$\frac{1}{f_{eq}} = \frac{1}{f_m} - \frac{2}{f_\ell}$$

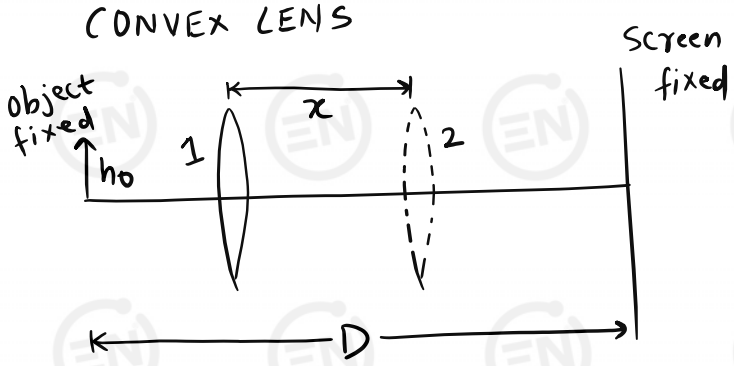
(1.) Put f_m, f_ℓ with sign.

(2.) If f_{eq} is +VE ⇒ Equivalent Convex mirror

If f_{eq} is -VE ⇒ Concave mirror

If f_{eq} is ∞ ⇒ Plane Mirror

23. DISPLACEMENT METHOD TO MEASURE FOCAL LENGTH OF CONVEX LENS



$$f = \frac{D^2 - x^2}{4D}$$

D: Distance between screen and object

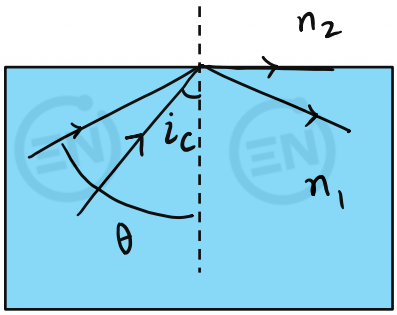
x: Distance between two positions of lens.

$$h_0 = \sqrt{h_1 h_2}$$

h_1: image height when LENS is at pos^n 1

h_2: when at pos^n 2

24. CRITICAL ANGLE AND TIR



(1.) Here $n_1 > n_2$

(2.) i_c: critical angle for which $r = 90°$

(3.) If $\theta > i_c \Rightarrow$ TIR

$$\sin i_c = \frac{n_2}{n_1} \quad \text{or} \quad i_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$

25. PRISM

(1.) A: ANGLE of PRISM, $A = r_1 + r_2$

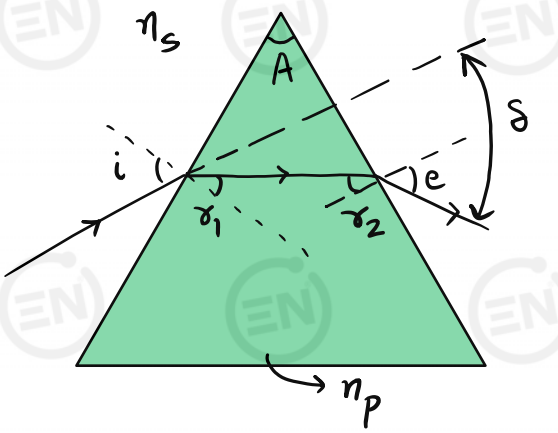
(2.) δ: ANGLE of Deviation

$$\delta = i + e - A$$

(3.) For δ to be minimum, δ_{min}

$i = e \Rightarrow r_1 = r_2 = r$

$$\therefore \frac{n_p}{n_s} = \frac{\sin\left(\frac{\delta_m + A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$



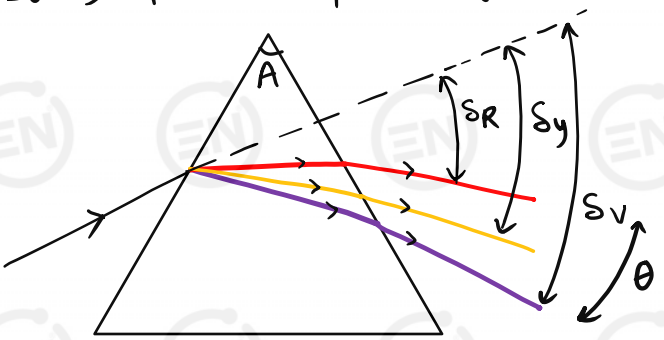
n_p: R.I of PRISM

n_s: RI of Surrounding

(4.) If A is very small (THIN PRISM), then

$$\delta = A\left(\frac{n_p}{n_s} - 1\right)$$

26. DISPERSIVE POWER(ω)



$\delta_R = A(n_R - 1)$, $\delta_y = A(n_y - 1)$, $\delta_V = A(n_V - 1)$

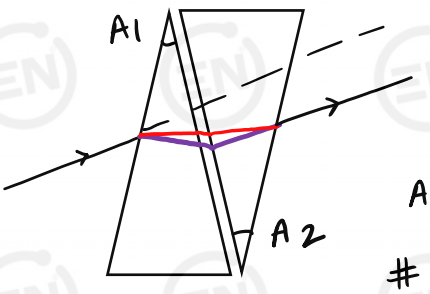
n_R, n_y, n_V are R.I. of medium for Red, yellow and violet color

(Cauchy's eqⁿ, $n = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} + \cdots$)

(a.) θ (Dispersion angle) $= \delta_V - \delta_R = \boxed{A(n_V - n_R)}$

(b.) ω (Dispersive power) $= \frac{\delta_V - \delta_R}{\delta_y} = \boxed{\frac{n_V - n_R}{n_y - 1}}$

27. CONDITION for DEVIATION WITHOUT DISPERSION



$\theta_{NET} = 0$

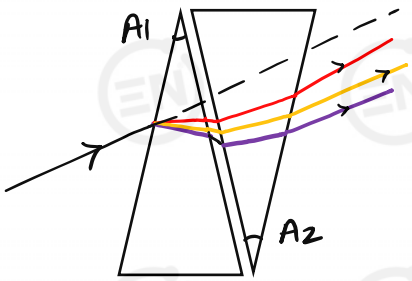
$\Rightarrow \theta_1 = \theta_2$

or

$A_1(n_{V_1} - n_{R_1}) = A_2(n_{V_2} - n_{R_2})$

Achromatic Prism Combination

28. CONDITION FOR DISPERSION WITHOUT DEVIATION



Here final emergent yellow is parallel to incident white light

So, $\delta_{net} = 0 \Rightarrow \delta_1 = \delta_2$

$\Rightarrow A_1(n_{y_1} - 1) = A_2(n_{y_2} - 1)$

Space to add concepts learnt from PYQs if any

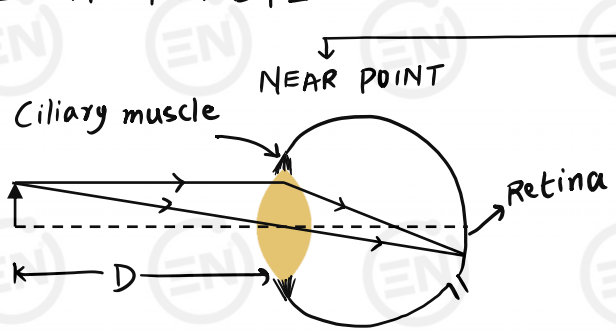
Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in OPTICAL INSTRUMENTS – PART 2

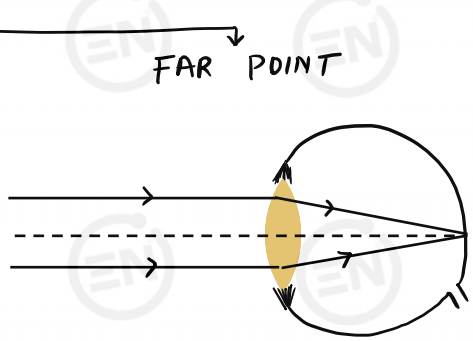
1.Human Eye
2.Eye Defects
3.Angular size
4.Simple Microscope
5.Compound Microscope
6.Refracting Telescope
7.Limit of Resolution & Resolving Power

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. HUMAN EYE



Least distance of Distinct Vision, D = 25cm (for normal eye)



For a normal eye far point is Infinity.

NOTE:
(a) Ciliary muscle changes eye lens curvature to change "f" so that sharp image is formed on RETINA.

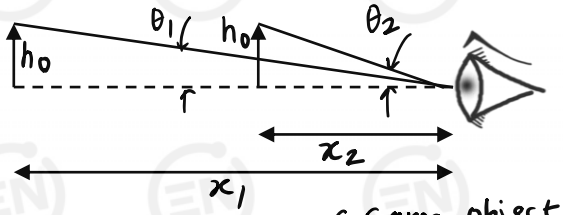
2. EYE DEFECTS

NAME	DEFECT	CORRECTIVE LENS
Myopia Nearsightedness	- Far object not clear - Rays converge before Retina	Concave Lens
Hypermetropia Farsightedness	- Near object not clear - Rays converge after Retina	Convex Lens
Presbyopia	Elderly person, generally is not able to read a book at about 25 cm distance from the eye	Bifocal Lens
Astigmatism	- Distorted image - generally occurs if the eye is myopic or hypermetropic.	Cylindrical Lens

old age issue of hardening of eye lens

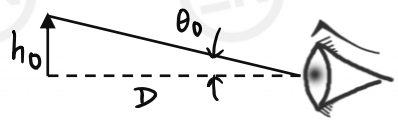
Cornea not spherical in shape

3. ANGULAR SIZE

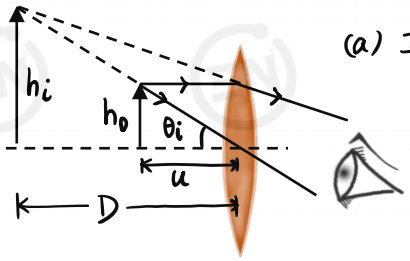


$\theta_1 = \frac{h_0}{x_1}$, $\theta_2 = \frac{h_0}{x_2}$ { Same object appears large if close to our eye }

4. SIMPLE MICROSCOPE



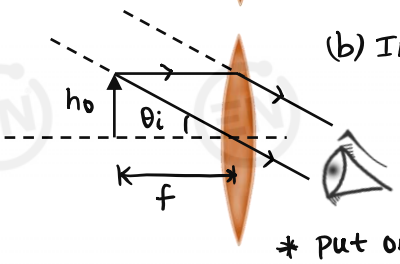
Angular size of object, $\theta_0 = \frac{h_0}{D}$



(a) Image at Near point, D

$\theta_i = \frac{h_0}{u}$, $u = \frac{Df}{D+f}$

$m = \frac{\theta_i}{\theta_0} = 1 + \frac{D}{f}$



(b) Image at Far Point, ∞

$\theta_i = h_0 / f$

$m = \frac{\theta_i}{\theta_0} = \frac{D}{f}$

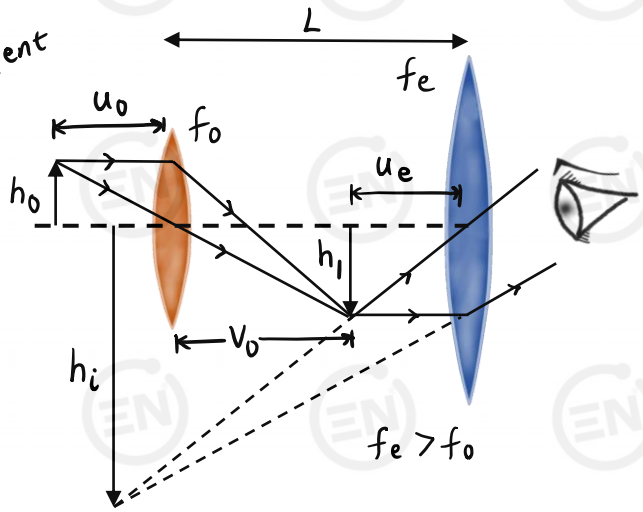
* put only magnitudes

5. COMPOUND MICROSCOPE

$m = m_o \times m_e$

Image at D: $m = \frac{V_o}{u_o}\left(1+\frac{D}{f_e}\right)$

Image at ∞ (normal Adjustment): $m = \frac{V_o}{u_o}\frac{D}{f_e}$



Generally $f_o \ll L$, $f_e \ll L$

$m = \frac{L}{f_o}\left(1+\frac{D}{f_e}\right)$ $\quad m = \frac{L}{f_o}\frac{D}{f_e}$

#Tube length (L)

(a) If image at D, $L = V_o + u_e$

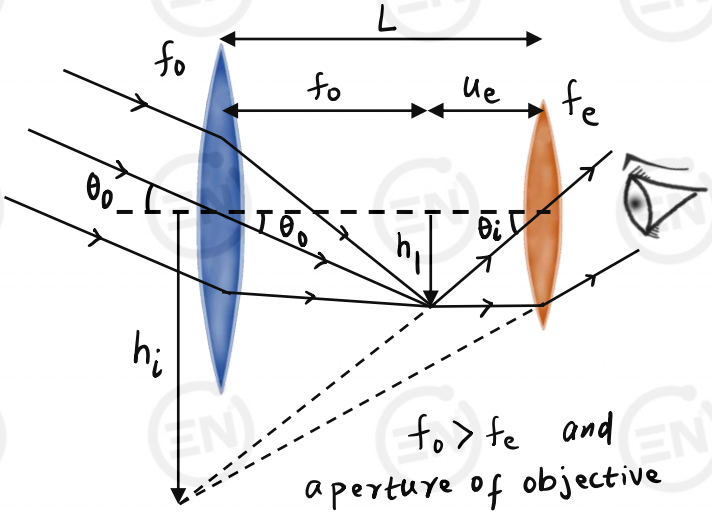
(b) If image at ∞, $L = V_o + f_e$

6. REFRACTING TELESCOPE

$\theta_o = h_1/f_o$, $\theta_i = h_1/u_e$, $u_e = \frac{Df_e}{D+f_e}$

Image at D: $m = \frac{\theta_i}{\theta_o} = \frac{f_o}{f_e}\left(1+\frac{f_e}{D}\right)$

Image at ∞: $\theta_i = h_1/f_e$, $\therefore m = \frac{\theta_i}{\theta_o} = \frac{f_o}{f_e}$



#Tube length

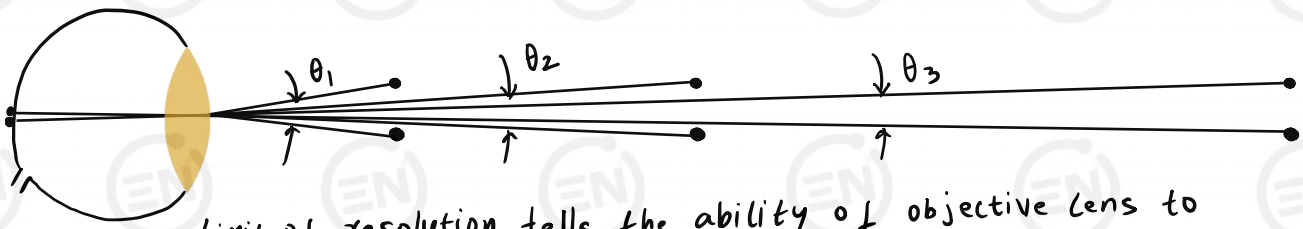
(a) If image at D, $L = f_o + u_e$

(b) If image at ∞, $L = f_o + f_e$

$f_o > f_e$ and aperture of objective is large to capture more light.

* In all formulae put only magnitudes.

7. LIMIT OF RESOLUTION, RESOLVING POWER

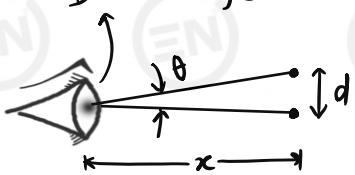


Limit of resolution tells the ability of objective lens to resolve two objects distinctively.

(a) There will be a minimum value of θ for which two objects are just resolved in image that θ is called "Limit of Resolution" and $\frac{1}{\theta}$ is "Resolving power"

7. LIMIT OF RESOLUTION, RESOLVING POWER (RP)

Eye



Limit of Resolution, $\theta = \frac{1.22\lambda}{D}$

$RP = 1/\theta$, $d = \theta x$

Telescope

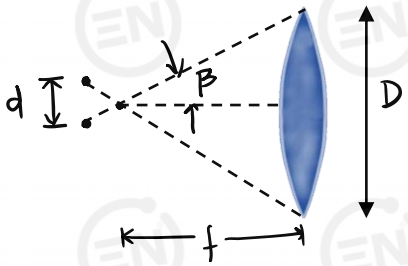
D = Aperture of Objective Lens

Two stars will be just resolved if,

$\# \theta = \frac{1.22\lambda}{D}$, $RP = \frac{D}{1.22\lambda}$

If distance of star is x, separation between them, $d = x\theta$

Microscope



$d_{min} = \frac{1.22\lambda}{2\mu \sin\beta}$

$RP = \frac{1}{d_{min}}$

(i) Numerical Aperture (NA) $= \mu \sin\beta$

(ii) μ: Refractive index of medium between Lens and object

Space to add concepts learnt from PYQs if any

Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in YDSE & INTERFERENCE– PART 1 (Wave Optics/EM Waves)

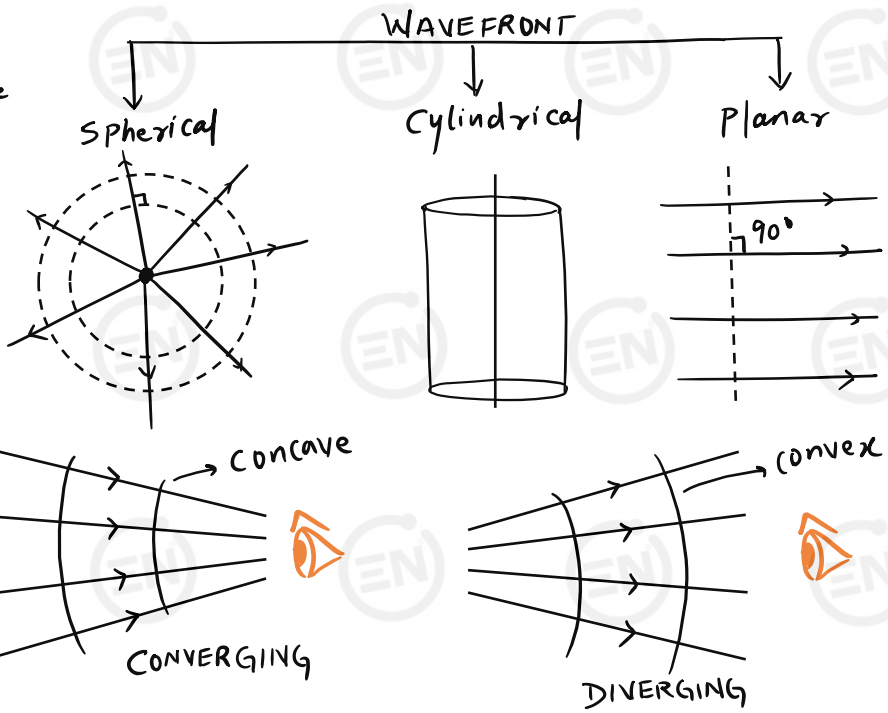
1. Wavefront
2. Net Amplitude and Intensity, Phase and Path difference relation
3. Constructive and Destructive Interference
4. YDSE
5. Position of Dark and Bright Fringes
6. Fringe Width and Angular Fringe Width
7. Effect on Fringe Width if YDSE Setup is in a Medium
8. Geometrical and Optical Path Length
9. Thin Film in YDSE
10. Shifting of Fringe in Oblique Incidence
11. White Light in YDSE
12. Slit Width effect in YDSE
13. Interference in Thin Films

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

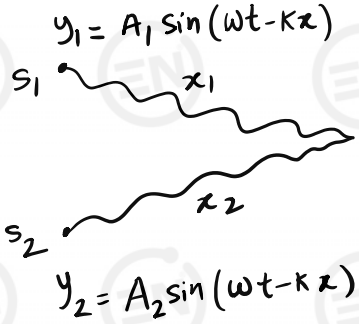
1. WAVEFRONT

When light propagates, the cross section where all particles oscillate in same phase is called Wavefront.

Wavefront is Normal to propagation direction



2. A_{net} and I_{net} IN INTERFERENCE (Coherent source, Same ω)



$y = y_1 + y_2 = A_1 \sin(\omega t - kx_1) + A_2 \sin(\omega t - kx_2)$

Phase difference, $\Delta\phi = k(x_2 - x_1) = k\Delta x$

$\Rightarrow \boxed{\Delta\phi = \frac{2\pi}{\lambda}\Delta x}$

$$A_{net} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\Delta\phi}$$

If $A_1 = A_2 = A$

$A_{net} = 2A\cos\frac{\Delta\phi}{2}$

($I = kA^2$)

$$I_{net} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\phi$$

If $I_1 = I_2 = I_0$

$I_{net} = 4I_0\cos^2\frac{\Delta\phi}{2}$

3. $\Delta\phi$, Δx, A_{net}, I_{net}

$$A_{net} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\Delta\phi}$$

$$I_{net} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\phi$$

INTERFERENCE

CONSTRUCTIVE

(i) $\Delta\phi = 2n\pi$

(ii) $2n\pi = \frac{2\pi}{\lambda}\Delta x$

$\Delta x = n\lambda$

(iii) $A_{net} = A_1 + A_2$

If $A_1 = A_2 = A$

$A_{net} = 2A$

(iv) $I_{net} = (\sqrt{I_1} + \sqrt{I_2})^2$

If $I_1 = I_2 = I_0$

$I_{net} = 4I_0$ (MAXIMA)

DESTRUCTIVE

(i) $\Delta\phi = (2n+1)\pi$

(ii) $(2n+1)\pi = \frac{2\pi}{\lambda}\Delta x$

$\Delta x = (2n+1)\frac{\lambda}{2}$

(iii) $A_{net} = A_1 - A_2$

If $A_1 = A_2 = A$

$A_{net} = 0$

(iv) $I_{net} = (\sqrt{I_1} - \sqrt{I_2})^2$

If $I_1 = I_2 = I_0$

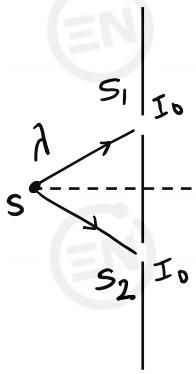
$I_{net} = 0$ (MINIMA)

4. YOUNG's DOUBLE SLIT EXPERIMENT (YDSE)

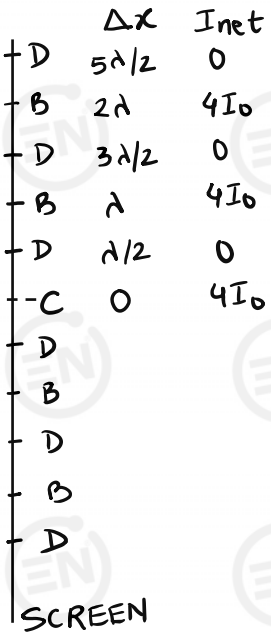
C: Central bright fringe

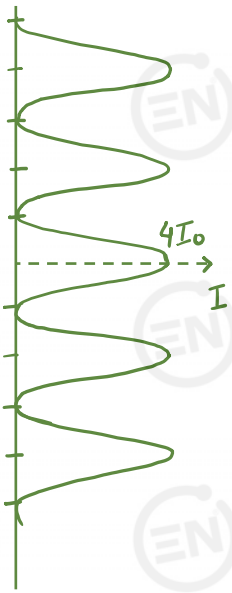
B: Bright fringe

D: Dark fringe

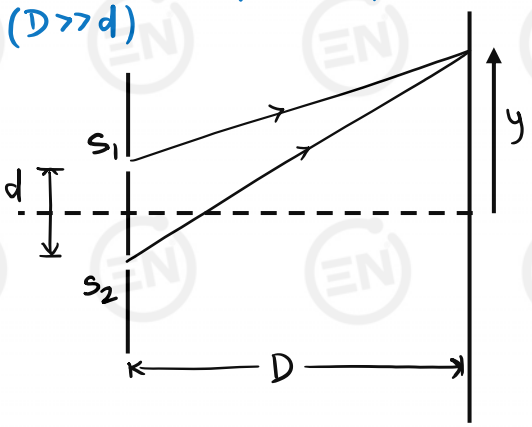


S_1 and S_2 are slits.





5. DISTANCE OF BRIGHT AND DARK FRINGES (D >> d)



Path difference at y, $\Delta x = \frac{yd}{D}$

BRIGHT FRINGE	DARK FRINGE
$\Delta x = n\lambda$	$\Delta x = (2n-1)\frac{\lambda}{2}$
$\Rightarrow \frac{y_n d}{D} = n\lambda$	$\Rightarrow \frac{y_n d}{D} = (2n-1)\frac{\lambda}{2}$
$\Rightarrow y_n = \frac{n\lambda D}{d}$	$\Rightarrow y_n = (2n-1)\frac{\lambda D}{2d}$

6. FRINGE WIDTH AND ANGULAR FRINGE WIDTH

↳ Distance between two successive bright or dark fringe.

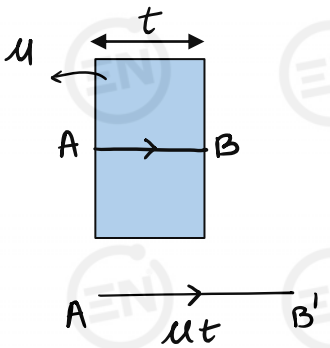
(a) $\beta = \frac{\lambda D}{d}$ (b) $\beta_\theta = \frac{\lambda}{d}$

7. β IF YDSE SETUP IS IN A MEDIUM OF REFRACTIVE INDEX μ.

If in air, $\beta = \frac{\lambda D}{d}$

In medium, $\beta' = \frac{\lambda D}{\mu d} = \frac{\beta}{\mu}$

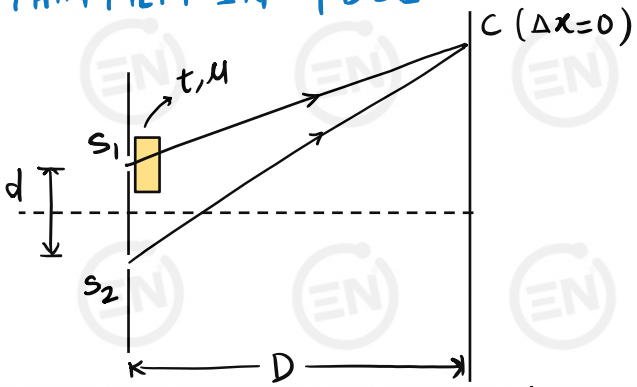
8. GEOMETRICAL AND OPTICAL PATH LENGTH



Geometrical path = AB = t

Optical path = AB' = μt

9. THIN FILM IN YDSE

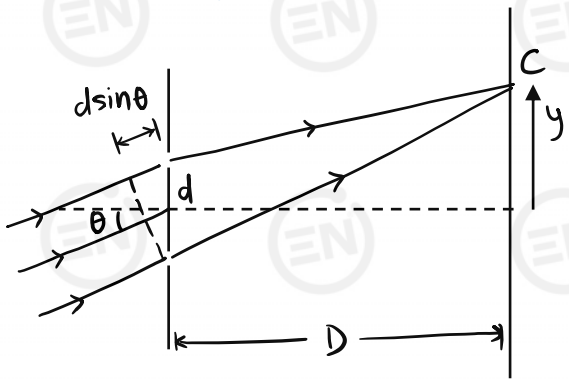


$$t(\mu-1) = \frac{yd}{D}$$

$$\Rightarrow y = \frac{tD(\mu-1)}{d}$$

↳ shift in fringe Pattern

10. SHIFTING OF FRINGE IN OBLIQUE INCIDENCE



FRINGE SHIFTS UP.

$$d\sin\theta = \frac{yd}{D}$$

$$\Rightarrow y = D\sin\theta$$

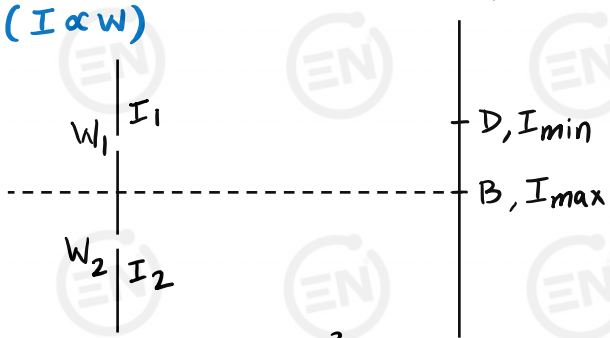
11. WHITE LIGHT IN YDSE

(1.) CENTRAL BRIGHT fringe is white colour

(2.) As you move a little away you see Reddish colour (violet destructive interference)

(3.) Move further away its bluish colour.

12. SLIT WIDTH EFFECT IN YDSE ($I \propto W$)



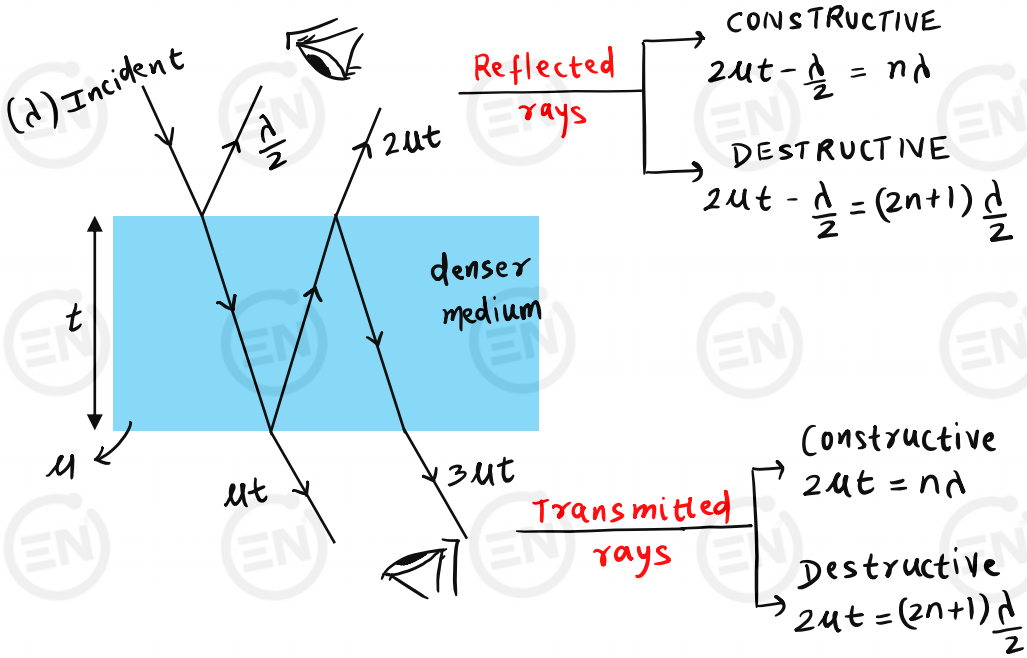
$$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

Here we don't see contrasting fringe pattern

13. INTERFERENCE IN THIN FILMS (normal incidence)

NOTE:
When reflection is from denser medium $\frac{\lambda}{2}$ path difference is added



Space to add concepts learnt from PYQs if any

Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in DIFFRACTION & POLARIZATION– PART 2 (Wave Optics/EM Waves)

1. Diffraction
2. Intensity variation in Diffraction from Single Slit
3. Fringe Pattern, Angular Position of Minima
4. Diffraction by Circular Aperture
5. Polarization of Light
6. Malus Law
7. Methods of Polarization of Light by
   a) Reflection of Light (Brewster's Law)
   b) Double Refraction
   c) Scattering
   d) Dichroism
8. Nicol Prism
9. Doppler's Effect in Light

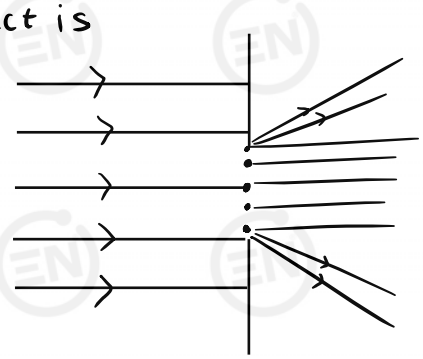
Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. DIFFRACTION

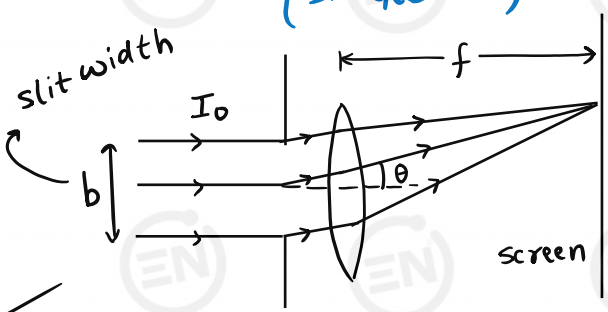
It is bending of Light around corners/edges.

If slit size is very Large, Diffraction effect is negligible.

But if small, effect is significant



2. INTENSITY IN DIFFRACTION (SINGLE SLIT)



$$I(\theta) = I_0 \frac{\sin^2\beta}{\beta^2}, \quad \beta = \frac{\pi b \sin\theta}{\lambda}$$

↳ MINIMA at $\beta = n\pi$

$\therefore n\pi = \frac{\pi b \sin\theta}{\lambda}$

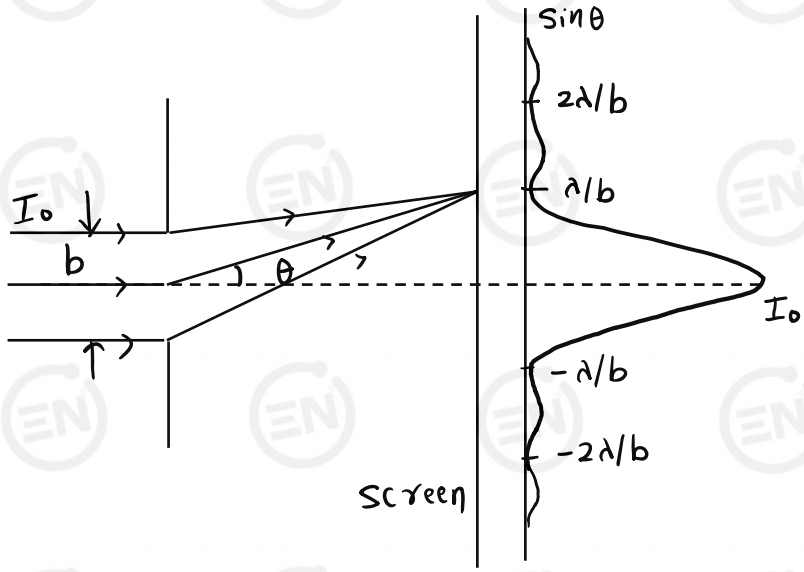
$\boxed{\sin\theta = \frac{n\lambda}{b}}$ and $\boxed{\theta = \frac{n\lambda}{b}}$ if θ is small.

3. FRINGE PATTERN

(a) for $\theta = 0^\circ$, $I = I_0$

(b) n^{th} MINIMA, $\sin\theta = \frac{n\lambda}{b}$ $n \neq 0$

(c) Unlike YDSE, here both fringe width and Intensity decreases as you move away from central Maxima.



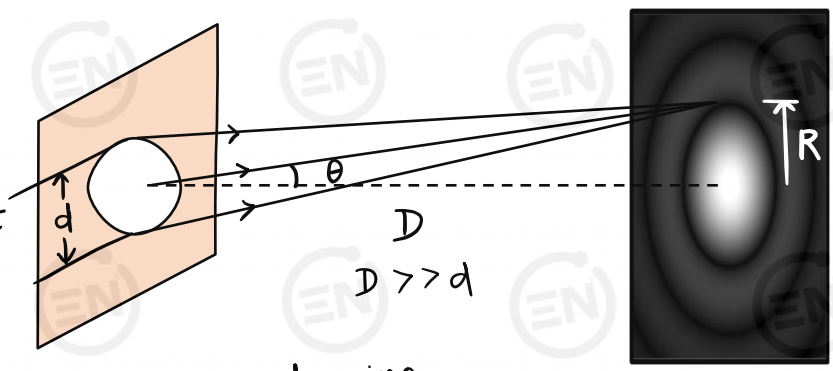
$$I(\theta) = I_0 \frac{\sin^2\beta}{\beta^2}, \quad \beta = \frac{\pi b \sin\theta}{\lambda}$$

4. DIFFRACTION BY CIRCULAR APERTURE

First MINIMA on screen,

$$\theta = 1.22\frac{\lambda}{d}$$

(a) RADIUS of 1st Dark fringe OR radius of central bright fringe, $R = \theta D = 1.22\frac{\lambda D}{d}$



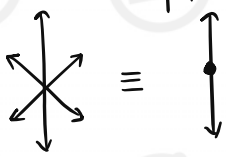
(b) If Light is converged using convex Lens at the screen placed at focal plane of Lens,

$$R = \theta f = 1.22\frac{\lambda f}{d}$$

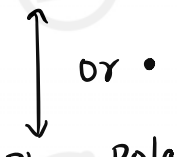
Circular fringes

5. POLARIZATION OF LIGHT

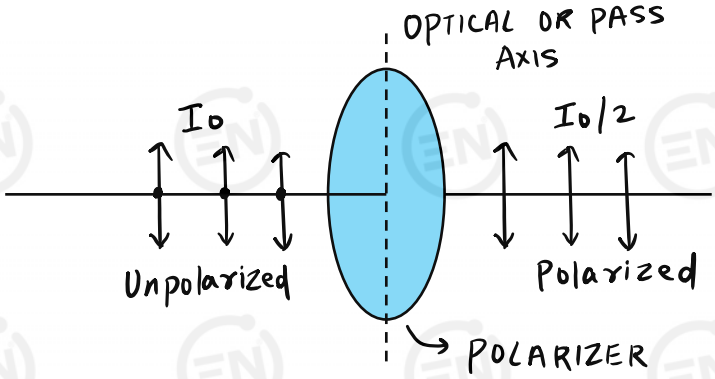
↳ Electric Field oscillating in one plane.



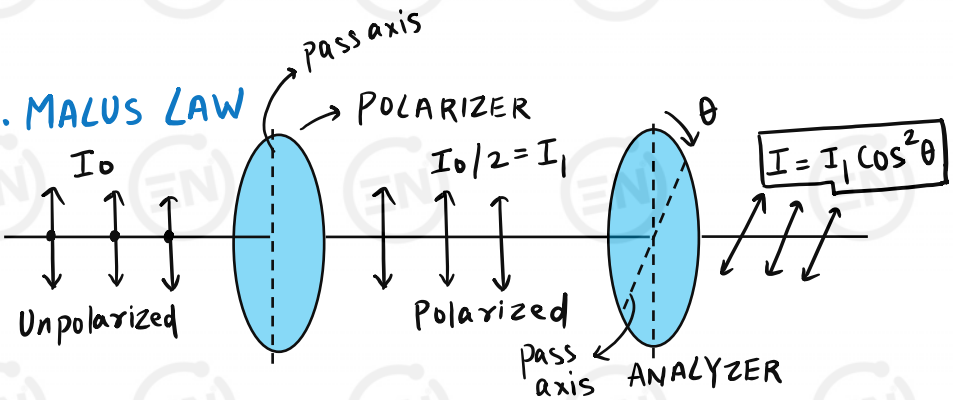
UNPOLARIZED LIGHT



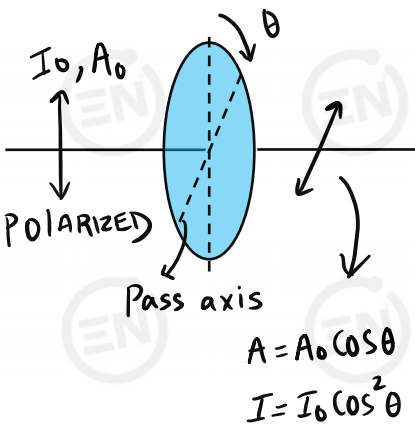
Plane Polarized Light



6. MALUS LAW



#Generalized

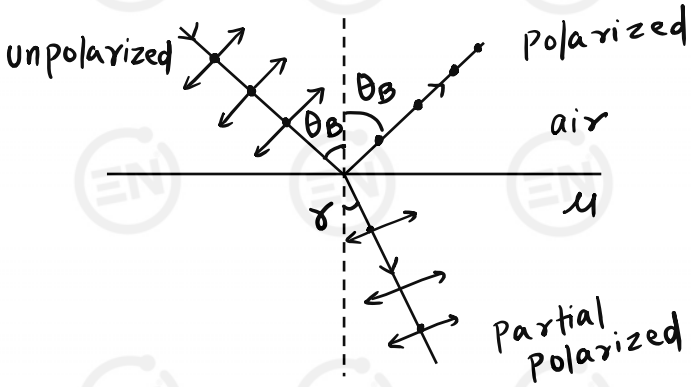


$A = A_0 \cos\theta$

$I = I_0 \cos^2\theta$

7. METHODS OF POLARIZATION OF LIGHT

(a) REFLECTION (BREWSTER'S LAW)



BREWSTER'S ANGLE (θ_B)

θ_B for which angle between reflected and refracted ray is 90°

$\therefore r = 90 - \theta_B$

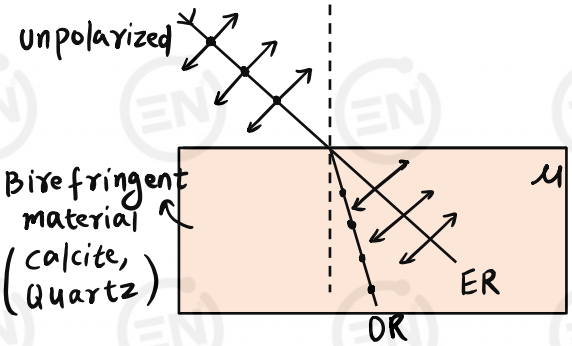
SNELL'S LAW:

$\sin\theta_B = \mu \sin(90-\theta_B)$

$\Rightarrow \boxed{\theta_B = \tan^{-1}\mu}$

Brewster's Law

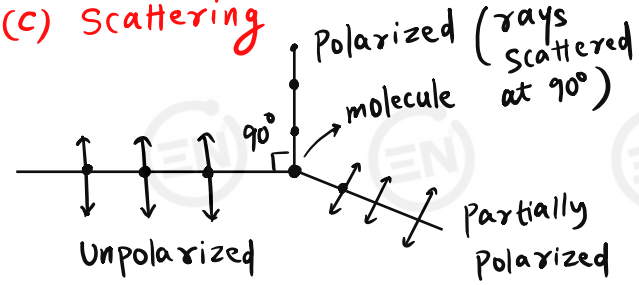
(b) Double Refraction



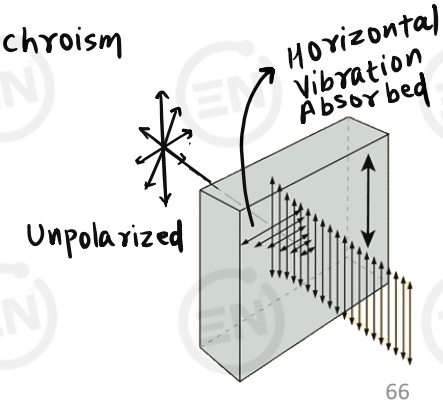
OR: Ordinary ray (Polarized normal to plane)

ER: Extraordinary ray (Polarized in plane)

(c) Scattering

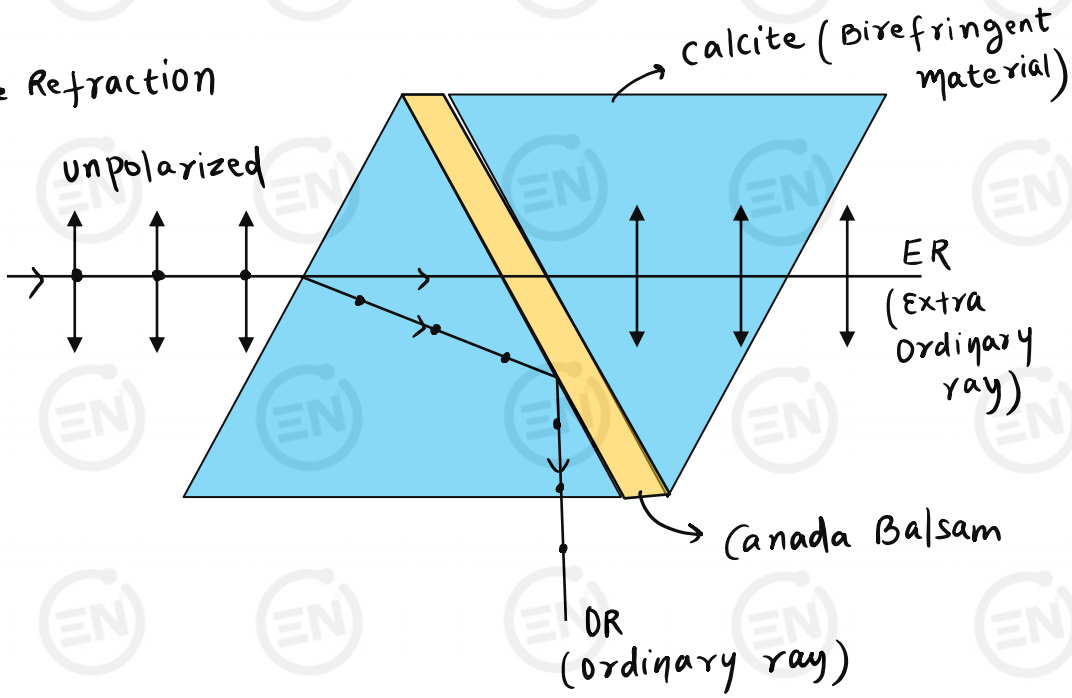


(d) Dichroism



8. NICOL PRISM

↳ uses Double Refraction



9. Doppler's Effect in Light

(1) $\frac{\Delta f}{f} = -\frac{V_{radial}}{c}$

$\Delta f = f_{app} - f$ (f is freq of source, f_{app} is observed freq)

V_{radial} (radial vel of source relative to observer)

NOTE: Valid if source speed is $<< c$

(2) $f\lambda = c \Rightarrow \frac{\Delta f}{f} = -\frac{\Delta\lambda}{\lambda}$ (for small changes in f & λ)

$\Rightarrow \frac{\Delta\lambda}{\lambda} = \frac{V_{radial}}{c}$ { use to find radial Velocity of distant Galaxy

Space to add concepts learnt from PYQs if any

Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in EM Waves – PART 3 (Wave Optics/EM Waves)

1. Types of EM Waves
2. Displacement Current
3. Ampere – Maxwell Law
   a. NCERT Solved Ex.
4. Maxwell's Equation
5. EM Waves equation and Key Points
   a) Relation among c, E and B
   b) Intensity of EM Waves
   c) Speed of EM Waves

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. TYPES OF EM WAVES

↳ Transverse waves
↳ Electric and magnetic field energy density is same.

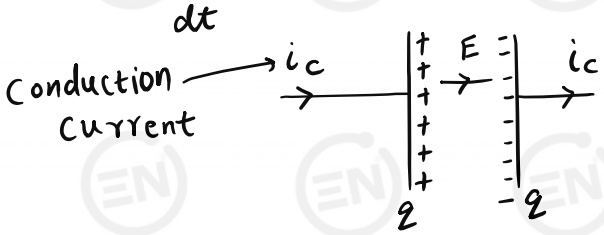
$u_B = \frac{B^2}{2\mu_0}$, $u_E = \frac{1}{2}\epsilon_0 E^2$

Type	Wavelength Range	Production	Detection
Radio	>0.1 m	Rapid acceleration and deceleration of electrons in aerials	Receiver's aerials
Microwave	0.1 m to 1 mm	Klystron valve or magnetron valve	Point contact diodes
Infra-red	1 mm to 700 nm	Vibration of atoms and molecules	Thermopiles, Bolometer, Infrared photographic film
Light	700 nm to 400 nm	Electrons in atom emit light when they move from one energy level to a lower energy level	The eye, photocells, photographic film
Ultraviolet	400 nm to 1 nm	Inner shell electrons in atoms moving from one energy level to lower level	Photocells, Photographic film
X-rays	1 nm to 0.001 nm	X-ray tubes or inner shell electrons	Photographic film, Geiger tubes
Gamma rays	< 0.001 nm	Radioactive decay of the nucleus	Photographic film, Geiger tubes

2. DISPLACEMENT CURRENT (i_d)

i_d is due to time varying electric field.

$i_d = \epsilon_0 \frac{d\phi_E}{dt}$, ϕ_E is electric flux



(a) If q changes, E changes

$\Rightarrow i_d = \epsilon_0 A \times \frac{1}{A\epsilon_0}\frac{dq}{dt} = \frac{dq}{dt}$

∴ In capacitor for time varying current conduction current is same as i_d.

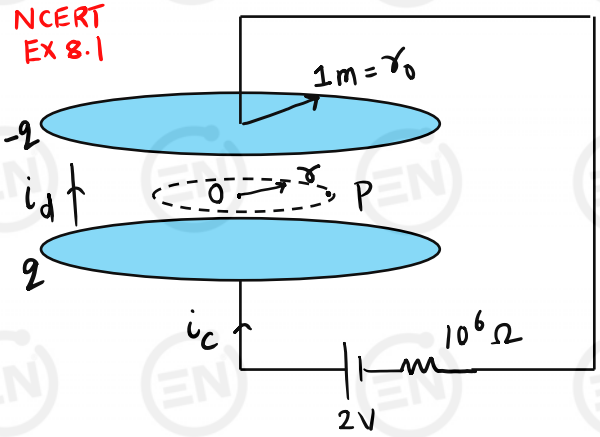
(b) Between plates $i_d \neq 0$, $i_c = 0$
Outside plates $i_c \neq 0$, $i_d = 0$

(c) i_d is uniform across plate cross-section

3. AMPERE'S LAW

$$\oint \bar{B}\cdot d\bar{l} = \mu_0 i_{enclosed} \longrightarrow i_c + \varepsilon_0 \frac{d\phi_E}{dt}$$

NCERT Ex 8.1



$C = 10^{-9}$ F, at t=0 charging starts. Find B at P (OP = 0.5m) at $t = 10^{-3}$ s

Soln^n Consider Ampere loop of $\frac{1}{2}$m radius (r), parallel to plates.

$$\oint \bar{B}\cdot d\bar{l} = \mu_0(i_c + i_d), \quad i_c = 0$$

$$\Rightarrow B \times 2\pi r = \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}, \quad \phi_E = E\pi r^2$$

$$\Rightarrow B \times 2\pi r = \mu_0 \varepsilon_0 \times \frac{1}{4\varepsilon_0}\frac{dq}{dt}$$

$$\Rightarrow \phi_E = \frac{q}{\varepsilon_0}\pi r^2 = \frac{q}{\pi r_0^2 \varepsilon_0}\pi r^2$$

$$\boxed{\phi_E = q/4\varepsilon_0}$$

$$\Rightarrow B\pi = \frac{\mu_0}{4} \frac{dq}{dt}, \quad 'q = q_0(1-e^{-t/RC})'$$

$$\Rightarrow B\pi = \frac{\mu_0}{4} \times \frac{CV}{RC} e^{-t/RC}$$

$$\Rightarrow B = 10^{-7} \times \frac{2}{10^6} e^{-1} = \boxed{7.35 \times 10^{-14}\ T}$$

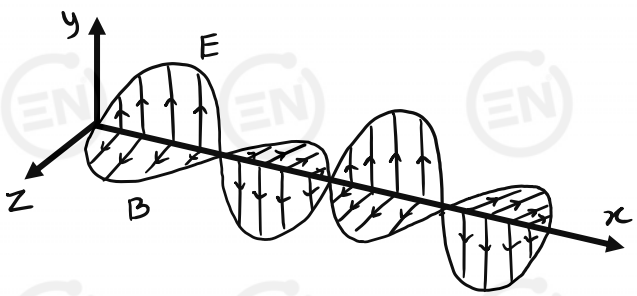
4. MAXWELL'S EQUATIONS

1. Gauss's Law in Electrostatics, $\Phi_E = \oint \bar{E}\cdot d\bar{A} = q_{in}/\varepsilon_0$
2. Gauss's Law for Magnetism, $\phi_B = \oint \bar{B}\cdot d\bar{A} = 0$ { due to closed loop lines of field. This explains why monopole can't exist
3. Faraday's Law, $\oint \bar{E}\cdot d\bar{l} = -\frac{d\phi_B}{dt}$
4. Ampere-Maxwell Law, $\oint \bar{B}\cdot d\bar{l} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$

5. EM WAVES EQUATION AND KEY POINTS

$\vec{E} = E_0 \sin(\omega t - kz)\,\hat{j}$

$\vec{B} = B_0 \sin(\omega t - kx)\,\hat{k}$



KEY POINTS

(a) RELATION AMONG $\hat{c}, \hat{E}, \hat{B}$ (unit vectors along direction of propagation, Electric field and magnetic field)

$\hat{c} = \hat{E} \times \hat{B}$, $\hat{B} = \hat{c} \times \hat{E}$, $\hat{E} = \hat{B} \times \hat{c}$

(b) FROM WAVE EQUATION

$\vec{E} = E_0 \sin(\omega t - kx)\hat{j}$ → E along $\hat{j}$, $E_{rms} = E_0/\sqrt{2}$

→ Wave propagation along +VE x-axis

(c) RELATION BETWEEN E_0 and B_0

$E_0 = cB_0$

(d) SPEED OF EM WAVES

$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ in free space

$c_{medium} = \frac{1}{\sqrt{\mu_r \mu_0 \varepsilon_r \varepsilon_0}} = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$

(e) INTENSITY

$I = \frac{1}{2}\varepsilon_0 c E_0^2 = \varepsilon_0 c E_{rms}^2$

NOTE: In above intensity due to Electric field

$= \frac{I}{2} = \frac{1}{2}\varepsilon_0 c E_{rms}^2$

Space to add concepts learnt from PYQs if any

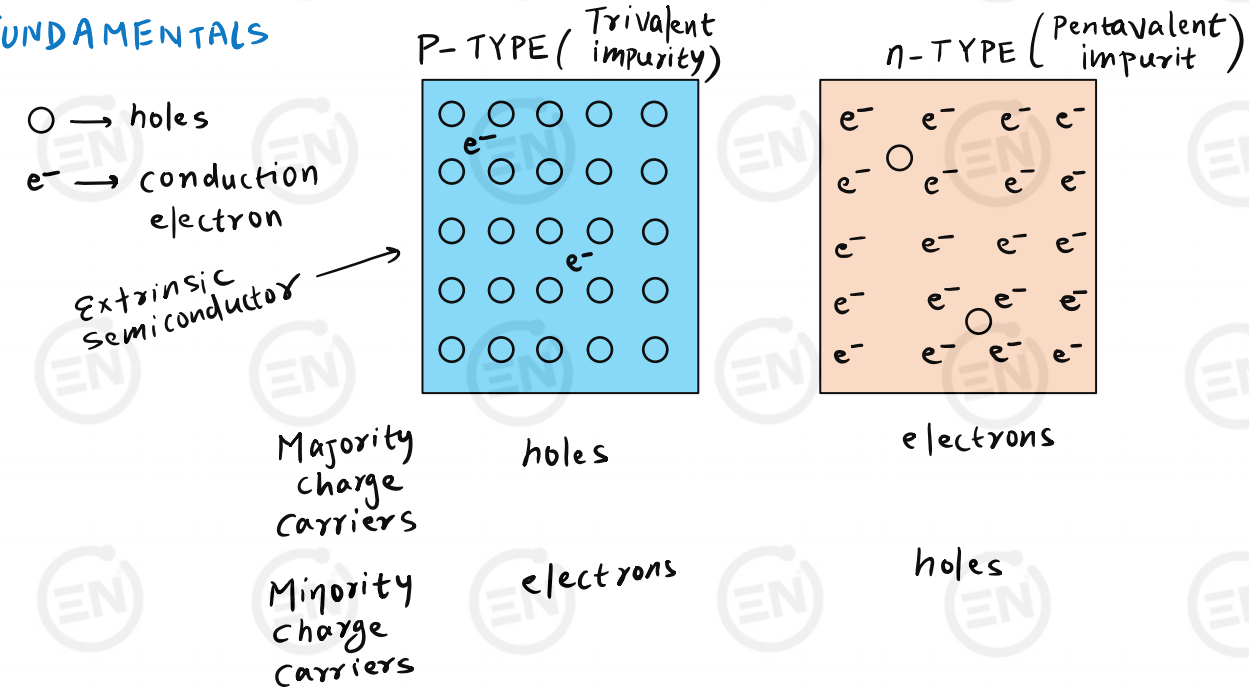
Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

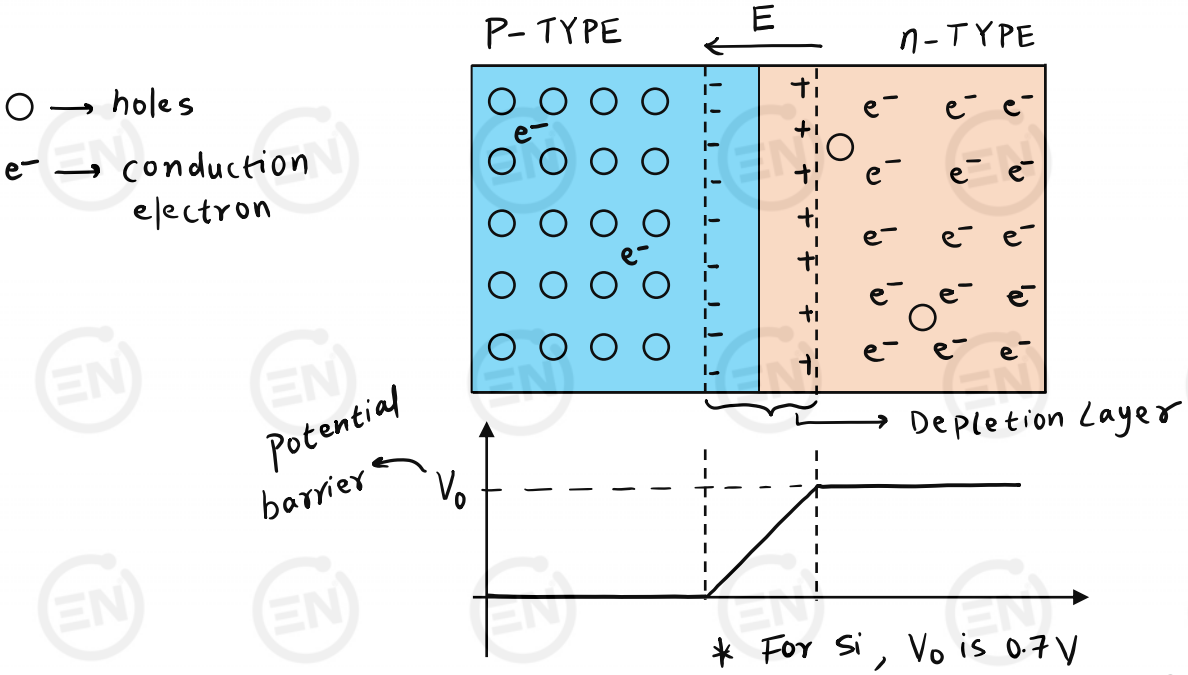
Topics to cover in ZENER DIODE – PART 1 (SEMICONDUCTORS)

1. Fundamentals of PN Diode
2. Forward and Reverse Biased
3. Zener Diode

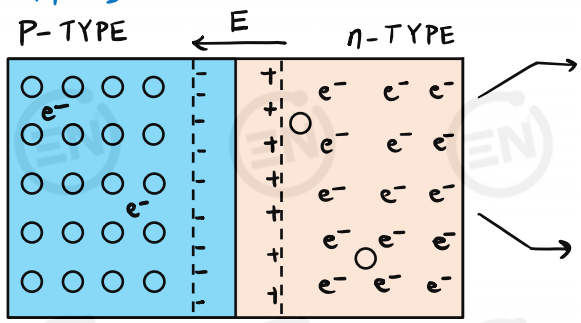
Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

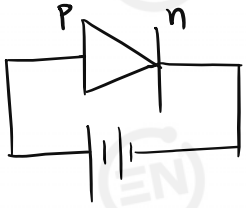
1. FUNDAMENTALS



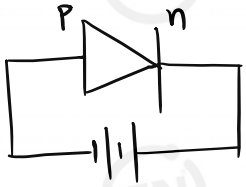


2. FORWARD AND REVERSE BIASED

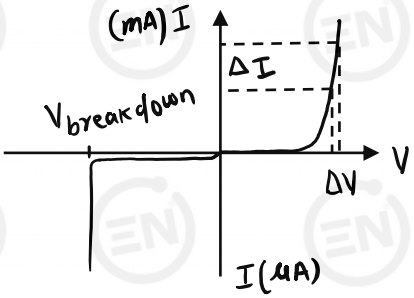




Forward biased
↳ current due to majority charge carriers.



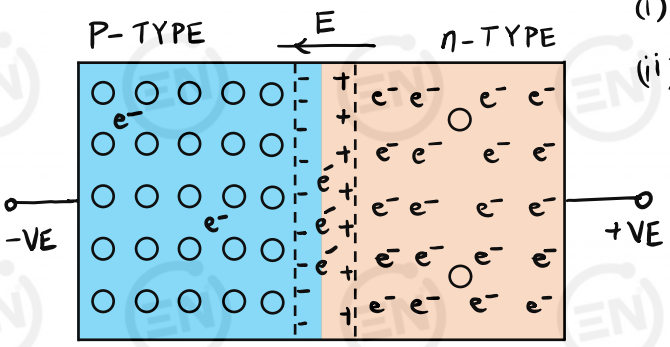
Reverse biase
↳ very small current due to minority charge carriers.



① For ideal diode Resistance is zero (FB) and infinite (RB).

② Dynamic Resistance,
$R = \frac{\Delta V}{\Delta I}$

3. ZENER DIODE



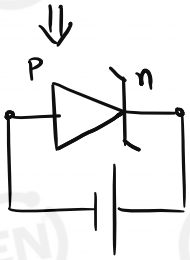
(i) It is heavily doped

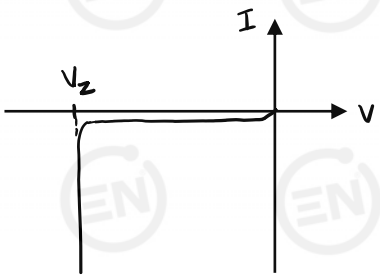
(ii) Depletion region very thin
⇒ E is very large.

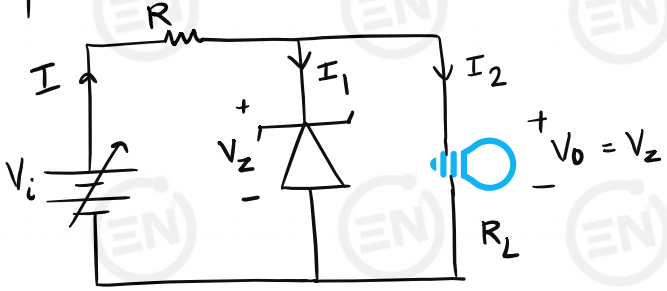
(iii) In RB state, E strength ↑ and e^- covalent bonds in depletion region breaks.

(iv) This e^- moves to n side and reverse current flows from n to P.

Zener Breakdown







① ↳ As p.d. across crosses V_z (breakdown Voltage), almost all current passes through diode.

② ↳ And constant V_z is across it

③ $I_2 = \frac{V_z}{R_L}$

④ $I = I_1 + I_2$

Space to add concepts learnt from PYQs if any

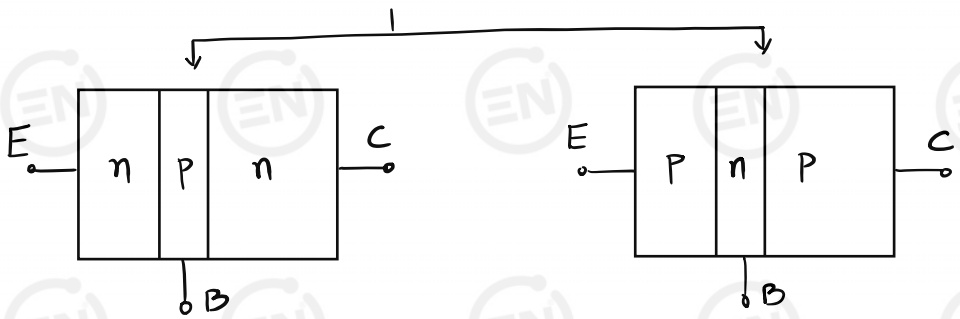
Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in TRANSISTORS – PART 2 (SEMICONDUCTORS)

1. Transistors Types
2. Symbol
3. Working of Transistors
4. DC Current Gain
5. Characteristic Curves
6. Transistor as Amplifier
7. CE Amplifier
8. Gain in CE Amplifiers

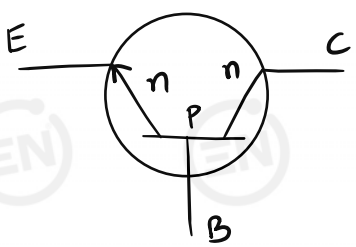
Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

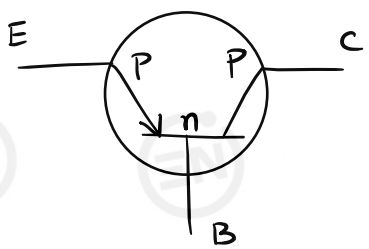
1. TRANSISTORS (3 terminal – 2 Junction Device)



	SIZE	DOPING
E, Emitter	Moderate	Highly Doped
B, BASE	Very thin	Lightly Doped
C, Collector	Large	Moderately Doped

2. CIRCUIT SYMBOL



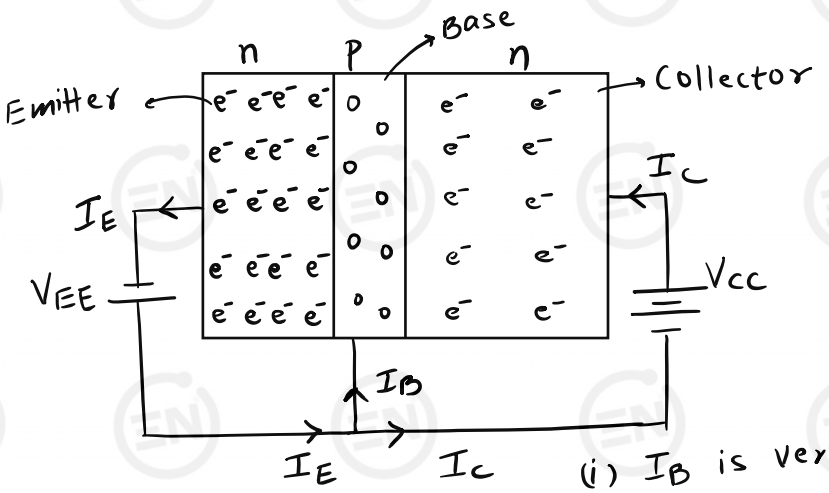


Arrow from P → n

3. WORKING

BE : forward biased

BC : Reverse biased



(i) I_B is very small (μA)

(ii) $I_E = I_C + I_B$

I_C is almost 0.95 to 0.98 I_E

4. DC CURRENT GAINS

↳ Base current Amplification factor, $\beta = \frac{I_C}{I_B}$

↳ Emitter current Amplification factor, $\alpha = I_C / I_E$

↳ $I_E = I_C + I_B \Rightarrow \frac{I_C}{\alpha} = I_C + \frac{I_C}{\beta}$

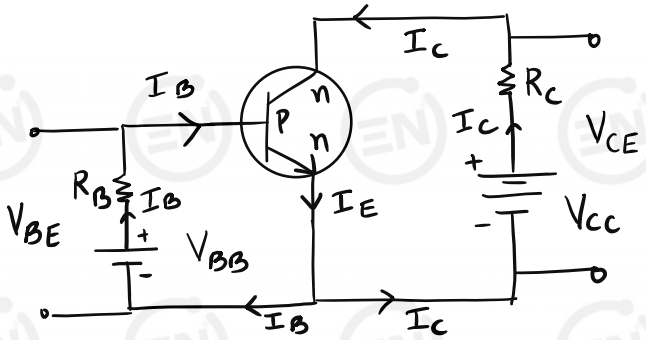
$$\Rightarrow \boxed{\alpha = \frac{\beta}{1+\beta}}$$

NOTE: $\because I_B << I_C$

$\Rightarrow \beta$ is large

and α is a LitHle smaller than 1.

5. CHARACTERISTIC CURVE



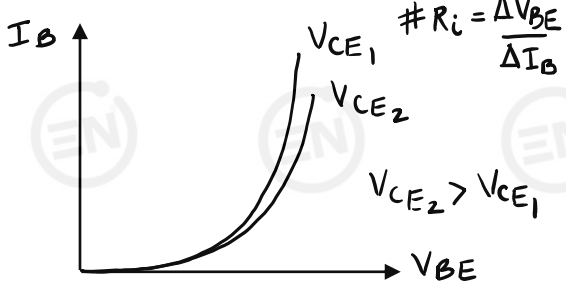
(1) $V_{CE} = V_{CC} - I_C R_C$

(2) $V_{BE} = V_{BB} - I_B R_B$

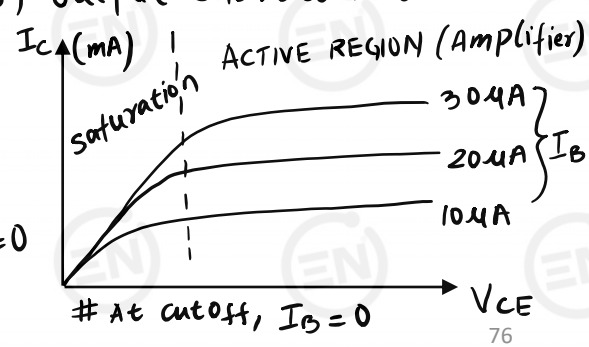
$$\# R_o = \frac{\Delta V_{CE}}{\Delta I_C}$$

At saturation $V_{CE} = 0$

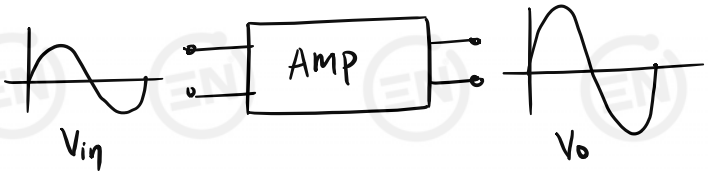
(a) Input characteristic



(b) output characteristic



6. TRANSISTOR AS AMPLIFIER

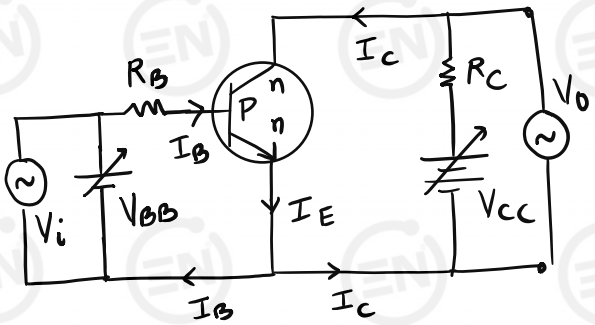


- Common Base Amplifier
- Common Emitter Amplifier
- Common Collector Amplifier

7. COMMON EMITTER AMPLIFIER

↳ 180° phase out.

8. GAIN IN CE AMPLIFIER



A small variation in input current causes large variation in output current.

(i) AC current gain, $\beta_{AC} = \frac{\Delta I_C}{\Delta I_B}$

(ii) AC voltage gain, $A_V = \frac{\Delta V_0}{\Delta V_i} = \frac{\Delta I_C R_0}{\Delta I_B R_i}$

$\Rightarrow A_V = \beta_{AC} \frac{R_0}{R_i}$

(iii) AC Power gain, $A_p = A_V \times \beta_{AC}$

$= \beta_{AC}^2 \frac{R_0}{R_i}$

Space to add concepts learnt from PYQs if any

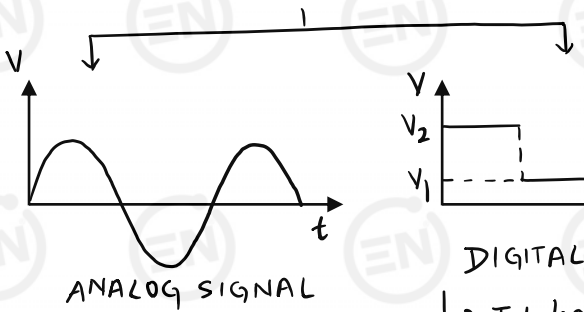
Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in LOGIC GATES – PART 3 (SEMICONDUCTORS)

1. Analog and Digital Signal
2. Logic Gates (Types)
3. NOT Gate
4. AND Gate
5. OR Gate
6. Rules of Boolean Algebra & De Morgan's Theorem
7. NAND Gate
8. NOR Gate
9. Exclusive Gates (XOR & XNOR)

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. ANALOG & DIGITAL SIGNAL



↳ It has two states
(1.) High → 1
(2.) LOW → 0

Logic gates are integral part of Digital Electronics

2. LOGIC GATES

↳ Electrical cicuits using logical relation between input and output voltages.

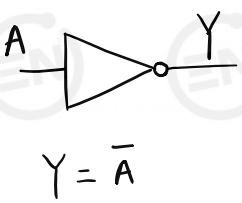
- NOT Gate
- AND Gate
- OR Gate

} BASIC GATES

- NAND Gate
- NOR Gate

} UNIVERSAL GATES

3. NOT GATE (Inversion Gate)



$Y = \bar{A}$

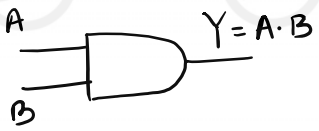
TRUTH TABLE

A	$Y = \bar{A}$
1	0
0	1

Truth Table : Relation between Input and Output.

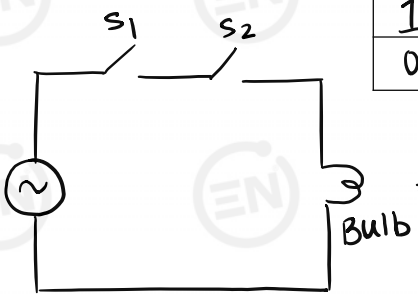
4. AND GATE

↳ Output high (1), if both input is high (1)
↳ Output Low (0), if either input is Low (0)



TRUTH TABLE

A	B	Y = A·B
1	0	0
0	1	0
1	1	1
0	0	0



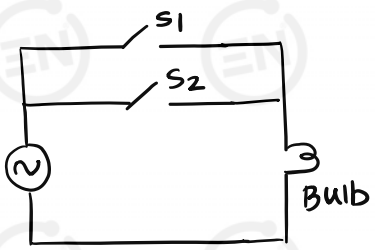
⇒ switch close : 1
switch open : 0
Bulb Glow = 1
else = 0

5. OR GATE

↳ Output high (1), if either input is high (1)
↳ Output Low (0), if both input Low (0)

TRUTH TABLE

A	B	Y = A+B
1	0	1
0	1	1
1	1	1
0	0	0



6. RULES OF BOOLEAN ALGEBRA & DE MORGAN'S THEOREM

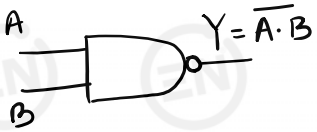
(a) $A + 0 = A$
(b) $A + A = A$
(c) $A \cdot A = A$
(d) $A \cdot \bar{A} = 0$
(e) $\bar{\bar{A}} = A$

DE MORGAN'S THEOREM

$\overline{A \cdot B} = \bar{A} + \bar{B}$ $\quad$ $\overline{A + B} = \bar{A} \cdot \bar{B}$

↳ Boolean Expressions

7. NAND GATE (AND + NOT)

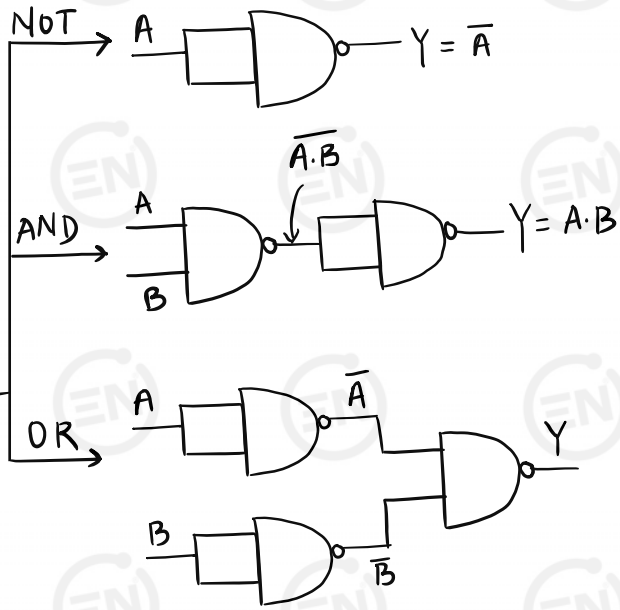


Inverts the AND Gate output.

TRUTH TABLE

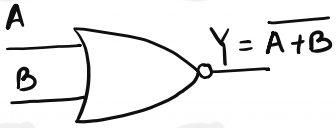
A	B	$Y = \overline{A \cdot B}$
1	0	1
0	1	1
1	1	0
0	0	1

NAND GATE (UNIVERSAL GATE)



$$\# Y = \overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = \boxed{A + B}$$

8. NOR GATE (OR + NOT)

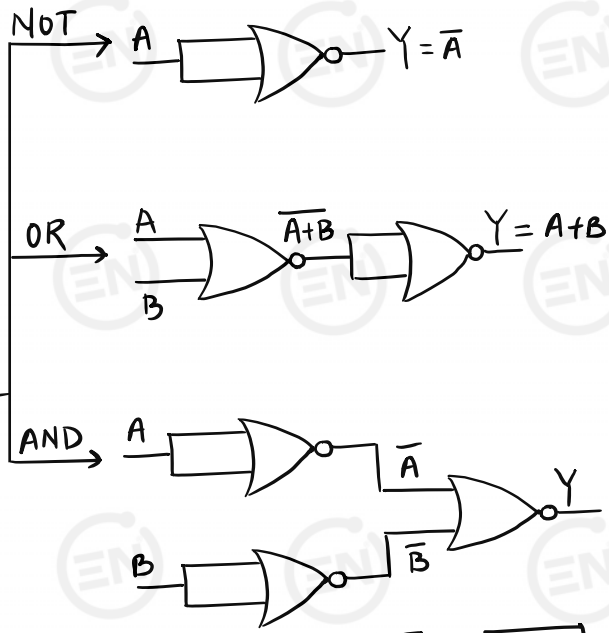


Inverts the AND Gate output.

TRUTH TABLE

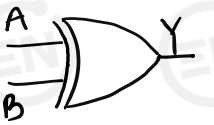
A	B	$Y = \overline{A + B}$
1	0	0
0	1	0
1	1	0
0	0	1

NOR GATE (UNIVERSAL GATE)



$$\# Y = \overline{\overline{A} + \overline{B}} = \overline{\overline{A}} \cdot \overline{\overline{B}} = \boxed{A \cdot B}$$

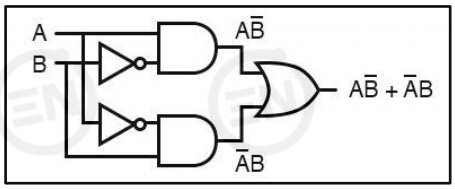
9. EXCLUSIVE GATES

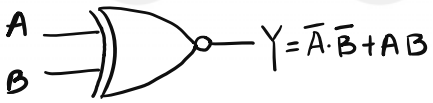


$Y = \bar{A} \cdot B + A \cdot \bar{B}$

XOR (Exclusive OR GATE)

A	B	Y
1	0	1
0	1	1
1	1	0
0	0	0





XNOR (EXCLUSIVE NOR GATE)

A	B	Y
1	0	0
0	1	0
1	1	1
0	0	1

Space to add concepts learnt from PYQs if any

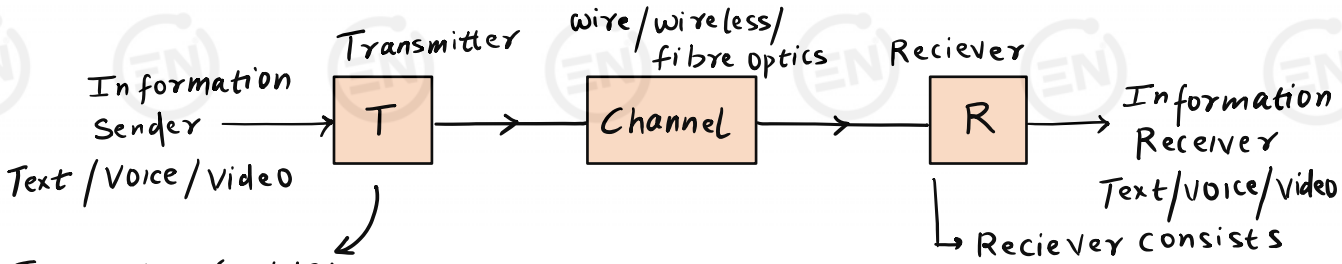
Note: To best use these 1st watch the video from "Revision Series Playlist" on Eduniti YouTube Channel (PYQs are also there for practice)

Topics to cover in COMMUNICATION SYSTEMS

1. Elements of Communication System
2. Noise, Attenuation & Amplification
3. Range and Repeater
4. Bandwidth of Signal and Transmission Medium
5. Transmission of EM Waves through Space
6. Propagation Modes for EM Waves
7. Problems in Transmission of Baseband Signals
8. Modulation of Baseband & Carrier Waves
9. Types of Modulation
10. Amplitude Modulation (AM)

Note: For video refer Revision Series Playlist on EDUNITI YouTube Channel

1. ELEMENTS OF COMMUNICATION SYSTEMS



Transmitter consists:

(a) Transducer
Converts information into electrical signal
Ex: Mic, fax

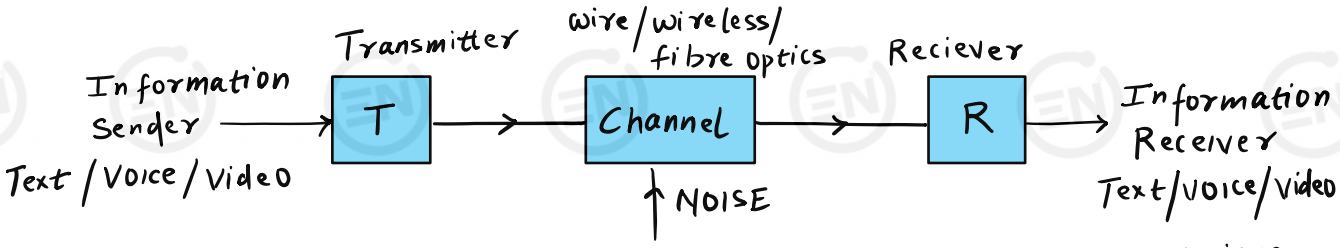
(b) Modulator
super position of signals

Reciever consists

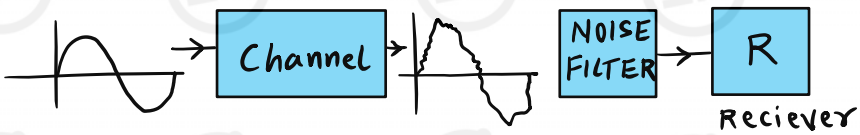
(a) Transducer
Converts electrical signal to understandable form
(Ex: Speaker, Monitor)

(b) Demodulator
Extracts information from carrier wave.

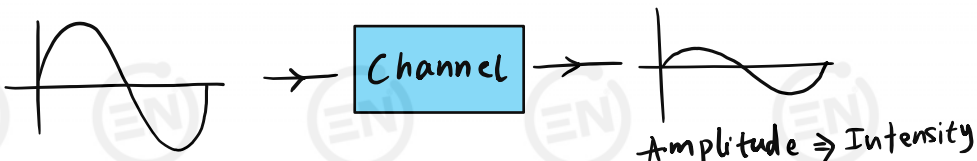
2. NOISE, ATTENUATION, AMPLIFICATION



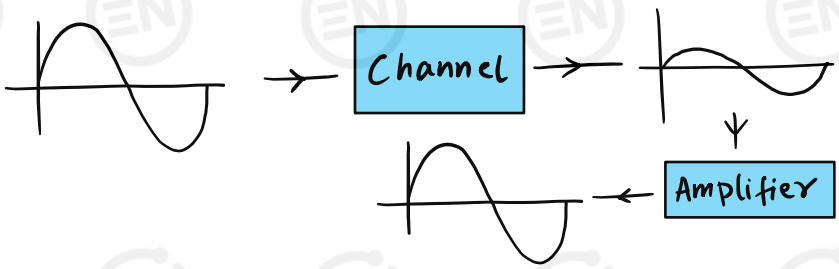
(a) NOISE : Unwanted signals disturbing transmission process.



(b) ATTENUATION : Loss of signal strength during propagation in channel.



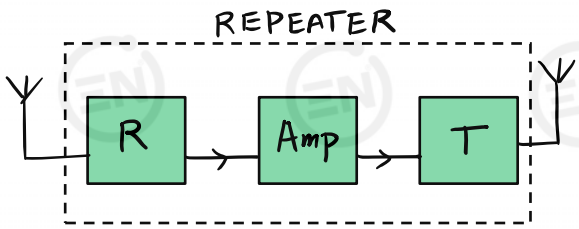
(c) Amplification : Amplitude of signal is increased

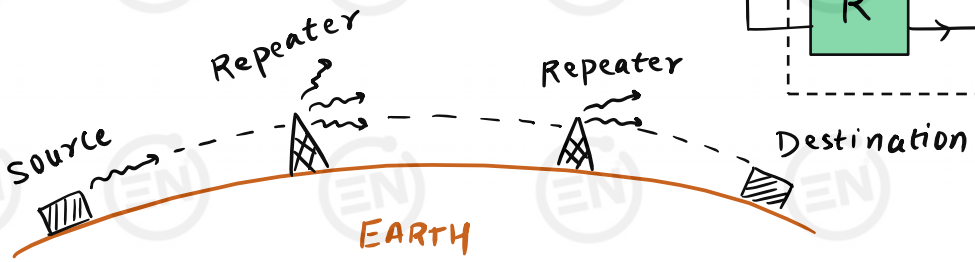


3. RANGE AND REPEATER

↳ Largest distance over which signal can be transmitted from source to destination with sufficient signal strength.

(i) REPEATER : TO increase range





4. BANDWIDTH OF SIGNALS AND TRANSMISSION MEDIUM

↳ Range of frequency used for information signal.

		Bandwidth
Speech signal	300 Hz – 3100 Hz	2800 Hz
Music signal	20 Hz – 20 kHz	20 KHz
Video signal		4.2 MHz
TV signal		6 MHz

↳ Operating frequency of Channels

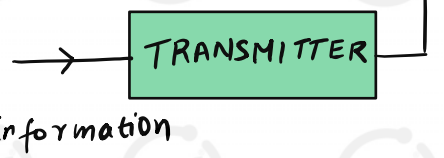
(i) Wire → Coaxiable cable, 10^9 to 10^{10} Hz

(ii) Free space → 10^3 Hz to 10^{10} Hz

(iii) optical Fibre → 10^{12} Hz to 10^{15} Hz

NUMBER OF CHANNELS $= \frac{\text{Total Bandwidth of Channel}}{\text{Bandwidth Needed Per Channel}}$

5. TRANSMISSION OF EM WAVES THROUGH SPACE



Power radiated, $P_{rad} \propto \left(\frac{l}{\lambda}\right)^2$ (l → Antenna length)

Antenna

↳ For optimum power transmission of signals, size of antenna $\geq \frac{\lambda}{4}$

Why high frequency signals for transmission?

$f \uparrow \Rightarrow \lambda \downarrow$

↓ Practical Antenna size

↓ $\lambda \downarrow \Rightarrow P_{rad} \uparrow$

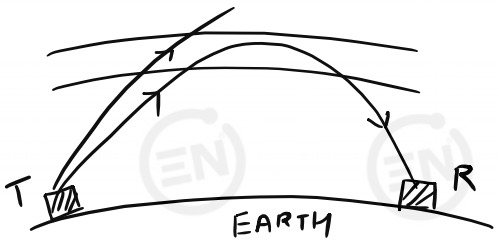
6. PROPAGATION MODES FOR EM WAVES

Antenna Ground Waves ①

→ The mode of propagation is called surface wave propagation and the wave glides over the surface of the earth

→ It is attenuated due to absorption of energy by earth surface.

Sky wave ②



↳ Long distance propagation done by Ionospheric Reflection. Ionosphere : 65 – 400 km

(a) 3 to 30 MHz is reflected

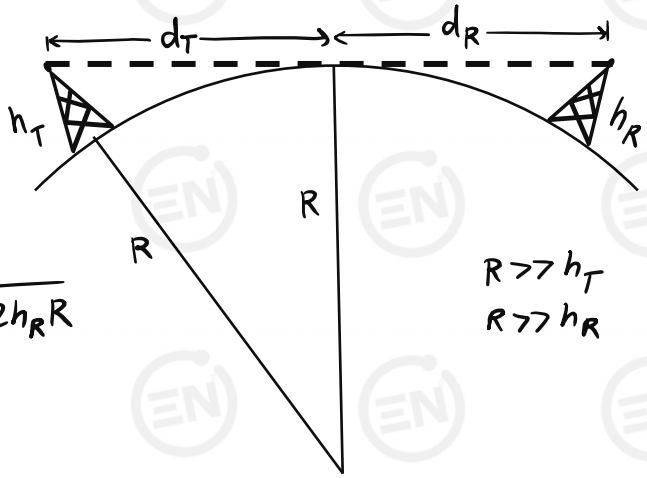
(b) $f > 30$ MHz penetrates ionosphere

↳ ③ Space Wave → for $f > 40MHz$ Line of sight Propagation is done.
→ LOS must be clear for Transmission and Reciever antenna.

$d_T = \sqrt{(h_T+R)^2 - R^2}$
$= \sqrt{h_T^2 + R^2 + 2h_TR - R^2}$
$\approx \boxed{\sqrt{2h_TR}}$

$d_R = \sqrt{2h_RR}$

∴ Max LOS distance $= d_T + d_R$
$= \sqrt{2h_TR} + \sqrt{2h_RR}$



$R >> h_T$
$R >> h_R$

7. PROBLEM IN TRANSMISSION OF BASEBAND SIGNALS (message signals)

$\ell \geq \frac{\lambda}{4}$ (Low frequency)

↳ Antenna size
↳ At low frequency, λ is too high (Ex: $f = 20kHz$, $\lambda = 15km$)
∴ Impractical antenna size.

Radiation power
$P_{rad} \propto \frac{1}{\lambda^2}$
↳ Low $f \Rightarrow \lambda \uparrow$
$\Rightarrow P_{rad} \downarrow$

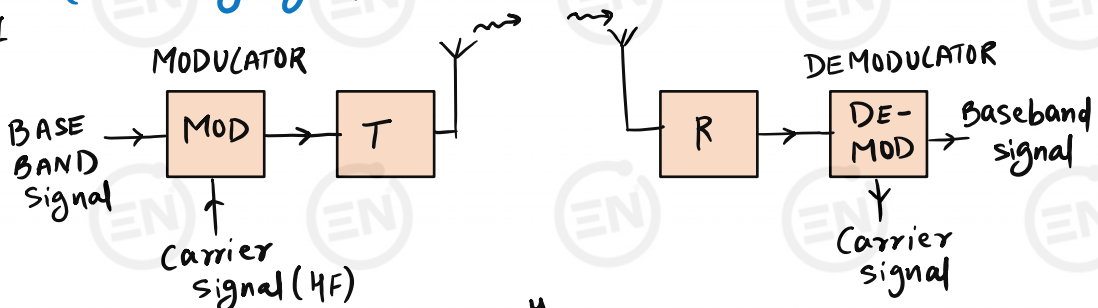
Mixing of signals
↳ If several voice Channels are broadcasted at same time, all signal gets mixed up.

Solution: by using communication at high frequencies and allotting a band of frequencies to each message signal for its transmission

8. MODULATION OF BASEBAND SIGNALS and Carrier Waves

↓ Superposition of Baseband with carrier wave

(modulating signal) (High frequency signal)



→ Sinusoidal

$y_c = A_c \sin(\omega_c t + \phi_c)$

→ Pulsating

9. TYPES OF MODULATION

$y_c = A_c \sin(\omega_c t + \phi_c)$ → During modulation: A_c, ω_c or ϕ_c can be controlled by modulating signal or information signal.

(1.) Amplitude Modulation (AM)
(2.) Frequency Modulation (FM)
(3.) Phase Modulation (PM)

AMPLITUDE MODULATION
↳ Amplitude of carrier wave is varied according to modulating signal.

10. AMPLITUDE MODULATION (AM)

Amplitude varies as per modulating signal.

$y_m = A_m \sin\omega_m t$ (Modulating signal) → AM MODULATOR ← $y_c = A_c \sin\omega_c t$ (carrier signal)

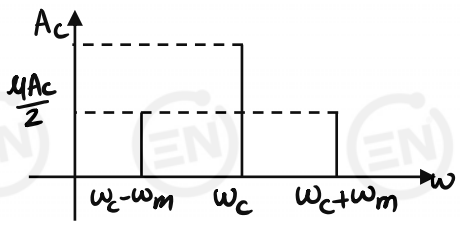
$\omega_c >> \omega_m$

→ $y = (A_c + A_m \sin\omega_m t)\sin\omega_c t$

$\Rightarrow y = A_c\left(1 + \frac{A_m}{A_c}\sin\omega t\right)\sin\omega_c t$

↳ Modulation index, $\mu = \frac{A_m}{A_c}$

$\Rightarrow y = A_c \sin\omega_c t + \frac{\mu A_c}{2}\cos(\omega_c - \omega_m)t - \frac{\mu A_c}{2}\cos(\omega_c + \omega_m)t$

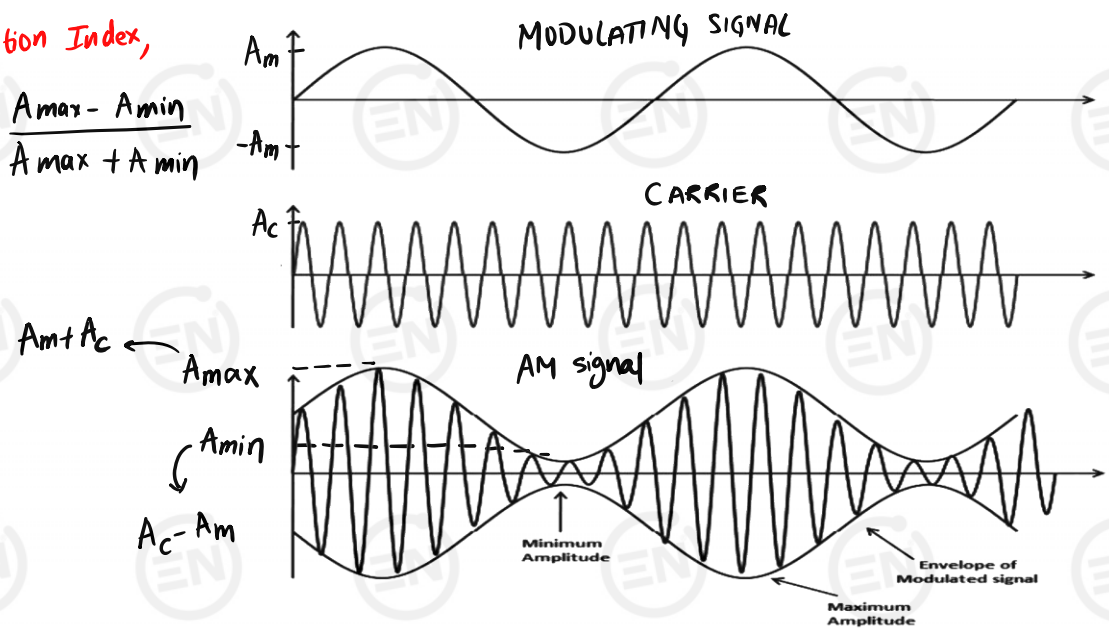


3 frequency:
(a) Carrier freq: ω_c
(b) Lower side freq: $\omega_c - \omega_m$
(c) Upper side freq: $\omega_c + \omega_m$

BANDWIDTH = $2\omega_m$ $(2f_m)$

NOTE: Modulation Index,

$\mu = \frac{A_m}{A_c} = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$



Space to add concepts learnt from PYQs if any

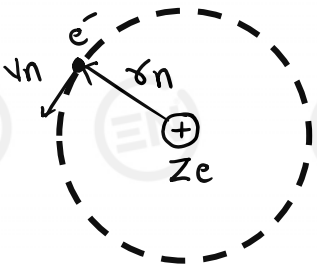
Topics to cover in Atomic Physics (PART 1 – Modern Physics)

1. Bohr's 3 Postulates
2. Radius, Speed, Angular Freq, Time Period & Energy in nth Orbit
3. Energy Level of Hydrogen type atom (1e system)
4. Excitation of atom
5. Wavelength of emitted radiation
6. Number of spectral lines
7. Hydrogen Spectral Series

Note: For video refer Revision Series Playlist in EDUNITI YouTube Channel

1. BOHR'S POSTULATES (for single e⁻ system)

1st



$\frac{Kze^2}{r_n^2} = \frac{mv_n^2}{r_n}$

2nd

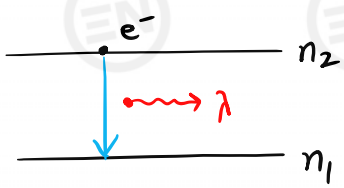
Bohr's Quantization

$mv_n r_n = \frac{nh}{2\pi}$

$(n \in I)$

3rd

Release of energy



$\frac{hc}{\lambda} = E_{n_2} - E_{n_1}$

2. BOHR'S MODEL (1e⁻ system)

$r_n \propto \frac{1}{m_e}$

→ Radius of n^{th} orbit, $r_n = \frac{n^2h^2}{4\pi^2 kze^2 m} = 0.529 \times \frac{n^2}{z}$ Å

→ Velocity in n^{th} orbit, $v_n = \frac{2\pi kze^2}{nh} = 2.18 \times 10^6 \times \frac{z}{n}$ m/s

→ $\omega_n = \frac{v_n}{r_n}$ $\omega_n \propto \frac{z^2}{n^3}$ rad/s

→ $T_n = \frac{2\pi}{\omega_n}$ $T_n \propto \frac{n^3}{z^2}$ s

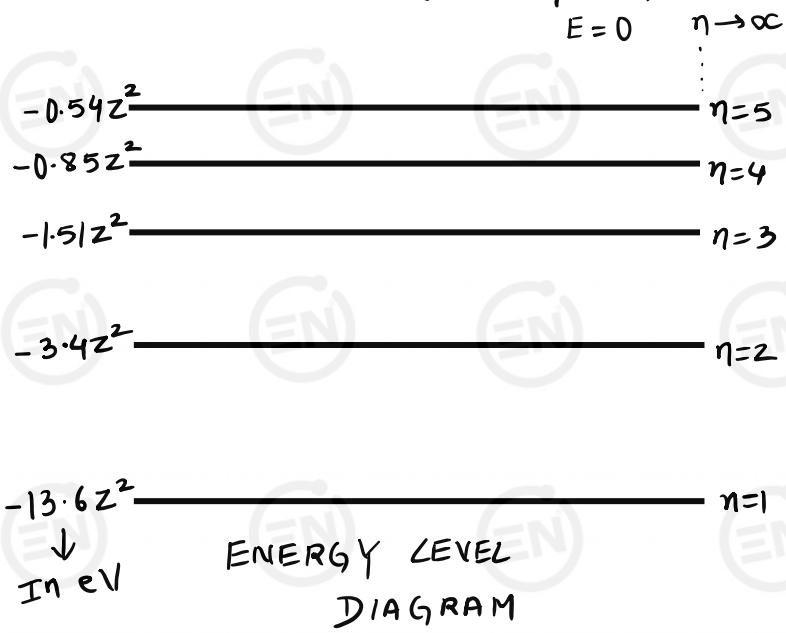
} Focus more on relations

→ $E_n = K_n + U_n = -\frac{Kze^2}{2r_n} = -13.6 \times \frac{z^2}{n^2}$ eV

3. ENERGY LEVEL OF HYDROGEN TYPE ATOM (1 e^- system)

$E_n = -13.6 \times \frac{Z^2}{n^2}$ eV

NOTE: Learn them for speed solving.



ENERGY LEVEL DIAGRAM

4. EXCITATION OF ATOM

↳ For e^- to absorb energy and excite from n_1 to n_2, the energy absorbed must be exactly equal to $E_{n_2} - E_{n_1}$

ex:

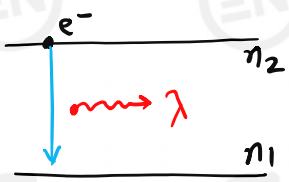
$n=3$ ——— -1.51 eV

$n=1$ ——— -13.6 eV

$\therefore E_3 - E_1 = 12.09$ eV

* Thus 12.09 eV of energy must be absorbed.

5. λ OF EMITTED RADIATION



$\frac{hc}{\lambda} = E_{n_2} - E_{n_1}$

$\Rightarrow \frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$, $R \sim 10^7\ m^{-1}$

↳ Rydberg's Constant

IMPORTANT:

(a) $\lambda = \frac{12430}{\Delta E\ (\text{in eV})}\ \text{Å}$ or $= \frac{1243}{\Delta E\ (\text{in eV})}\ \text{nm}$

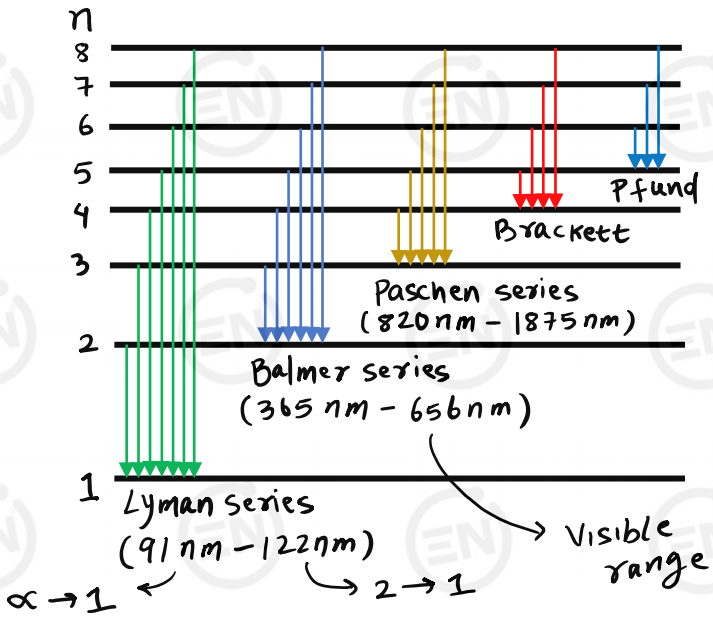
(b) $\Delta E = \frac{12430}{\lambda(\text{Å})}\ \text{eV}$

6. NUMBER OF SPECTRAL LINES

↳ possible number of photon energies emitted due to de excitation of e^- from $n = n_2$ to $n = 1$ state

$$= {}^{n}C_2 = \frac{n(n-1)}{2}$$

7. HYDROGEN SPECTRAL SERIES



Space to add concepts learnt from PYQs if any

Topics to cover in Photoelectric Effect (PART 2 – Modern Physics)

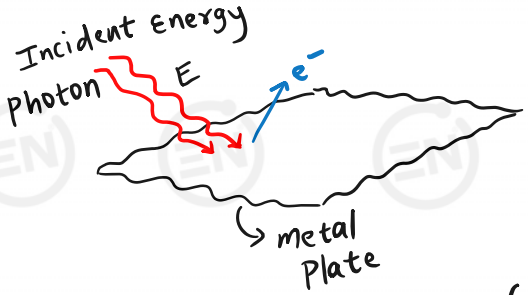
1.Electron Emission (Work Function & Types of Emission)
2.Photoelectric Emission
3.Effect of Intensity and Temperature on Photoelectric Effect
4.Stopping Potential & Graphs
Note: For video refer Revision Series Playlist in EDUNITI YouTube Channel

1. ELECTRON EMISSION

↳ Work function, ϕ (Minimum energy required to eject e^- from metal surface)

TYPES	ENERGY SOURCE
Photoelectric Emission	Electromagnetic radiation
Thermionic Emission	Heat
Field Emission	External Electric Field
Secondary Emission	Collision (bombarding α-Particle)

2. PHOTOELECTRIC EMISSION



(a) Threshold frequency (ν_{th}), Threshold wavelength (λ_{th})

$$\phi = h\nu_{th} = \frac{hc}{\lambda_{th}}$$

ν_{th}: minimum freq. to start photoelectric effect.

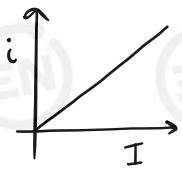
(b) If $\nu > \nu_{th}$ $(E > \phi)$

e^- comes out with V_{max}, $\frac{1}{2}mv_{max}^2 = E - \phi$

$\Rightarrow K_{max} = h\nu - h\nu_{th}$

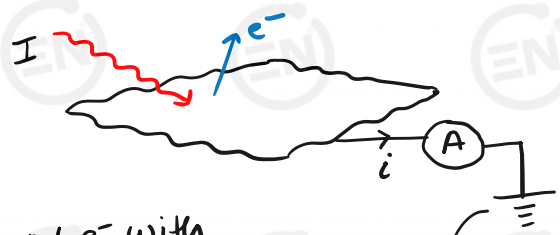
NOTE: e^- may come out with $V < V_{max}$ if it collides with other e^-.

3. EFFECT OF INTENSITY (I) and TEMP° (T) ON PHOTOELECTRIC EFFECT



(a) If I ↑ ⇒ i also ↑

(b) If Temp° ↑ ⇒ No effect



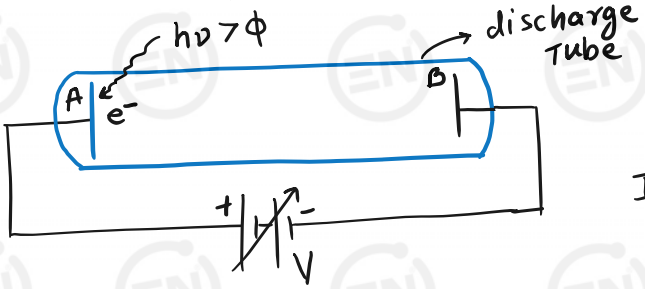
Grounded to keep plate neutral.

NOTE: ↑ I doesn't ↑ K.E of e⁻ with which it comes out.

$[K = h\nu - \phi]$

K depends on ν of incident energy.

4. STOPPING POTENTIAL



(a) e⁻ with V_{max} moves to B

(b) Battery does −ve work

(c) so energy at B = $K_{max} - eV$

If for a $V = V_0$, $K_{max} - eV_0 = 0$

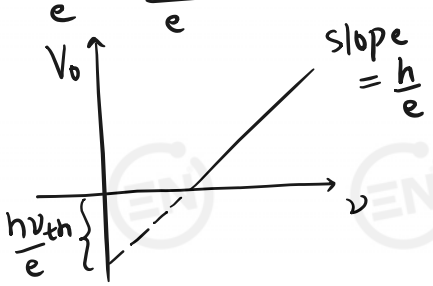
⇒ V_0 is stopping potential

$\therefore$ $\boxed{eV_0 = K_{max}}$

(d) $V_0 = \frac{K_{max}}{e} = \frac{E - \phi}{e} = \frac{h\nu}{e} - \frac{h\nu_{th}}{e}$

$\boxed{V_0 = \frac{h}{e}\nu - \frac{h\nu_{th}}{e}}$

Einstein Photoelectric equation



Space to add concepts learnt from PYQs if any

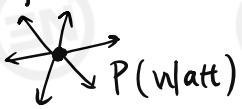
Topics to cover in Dual Nature of Light (PART 3 – Modern Physics)

1.Photon Flux & Photon Density
2.Wave Particle Duality
3.De Broglie's Hypothesis
4.Radiation Force and Radiation Pressure (Projected Area Concept)
5.Atom Recoil During De-Excitation

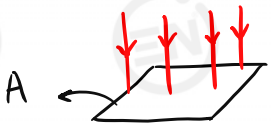
Note: For video refer Revision Series Playlist in EDUNITI YouTube Channel

1. PHOTON FLUX / PHOTON DENSITY

↳ Number of photons emitted /sec, $N = \frac{P}{hc/\lambda} = \boxed{\frac{P\lambda}{hc}}$



↳ PHOTON FLUX, ϕ_P (no. of photons per sec per unit Area)



$\phi_P = \frac{N}{A} = \frac{1}{A} \times \frac{P\lambda}{hc} = \boxed{\frac{I\lambda}{hc}}$

↳ PHOTON DENSITY, $\rho_N = \frac{\phi_P}{c} = \boxed{\frac{I\lambda}{hc^2}}$ { Just put I to get ρ_N

Ex: P ..... r (POINT SOURCE) $I = \frac{P}{4\pi r^2}$

2. WAVE PARTICLE DUALITY

PARTICLE NATURE
(a) Treated as photon

(b) Energy, $E = pc$ (p: momentum)

WAVE NATURE
Treated as Em waves

$E = \frac{hc}{\lambda}$

$\therefore pc = \frac{hc}{\lambda}$

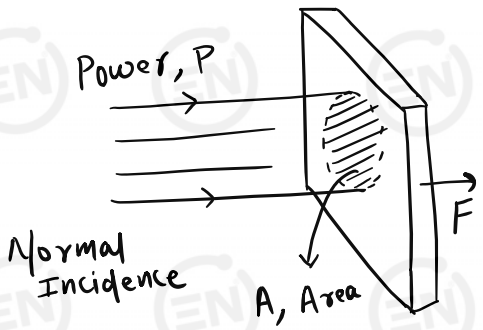
$\Rightarrow \boxed{p = \frac{h}{\lambda}}$ → photon momentum

3. DE BROGLIE'S HYPOTHESIS

(If Light behaves as particle then Physical particle too can behave as waves)

$\lambda = \frac{h}{p}$ or $\boxed{\lambda = \frac{h}{mv}}$

4. RADIATION FORCE / PRESSURE



(a) Mostly body is black body (absorbs all light)

(b) Momentum is tranferred to body

(c) Thus body experiences FORCE (RADIATION FORCE)

F = no of photons/sec × momentum change

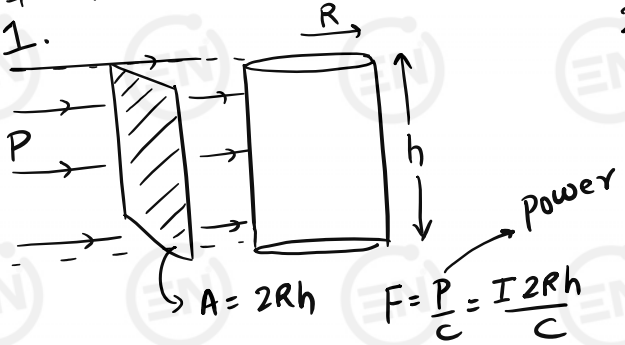
$$= \frac{P\lambda}{hc} \times \frac{h}{\lambda} = \boxed{\frac{P}{c}}$$ { P: Power, P = IA

Radiation Pressure $= \frac{F}{A} = \frac{P/c}{A} = \boxed{\frac{I}{c}}$

NOTE: If surface is perfectly reflective, $F = \frac{2P}{c}$, Pressure $= 2I/c$

PROJECTED AREA

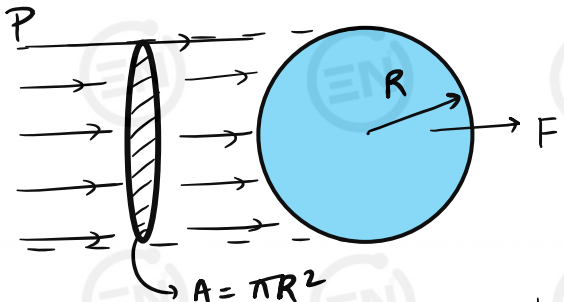
1.



$F = \frac{P}{c} = \frac{I\,2Rh}{c}$ (Power)

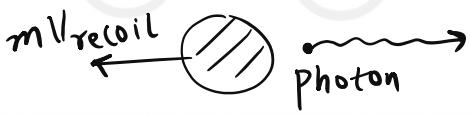
(Assuming Black Body)

2.



$F = \frac{P}{c} = \frac{I\pi R^2}{c}$ { Black Body

5. ATOM RECOIL DURING DE-EXCITATION



$\lambda = \frac{12430\ A^\circ}{\Delta E\ (\text{in eV})}$

Hydrogen type atom (1 e^- system)

$\therefore\ mV_{recoil} = \frac{h}{\lambda}$

$\Rightarrow \boxed{V_{recoil} = \frac{h}{\lambda m}}$

↳ m: mass of atom

λ: wavelength of Photon

h: Planck's Constant

Space to add concepts learnt from PYQs if any

Topics to cover in Radioactivity (PART 4 – Modern Physics)

1. Radioactivity (activity & units)
2. Radioactive Decay Law
3. Half Life Time
4. Mean Life Time
5. Simultaneous Decay Equation
6. Radioactive Series

Note: For video refer Revision Series Playlist in EDUNITI YouTube Channel

1. RADIOACTIVITY (ACTIVITY, UNITS)

(a) Unstable nucleus disintegrate spontaneously.

(b) This phenomena of disintegration is called "ACTIVITY", A_c
Also known as decay rate
$A_c = -dN/dt$

$X \longrightarrow Y$
Parent Nucleus → Daughter Nucleus

(c) UNIT of Activity is dps (decay per sec)

- 1 Bq (Becquerel) = 1 dps
- 1 Ci (Curie) = 3.7×10^{10} dps = 3.7×10^{10} Bq
- 1 Ru (Rutherford) = 10^6 dps

2. RADIOACTIVE DECAY LAW ($X \xrightarrow{\lambda} Y$)

X → ACTIVE NUCLEI

(a) Activity $\propto$ Number of Active nuclei

$-\frac{dN}{dt} \propto N \Rightarrow -\frac{dN}{dt} = \lambda N$ {Activity, $A_c = \lambda N$}

λ → Decay Constant (tells how fast decay occurs)

Integrate

Radioactive decay eqn: $N = N_0 e^{-\lambda t}$

(i) N_0 is NO of Nuclei at $t=0$

(ii) $A_c = A_{c0} e^{-\lambda t}$

3. HALF LIFE TIME (T)

↳ time taken to become half

At $t=0, N=N_0$

$t=T, N=\frac{N_0}{2}$

$\Rightarrow \frac{N_0}{2} = N_0 e^{-\lambda T}$

$\Rightarrow \boxed{T = \frac{\ln 2}{\lambda}}$ or $\frac{0.693}{\lambda}$

NOTE: Radioactive decay eqn in terms of T:

$N = N_0 e^{-\lambda t} \Rightarrow N = N_0 e^{-\frac{\ln 2}{T} t}$

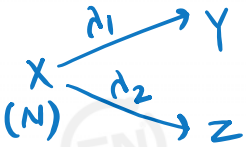
$\boxed{N = N_0 (2)^{-t/T}}$

and, $A_c = A_{c0} (2)^{-t/T}$

4. MEAN LIFE TIME

$\boxed{\tau = \frac{1}{\lambda}}$

5. SIMULTANEOUS DECAY



$A = \lambda_1 N + \lambda_2 N$

$\Rightarrow -\frac{dN}{dt} = N(\lambda_1 + \lambda_2)$

$\boxed{N = N_0 e^{-(\lambda_1+\lambda_2)t}}$

6. RADIOACTIVE SERIES

$A_1 \xrightarrow{\lambda_1} A_2 \xrightarrow{\lambda_2} A_3 - - - - A_n$

↳ Stable end product

Artificial as $T_{1/2}$ is too Low and so don't exist in nature

	SERIES	PARENT	END PRODUCT
4n	THORIUM	$_{90}Th^{232}$	$_{82}Pb^{208}$
4n+1	NEPTUNIUM	$_{93}Np^{237}$	$_{83}Bi^{209}$
4n+2	URANIUM	$_{92}U^{238}$	$_{82}Pb^{206}$
4n+3	ACTINIUM	$_{92}U^{235}$	$_{82}Pb^{207}$

Space to add concepts learnt from PYQs if any

Topics to cover in Nuclear Physics (PART 5 – Modern Physics)

1. How to write a Nuclear Reaction
2. Nuclear Force
3. Nucleus Size & Stability of Heavy Nucleus
4. Nuclear Binding Energy
5. Binding Energy Per Nucleon
6. Nuclear Fusion & Fission
7. Alpha, Beta & Gamma decay

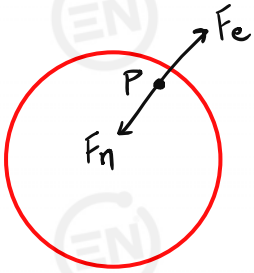
Note: For video refer Revision Series Playlist in EDUNITI YouTube Channel

1. HOW TO WRITE A NUCLEAR REACTION

$C + c \longrightarrow D + d$

- C: Initial nuclei
- c: Attacking Particle
- D: END NUCLEI
- d: emitting Particle (RADIATION)

$C(c,d)D$



2. NUCLEAR FORCE

$^{A}_{Z}X$

Mass no. ← A

Atomic no. ← Z

→ Strong attractive short range force.

$F_{nn} = F_{np} = F_{pp}$

(i) F_e : repulsive electrostatic force

F_n : net attractive nuclear force

$F_e = F_n \Rightarrow$ Stable Nucleus

3. NUCLEUS SIZE AND STABILITY OF HEAVY NUCLEUS

↳ Size of Nucleus ∝ Atomic mass

$\Rightarrow \frac{4}{3}\pi R^3 \propto A \Rightarrow \boxed{R = R_0 A^{1/3}}$

R_0 : fermi-const. $R_0 \sim 10^{-15}$ m

↳ If $R \uparrow \Rightarrow F_n \downarrow$

So, Nucleus gets unstable $\Rightarrow$ Decay starts

$X \longrightarrow Y +$ Radiation

4. NUCLEAR BINDING ENERGY

the energy required to separate an atomic nucleus completely into its constituent protons and neutrons, or, equivalently, the energy that would be liberated by combining individual protons and neutrons into a single nucleus.

$Z\,p + (A-Z)\,n \longrightarrow {}^{A}_{Z}X + E_b$ (BINDING ENERGY)

NOTE: When reactants combine to form stable product, THERE IS MASS LOSS called "MASS DEFECT"

$\boxed{\Delta m = Z m_p + (A-Z) m_n - M_X}$

and, $E_b = \Delta m c^2$ { Δm if in AMU, 1 AMU = 1.66×10^{-27} kg

or

$\boxed{E_b = \Delta m \text{ (in AMU)} \times 931.5 \text{ MeV}}$

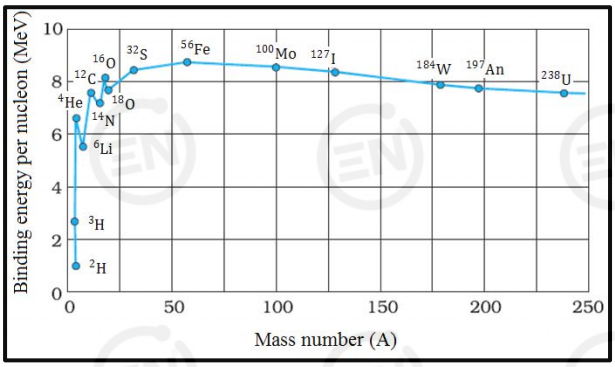
5. BINDING ENERGY PER NUCLEON

↳ $_{Z}^{A}X$ A is no of nucleons

NOTE: E_b is lower for $A<30$ and $A>170$

BE/nucleon $= \frac{E_b}{A}$

↓ tells how stable is a nucleus



6. NUCLEAR FISSION AND FUSION

↳ $A + B \longrightarrow C + \Delta E_1$ (JOINING of nucleus)

↓

$X \longrightarrow Y + Z + \Delta E_2$ (splitting)

(i) In both Energy released

(ii) $\Delta E_1 = (m_A + m_B - m_C)c^2$

$\Delta E_2 = (m_X - m_Y - m_Z)c^2$

*NOTE: Energy released (+VE) or even supplied (-VE) is called Q-Value

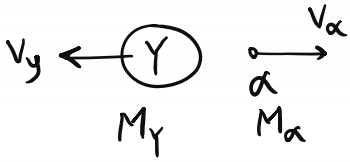
$Q = \Delta m c^2$

7. Alpha, BETA AND GAMMA DECAY

(a) α-decay ($_2^4He$ Nucleus)

$$_{Z}^{A}X \longrightarrow {}_{Z-2}^{A-4}Y + {}_{2}^{4}\alpha + E, \quad Q = (M_X - M_Y - M_\alpha)c^2$$

NOTE: THIS energy is released in form of kinetic energy.



$M_Y V_Y = M_\alpha V_\alpha$ — (i), $Q = \frac{1}{2}M_Y V_Y^2 + \underbrace{\frac{1}{2}M_\alpha V_\alpha^2}_{K_\alpha}$ — (ii)

from (i) and (ii):

$$K_\alpha = \frac{Q M_Y}{M_\alpha + M_Y} = \boxed{\frac{Q(A-4)}{A}}$$

(b) Beta Decay (e^- or e^+)

↓ electron ↳ positron

$\nu \rightarrow$ neutrino

$\bar{\nu} \rightarrow$ antineutrino

β^+ Decay

$p \longrightarrow n + e^+ + \nu$

$$_{Z}^{A}X \longrightarrow {}_{Z-1}^{A}Y + e^+ + \nu$$

$\Delta m = M_X - M_Y - 2m_e$

β^- Decay

$n \longrightarrow p + e^- + \bar{\nu}$

$$_{Z}^{A}X \longrightarrow {}_{Z+1}^{A}Y + e^- + \bar{\nu}$$

$\Delta m = M_X - M_Y$

K capture

$p + e^- \longrightarrow n + \nu$

$$_{Z}^{A}X \longrightarrow {}_{Z-1}^{A}Y + \nu$$

$\Delta m = M_X - M_Y$

(C) Gamma Decay (EM Radiation) (In K capture, e^- is captured from K-shell)

$X \longrightarrow Y^* + \alpha$, $Y^* \longrightarrow Y + \gamma$

Y^* : Excited state

$\Rightarrow X \longrightarrow Y + \alpha + \gamma$

NOTE: It can happen even for β-decay

Space to add concepts learnt from PYQs if any

Topics to cover in X-Rays (PART 6 – Modern Physics)

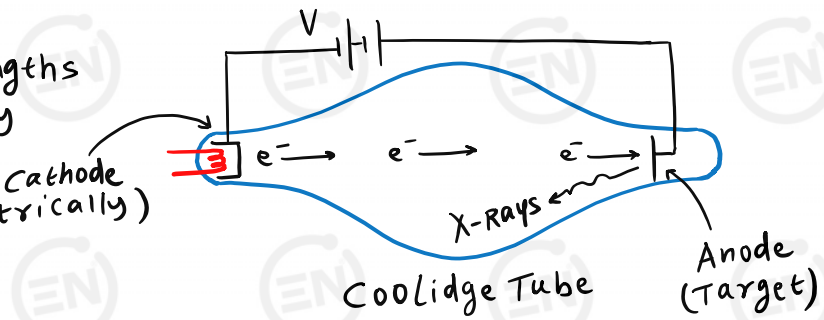
1.Soft and Hard X-Rays
2.Production of X-Rays : Coolidge Tube
3.Continuous X-Ray Production
4.Characteristic X-Ray Production
5.Complete Spectrum
6.Moseley's Law

Note: For video refer Revision Series Playlist in EDUNITI YouTube Channel

1. X-RAYS ($\approx 1\,A^\circ$)

- Soft X-RAYS
  - High wavelength
  - Low energy
- Hard X Rays
  - Low wavelengths
  - High Energy

2. PRODUCTION OF X-RAYS (X-ray tubes)



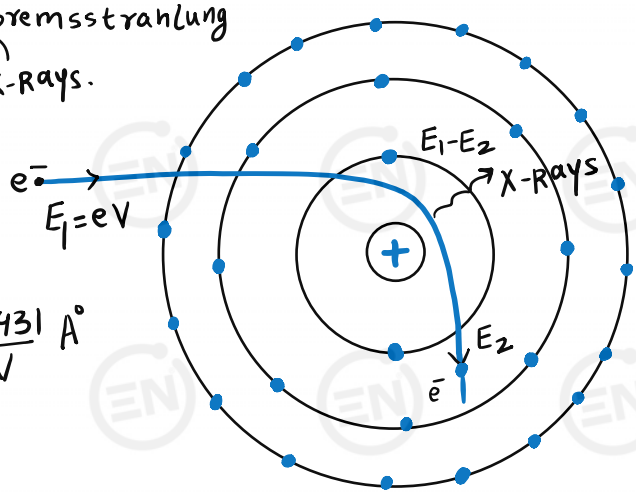
X-Rays are produced by incidence of accelerated e^- on target material.

- Continuous X-Rays (Bremsstrahlung)
- Characteristic X-Rays

3. CONTINUOUS X-RAY

This Phenomena is called Bremsstrahlung

↳ Deceleration of e^- when deflected by atomic nucleus causes production of X-Rays.

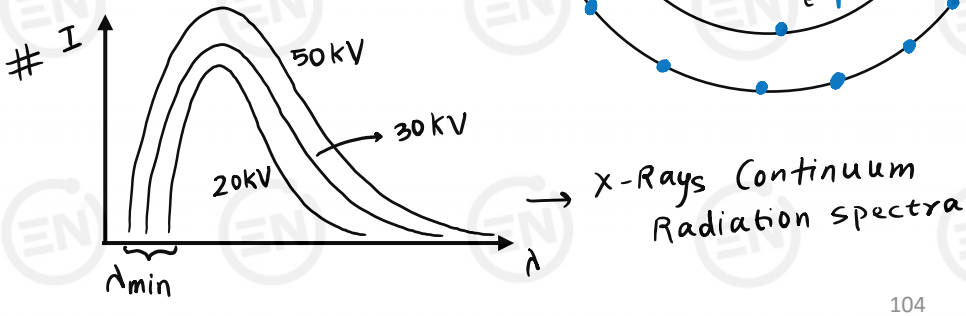


(a) Energy of X-rays, $E = E_1 - E_2$

$E_{max} = E_1 = eV \quad (E_2 = 0)$

$\therefore$ Cutoff Wavelength of X-Ray,

$$\# \lambda_{min} = \frac{hc}{E_{max}} = \frac{hc}{eV} = \frac{12431}{V}\,A^\circ$$



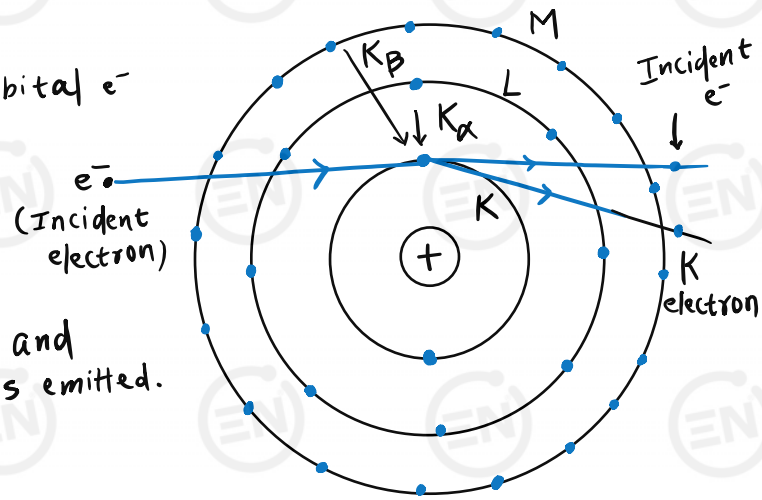
4. CHARACTERISTIC X-RAY

(i) some incident e^- knocks off orbital e^- of K, L, M .. shell.

(ii) If eV > Binding Energy of "K shell e^-", only then it is removed.

(iii) e^- from L, M, N.. can jump to K and during this photon (x-ray) is emitted.

$\lambda = \frac{hc}{\Delta E}$



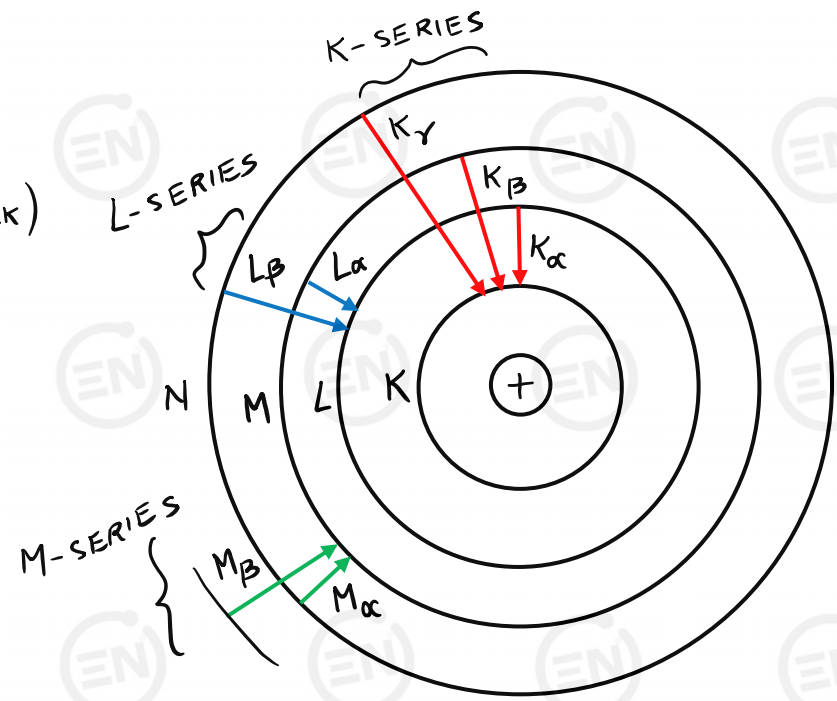
(a) K_α X-Ray → If e^- jumps from L → K
(b) K_β → e^- jumps from M → K
(c) K_γ → e^- jumps from N → K

} K-Series

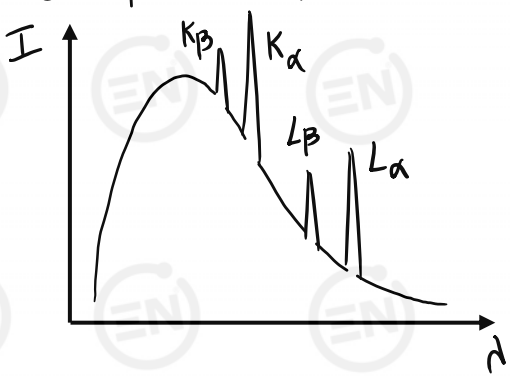
NOTE:

(i) If you compare K_β and K_α

$\lambda_{K_\beta} < \lambda_{K_\alpha}$ $(\because \Delta E_{MK} > \Delta E_{LK})$



5. COMPLETE SPECTRUM



K_β : M → K
K_α : N → K
L_β : N → L
L_α : M → L

6. MOSELEY'S LAW $\left(\sqrt{\nu} = a(z-\sigma)\right)$

λ of characteristic X-Rays :

$$\frac{1}{\lambda} = R(z-\sigma)^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$z-\sigma$ → effective atomic no

$$\therefore \nu = \frac{c}{\lambda}$$

$$\Rightarrow \nu = Rc(z-\sigma)^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

For K-Series
$\sigma = 1$
$R : 10^7\ m^{-1}$
↳ Rydbergs const.

$$\therefore \sqrt{\nu} = a(z-\sigma)$$

$$a = \sqrt{Rc\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)}$$

↳ square root of frequency is linearly proportional to Atomic number.

Space to add concepts learnt from PYQs if any